

Operator Methods for the Weil Criterion: Q3

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Abstract

Background: The Riemann Hypothesis (RH) is equivalent, by Weil, to the nonnegativity of the quadratic functional $Q^*(t; \cdot)$ on an explicit cone of even, compactly supported test functions. Establishing $Q^*(t; \Phi) \geq 0$ on the full Weil class requires a precise chain of analytic inputs: normalization, local density, continuity, a Toeplitz–symbol bridge, control of the prime contribution, and a compact-by-compact limit.

Main result: We present a self-contained operator-theoretic proof of the analytic module chain (T0)+(A1′)+(A2)+(A3)+(RKHS). Starting from the Guinand–Weil normalization (T0), we construct Fejér×heat dictionaries that are dense in each compact window $[-K, K]$ (A1′) and obtain Lipschitz control of $Q^*(t; \cdot)$ (A2). The Toeplitz bridge (A3) provides a positive symbol margin via Szegő–Böttcher theory and an explicit modulus of continuity. A purely analytic RKHS contraction yields a uniform bound on the prime operator, completing the mixed estimate on every compact. Density and continuity then propagate positivity from each W_K to the full Weil class.

Conclusion: Combining these ingredients we obtain $Q^*(t; \Phi) \geq 0$ for every Φ in the Weil cone $\mathcal{W}$ (even, nonnegative tests generated by Fejér×heat windows), and by Weil’s positivity criterion this yields the Riemann Hypothesis within our normalization.

1 Introduction

Background and motivation

We establish that a canonical quadratic form on the Weil test class is nonnegative under the analytic module stack (T0)+(A1′)+(A2)+(A3)+(RKHS); by the Weil criterion this suffices to deduce the Riemann Hypothesis. The entire argument is analytic: every bound is established on paper from explicit inequalities, the parameters are given in closed form, and the uniform bridge avoids K -dependent schedules. No numerical tables or automated certificates enter the main proof.

Main result

Theorem 1.1 (Main result, informal). *Let $Q^*(t; \cdot)$ be the quadratic form fixed in Section 6 on the Weil class $\mathcal{W}$. Then*

$$Q^*(t; \Phi) \geq 0 \quad \text{for all } \Phi \in \mathcal{W},$$

Via Theorem 12.1 (the Weil criterion) this positivity is equivalent to the Riemann Hypothesis.

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Program diagram

Our proof is organized as a controlled-limit program: a small number of analytic bridges are composed, and each bridge is driven by explicit parameter schedules along a compact exhaustion. This “limits with a bridge” architecture mirrors the standard programmatic style in mathematical physics, where theorems are structured around robust intermediate objects and stable limit transitions (compare the two-step limiting diagram in Deng–Hani–Ma [11]). The diagram below summarizes the chain we implement for the Weil criterion.

$$\begin{array}{c}
\text{Weil criterion} \Leftarrow \text{Weil positivity on } \mathcal{W} \\
\uparrow \\
\text{PSD on each } W_K \Leftarrow \text{Toeplitz barrier} + \text{uniform RKHS cap} \\
\uparrow \\
\text{cone density} + \text{Lipschitz control (uniform A3 bridge)}
\end{array}$$

The proof organises around three analytic modules.

Archimedean bridge

(A3) *Archimedean Toeplitz barrier.* The Toeplitz component $T_M[P_A]$ in the Rayleigh decomposition of $Q^*(t; \Phi_{B,t})$ is bounded from below by the explicit uniform Archimedean floor c_* (Lemma 9.19), up to the controllable Lipschitz loss $C \omega_{P_A}(1/(2M))$. Szegő–Böttcher asymptotics together with an explicit modulus of continuity for P_A yield

$$\lambda_{\min}(T_M[P_A]) \geq c_* - C \omega_{P_A}\left(\frac{1}{2M}\right),$$

as developed in Section 9.

Prime contraction

(RKHS) *Uniform prime contraction without tables.* The prime contribution is encoded by the RKHS operator $T_P^{\text{RKHS}}(t)$ supported on the nodes $\xi_n = \frac{\log n}{2\pi}$; in the Weil functional we use the one-sided weights $w_Q(n) = 2\Lambda(n)/\sqrt{n}$, while in the RKHS analysis we keep the undoubled operator weights $w_{\text{RKHS}}(n) = \Lambda(n)/\sqrt{n}$. Section 10.5 develops a *tables-free* uniform cap from the closed form $\rho(t)$; fixing $t = t_{\text{crit}}$ (Corollary 9.22) ensures $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq c_*/4$ once $c_* > 0$ is established. This cap is independent of K and suffices for the uniform A3 bridge. The adaptive Gram-geometry and early/tail routes are archived for reference in Appendix G.

Compact extension

(A1' + A2) *Density and continuity.* The uniform A3 bridge yields $Q^*(t; \Phi) \geq 0$ on the Fejér×heat generator cone inside each W_K . By density (A1') and continuity (A2), this extends to all of W_K , and taking the union over K gives $Q^*(t; \Phi) \geq 0$ on the full Weil cone $\mathcal{W}$.

Outline of the proof

Combining the Toeplitz barrier and the uniform RKHS cap yields, on each W_K ,

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq c_* - C \omega_{P_A}\left(\frac{1}{2M}\right) - \rho(t).$$

Selecting $M \geq M_0^{\text{unif}}$ gives $C \omega_{P_A}(1/(2M)) \leq c_*/2$, and fixing $t = t_{\text{crit}}$ yields $\rho(t) \leq c_*/4$, so whenever $c_* > 0$ one has

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \frac{c_*}{4} > 0.$$

Hence $Q^*(t; \Phi) \geq 0$ on the Fejér×heat cone inside W_K . Density (A1') and continuity (A2) extend this to all of W_K , and the union over K gives $Q^*(t; \Phi) \geq 0$ on $\mathcal{W}$, proving Theorem 12.4. Finally Section 12 applies Theorem 12.1 to convert this positivity into the Riemann Hypothesis.

What is new

Two features distinguish the present work.

1. **A consistent symbol definition.** The Archimedean symbol is defined once via periodization of the Fejér×heat window, and its Lipschitz modulus is proved directly for that same object (Lemma 9.11), eliminating mixed definitions.
2. **A uniform prime cap.** The RKHS bound is set at $t = t_{\text{crit}}$ and is independent of K , removing all K -dependent schedules from the main proof chain.

Organization of the paper

Section 6 recalls the Weil class, the quadratic form $Q^*(t; \cdot)$, and the Guinand–Weil normalization. Section 9 establishes the Archimedean Toeplitz barrier (A3). Section 10.5 develops the uniform RKHS prime cap at the fixed scale $t = t_{\text{crit}}$. Section 12 links compact positivity to the full Weil class and states the main theorem together with its Weil corollary. Appendix G archives the legacy K -dependent branches and reproducibility data.

Notation

We write Λ for the von Mangoldt function, $\xi_n = \frac{\log n}{2\pi}$ for the sampling nodes, $w_Q(n) = 2\Lambda(n)/\sqrt{n}$ for the weights inside the Weil functional, and $w_{\text{RKHS}}(n) = \Lambda(n)/\sqrt{n}$ (with $w_{\max} = \sup_n w_{\text{RKHS}}(n) \leq 2/e$) for the operator analysis. The heat kernel is $k_t(x, y) = \exp(-\frac{(x-y)^2}{4t})$. Compact windows are denoted $W_K = [-K, K]$, and $\mathcal{W} = \bigcup_{K>0} W_K$ is the Weil cone. Complete conventions appear in Section 4.

Analytic modules at a glance

Stage legend. (T0) fixes the Guinand–Weil normalization of the Weil functional. (A1') proves density of the Fejér×heat generator cone on each compact, and (A2) supplies Lipschitz continuity so that positivity propagates from the generators to all even nonnegative tests. (A3) is the Toeplitz bridge: it splits $Q^*(t; \Phi_{B,t})$ into an Archimedean Toeplitz symbol and a finite-rank prime block with explicit lower bounds on $\lambda_{\min}$. The prime contribution is controlled by the uniform RKHS cap in Section 10.5. Legacy K -dependent chains (MD/IND/AB and T5) are archived in Appendix G.

Dependency map for the analytic chain

Module	Key statement	Consumed by
T0	Proposition 6.1 (Guinand–Weil normalization)	Theorem 12.4, Theorem 12.2
A1′	Theorem 7.3 (Density on W_K)	Theorem 12.4
A2	Lemma 8.3 / Corollary 8.4 (Lipschitz control)	Theorem 12.4
A3	Theorem 9.35 (Uniform A3 bridge)	Theorem 12.4
RKHS	Corollary 9.22 (Uniform prime cap)	Theorem 12.4
MAIN	Theorem 12.4 (Weil positivity on W)	Theorem 12.2
WEIL	Theorem 12.1 (Weil criterion)	Theorem 12.2

Assumption stack. When we write “under (T0) + (A1′) + (A2) + (A3) + (RKHS)” we mean precisely the data enumerated above: a fixed normalization, cone density, Lipschitz control, the mixed Toeplitz lower bound, and the uniform RKHS cap. Legacy K -dependent chains are archived and not used in the main proof.

Verification aids. Appendix C records reproducibility collateral; Appendix F records an analytic framework and optional numerical exploration. These materials are not used as premises in any theorem. All bounds invoked in the main chain are proved in-line from explicit analytic estimates.

1.1 Contemporary Context and Inspiration

This work was inspired by several recent developments in analytic number theory, computational complexity, and mathematical logic:

- **Analytic criteria.** Li’s positivity sequence [20] and the Jensen polynomial programme of Griffin–Ono–Rolen–Zagier [15] give logically equivalent restatements of RH; both inspire our insistence on keeping every cone generator and Lipschitz bound explicit.
- **Zero-density breakthroughs.** The new Dirichlet-polynomial bounds of Guth and Maynard [17] illustrate how much can be gained by encoding the zeta problem as a spectral estimate, a viewpoint we adopt through the Toeplitz bridge.
- **Near-miss invariants.** Rodgers and Tao’s work on the de Bruijn–Newman constant [28] shows that RH may be “barely true”, motivating a watchdog table that certifies every slack we introduce along the chain.
- **Geometric and noncommutative ideas.** Fesenko’s two-dimensional adelic programme [13] and the Connes–Marcolli noncommutative approach [9] highlight how positivity hinges on careful operator factorizations, reinforcing our choice to stay within verifiable Toeplitz/RKHS settings.
- **Physical operator heuristics.** PT-symmetric constructions such as Bender–Brody–Müller [3] keep the Hilbert–Pólya dream alive; our framework aims to supply the missing rigorous operator inequalities.
- **Geometric flows and smoothing.** Perelman’s Ricci-flow programme [25, 26] shows how parabolic averaging can enforce global structure; we mirror that philosophy by pairing Fejér kernels with heat-flow smoothing in the Toeplitz bridge.
- **Programmatic limit architectures.** Deng–Hani–Ma [11] present a two-step limiting program (Newtonian dynamics $\rightarrow$ Boltzmann $\rightarrow$ fluid equations) that highlights how long-time control stabilizes successive limits; our compact-by-compact transfer plays an analogous “bridge” role for the Weil criterion.

- **Trace-formula-to-gap pipelines.** Anantharaman–Monk [1] use length-spectrum data and trace-formula technology to obtain asymptotic spectral gaps for random hyperbolic surfaces, a context that reinforces why spectral/trace language is a natural narrative for Weil-criterion positivity (here “Weil–Petersson” is unrelated to the Weil criterion).
- **Massive computations.** Platt and Trudgian’s verification of RH up to $3 \cdot 10^{12}$ [27], together with surveys like Conrey’s [10], emphasise the need for transparent, audit-friendly proofs rather than ever-larger numerics.
- **Cautionary analyses.** Cairo’s audit of proposed counterexamples [8] underlines how fragile heuristic arguments can be; we therefore keep every analytic assumption explicit and machine-checkable.

While these works influenced our methodology, our approach is fundamentally distinct: we construct a self-contained, verifiable chain from Toeplitz positivity to Weil positivity, with all critical steps amenable to formal verification.

2 Positioning and Scope

This work introduces a quantitative, modular operator framework for the Weil criterion that transfers positive semidefiniteness (PSD) of structured Toeplitz forms to nonnegativity of the Weil functional on the full test class via symbol regularity, RKHS contraction, and compact-by-compact limits. The scope and boundaries are as follows.

- **What this is:** A unified blueprint with explicit constants (modulus of continuity of the symbol, RKHS Gram tail, node spacing, tail cutoffs) that composes into a global positivity statement for $Q^*(t; \cdot)$.
- **What this is not:** No claim of new zero-free regions, density results for zeta zeros, or numerical hypotheses about zeros. The pathway works entirely through the Weil criterion.
- **Modularity:** Local improvements (sharper symbol modulus, tighter spacing/tail estimates, smaller effective weights) increase the contraction slack and propagate to strengthen $Q^*(t; \Phi) \geq 0$ on the Weil class.
- **Test class:** Even, nonnegative, compactly supported frequency tests. On $W_K = [-K, K]$ we denote by $\mathcal{W}_K$ the Fejér×heat cone, and by

$$\mathcal{W} := \bigcup_{K>0} \mathcal{W}_K$$

the **Weil cone**; density and continuity are always invoked inside this cone before taking the inductive limit. In the mainline proof the Fejér×heat atoms are fixed at $\tau = 0$; shifted variants appear only in the legacy appendix.

- **Verification path:** Sections 6–10.5 supply the fully written proofs for the main modules, with Appendix C recording auxiliary machine-checks and Appendix G archiving legacy branches.
- **Computation:** Symbol scans and PSD checks are reproducibility aids only; they do not enter the logical core of the proofs.

Bridge summary. We split $Q^*(t; \Phi_{B,t})$ via the Rayleigh identification into a Toeplitz block and a Rayleigh prime block, i.e., $T_M[P_A](t, B, M)$ versus $T_P^{\text{Ray}}(t, M)$. The symbol barrier yields $\lambda_{\min}(T_M[P_A]) \geq c_* - C \omega_{P_A}(1/(2M))$, where $c_* = \frac{11}{10}$ is the **uniform Archimedean floor** from Lemma 9.19 (for $t = t_{\text{crit}} = \frac{3}{20} = 0.15$, $B_{\min} = 3$), valid on the full circle $\mathbb{T}$ and independent of K . The prime norm is bounded in the Arch-induced RKHS by the uniform cap

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho(t),$$

at the same scale t (Corollary 9.22). Thus

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \min P_A - C \omega_{P_A}(1/(2M)) - \rho(t),$$

closing the bridge module and feeding the remaining steps.

3 Global Hypotheses

For reference we collect the global hypotheses used in the closure section. Each item is proved in the indicated place and recorded explicitly so that Theorem 12.4 and the Weil linkage (Section 12) invoke a single hypothesis list.

(H1) (T0) — Guinand–Weil normalization of $Q^*(t; \cdot)$ (Proposition 6.1).

(H2) (A1') — Density of the Fejér×heat cone on every W_K (Theorem 7.3).

(H3) (A2) — Lipschitz continuity of $Q^*(t; \cdot)$ on each W_K (Lemma 8.3 and Corollary 8.4).

(H4) (A3) — Toeplitz bridge with the explicit uniform floor $c_* > 0$ (Lemma 9.19), RKHS cap $\rho(t) \leq c_*/4$ at $t = t_{\text{crit}}$, and uniform discretisation threshold M_0^{unif} (Theorem 9.35).

(H5) (RKHS) — prime contraction via the uniform RKHS cap at $t = t_{\text{crit}}$ (Corollary 9.22).

Sections 6–10.5 establish (H1)–(H5); the closure Theorem 12.4 assumes precisely these hypotheses, and Theorem 12.2 invokes (H1)–(H5) together with Weil's criterion.

4 Notation and Conventions

On the frequency axis we write $\xi = \eta/(2\pi)$. In the Toeplitz bridge we work on the unit circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with fundamental domain $[-\frac{1}{2}, \frac{1}{2}]$. The Archimedean density is

$$a(\xi) = \log \pi - \Re \psi\left(\frac{1}{4} + i\pi\xi\right), \quad a_*(\xi) = 2\pi a(\xi),$$

and prime nodes are at $\xi_n = \frac{\log n}{2\pi}$ with symmetric placement $\pm\xi_n$. We distinguish two weight conventions:

$$\begin{aligned} w_Q(n) &= \frac{2\Lambda(n)}{\sqrt{n}} && \text{(the one-sided weight inside } Q^*(t; \Phi)), \\ w_{\text{RKHS}}(n) &= \frac{\Lambda(n)}{\sqrt{n}} && \text{(the operator weight on } W_K). \end{aligned}$$

Evenization lets us pass freely between them: doubling w_{RKHS} on $\xi_n > 0$ gives w_Q , while placing both $\pm\xi_n$ leaves w_{RKHS} unchanged. All RKHS and operator bounds below use w_{RKHS} ; we abbreviate $w_{\max} := \sup_n w_{\text{RKHS}}(n) \leq 2/e$. Throughout we use

$$Q^*(t; \Phi) = \int_{\mathbb{R}} a_*(\xi) \Phi(\xi) d\xi - \sum_{n \geq 2} w_Q(n) \Phi(\xi_n),$$

with the single heat scale $t = t_{\text{crit}}$ in the main line; Section 6 records the exact crosswalk to the Guinand–Weil form. (We call Q^* “quadratic” only because $\Phi = g * g^\vee$; as a functional of Φ it is linear.) Notational summaries and parameter tables are collected in Appendix A.

Quick reference for reviewers

Architecture. $T_0 \rightarrow A1' \rightarrow A2 \rightarrow A3 \rightarrow \text{RKHS} \rightarrow \text{Main closure}$. **Goal:** $Q^*(t; \Phi) \geq 0$ on the Weil cone $\mathcal{W}$.

Single scale. The same $t = t_{\text{crit}}$ controls the symbol modulus ω_{P_A} (A3) and the prime cap $\|T_P^{\text{RKHS}}(t)\|$ (RKHS).

Uniform margins. Global Arch floor $c_* = \frac{11}{10}$ is supplied explicitly by Lemma 9.19 (with $t = t_{\text{crit}} = 3/20 = 0.15$, $B_{\min} = 3$). The prime cap is set by $\rho(t) \leq c_*/4$ at $t = t_{\text{crit}}$, ensuring $\|T_P^{\text{RKHS}}(t)\| \leq c_*/4$. Budget split: $C_{\text{SB}}\omega_{P_A}(1/(2M)) \leq c_*/2$ and $\|T_P^{\text{Ray}}(t, M)\| \leq c_*/4 \Rightarrow \lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq c_*/4$.

Uniform constants. $t_{\text{crit}} = 3/20 = 0.15$, $B_{\min} = 3$, $c_* = \frac{11}{10}$; see Appendix A for M_0^{unif} .

5 Q^* parameter contract

Symbol	Meaning	Space/parameters
t	single heat scale	$t_0 = 1/(16\pi^2 t)$
$\Phi_{B,t}$	Fejér×heat window ($\tau = 0$)	bandwidth B , heat scale t
$Q^*(t; \Phi)$	Weil functional with explicit t	Φ on a compact window W_K
$T_M[P_A](t, B, M)$	Toeplitz block for P_A	$\mathbb{C}^{2M+1}$
$T_P^{\text{Ray}}(t, M)$	prime sampling block (Rayleigh)	$\mathbb{C}^{2M+1}$
$T_P^{\text{RKHS}}(t)$	prime operator on RKHS	$\mathcal{H}_t$, cap $\rho(t)$

WRONG: T_P , Q , ρ (naked, ambiguous).

RIGHT: $T_P^{\text{Ray}}(t, M)$ on $\mathbb{C}^{2M+1}$, $T_P^{\text{RKHS}}(t)$ on $\mathcal{H}_t$, $Q^*(t; \Phi)$, $\rho(t)$.

6 Normalization (T0)

6.1 Fourier normalization adjustments

We fix

$$\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(t) e^{-2\pi i t \xi} dt, \quad \varphi(t) = \int_{\mathbb{R}} \widehat{\varphi}(\xi) e^{2\pi i t \xi} d\xi, \quad (6.1)$$

and use the Lebesgue measure $d\xi$ on the frequency side. For even test functions, all identities are taken in the cosine form. We write $Q^*(t; \cdot)$ for the T0-normalized Weil functional; the parameter t is a bookkeeping label for the scale carried by the test.

Proposition 6.1 (T0' — Guinand–Weil matching). *Under Convention 6.1, the repository normalization $Q^*(t; \varphi)$ matches the classical Guinand–Weil functional [16, 33] after the change of variables $\eta = 2\pi\xi$:*

$$Q^*(t; \varphi) = Q_{\text{GW}}(\varphi) \quad \text{with} \quad \eta = 2\pi\xi, \quad d\eta = 2\pi d\xi. \quad (6.2)$$

Proof. Make the substitution $\eta = 2\pi\xi$ in all frequency integrals (see [29, Ch. 2]); by evenness the sine parts vanish and the cosine parts coincide. The Jacobian $d\eta = 2\pi d\xi$ is absorbed by the fixed normalization of $\widehat{\varphi}$. $\square$

Lemma 6.2 (T0: Q normalization crosswalk). *Let $\varphi_{\text{GW}} \in C_c(\mathbb{R})$ be even and nonnegative on the Guinand–Weil frequency axis $\eta \in \mathbb{R}$. Define*

$$Q_{\text{GW}}(\varphi_{\text{GW}}) := \int_{\mathbb{R}} \left(\log \pi - \Re \psi \left(\frac{1}{4} + \frac{i\eta}{2} \right) \right) \varphi_{\text{GW}}(\eta) d\eta - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\varphi_{\text{GW}}(\log n) + \varphi_{\text{GW}}(-\log n)). \quad (6.3)$$

On our (repository) frequency axis $\xi := \eta/(2\pi)$, define the even window $\varphi(\xi) := \varphi_{\text{GW}}(2\pi\xi)$, nodes $\xi_n := \frac{\log n}{2\pi}$, and the Archimedean densities

$$\boxed{a(\xi) := \log \pi - \Re \psi \left(\frac{1}{4} + i\pi\xi \right), \quad a_*(\xi) := 2\pi a(\xi)}. \quad (6.4)$$

Then the repository's quadratic functional

$$Q^*(t; \varphi) := \int_{\mathbb{R}} a_*(\xi) \varphi(\xi) d\xi - \sum_{n \geq 2} w_{\text{Q}}(n) \varphi(\xi_n) \quad (6.5)$$

coincides with Q_{GW} evaluated at φ_{GW} , i.e.

$$Q^*(t; \varphi) = Q_{\text{GW}}(\varphi_{\text{GW}}), \quad \eta = 2\pi\xi, \quad \varphi_{\text{GW}}(\eta) = \varphi(\eta/2\pi). \quad (6.6)$$

In operator or RKHS estimates we use the undoubled weights $w_{\text{RKHS}}(n)$; the evenization doubling appears only in the $Q^(t; \cdot)$ functional.*

Proof. Change variables $\eta = 2\pi\xi$ in the Archimedean integral: $d\eta = 2\pi d\xi$ and $\psi(\frac{1}{4} + \frac{i\eta}{2}) = \psi(\frac{1}{4} + i\pi\xi)$. Hence

$$\int_{\mathbb{R}} \left(\log \pi - \Re \psi \left(\frac{1}{4} + \frac{i\eta}{2} \right) \right) \varphi_{\text{GW}}(\eta) d\eta = \int_{\mathbb{R}} 2\pi \left(\log \pi - \Re \psi \left(\frac{1}{4} + i\pi\xi \right) \right) \varphi(\xi) d\xi. \quad (6.7)$$

For the prime term, $\varphi_{\text{GW}}(\pm \log n) = \varphi(\pm \xi_n)$ with $\xi_n = \frac{\log n}{2\pi}$. Since φ is even, $\varphi(\xi_n) + \varphi(-\xi_n) = 2\varphi(\xi_n)$. Thus

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (\varphi_{\text{GW}}(\log n) + \varphi_{\text{GW}}(-\log n)) = \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \varphi(\xi_n). \quad (6.8)$$

Combining the two identities yields $Q^*(t; \varphi) = Q_{\text{GW}}(\varphi_{\text{GW}})$, as claimed; the properties of the digamma function used here follow from [22, §5.2]. $\square$

Remark. (i) The choice of doubling the prime weights $w(n) = 2\Lambda(n)/\sqrt{n}$ at positive nodes $\xi_n > 0$ is equivalent to placing unit weights at both $\pm\xi_n$; evenness of φ makes the two conventions identical. (ii) If one prefers to keep $a(\xi)$ without the Jacobian factor 2π , then the same equality holds with $Q^*(t; \varphi)$ written as $\int (2\pi a) \varphi d\xi - \sum 2\Lambda(n)/\sqrt{n} \varphi(\xi_n)$; Lemma 6.2 records the canonical a_* that directly matches the Guinand–Weil form under $\eta = 2\pi\xi$. (iii) The digamma identities used throughout are tabulated in the NIST Digital Library of Mathematical Functions [22].

Lemma 6.3 (Invariance under normalisation conventions). *Different choices of Fourier-transform normalisations and node indexing yield equivalent formulations of the Weil positivity criterion. Specifically:*

- (a) *Switching from the unitary normalisation $\widehat{\Phi}(\xi) = \int \Phi(x) e^{-2\pi i x \xi} dx$ to the measure $\widehat{\Phi}'(\eta) = \int \Phi(x) e^{-i\eta x} dx$ with $\eta = 2\pi\xi$ induces the density rescaling $a^*(\xi) = 2\pi a(\xi)$ and preserves the form of $Q^*(t; \cdot)$.*
- (b) *Replacing the node sequence $\xi_n = \log n/(2\pi)$ by $\pm \log n/(2\pi)$ preserves the symmetry of the sampling operator and the archimedean/prime decomposition.*
- (c) *The quadratic form $Q^*(t; \phi)$ defined via the Guinand–Weil convention coincides with $Q_{GW}(\phi_{GW})$ when test functions are converted via the measure factor.*

In particular, the positivity of $Q^(t; \cdot)$ is independent of these technical choices.*

Proof. Each rescaling is a linear change of variable that preserves the spectral gap and the compact-by-compact structure. The node-symmetry $\pm \log n/(2\pi)$ is already built into the Guinand–Weil formalism; see [33], §16. The measure conversion $a^*(\xi) = 2\pi a(\xi)$ follows from the Jacobian of the coordinate change $\eta = 2\pi\xi$. $\square$

Transition. With the normalization T0 established, we now verify local density of the Fejér $\times$ heat cone on each compact in Section 7.3.

6.2 AD Normalization (Unitary FT + L2 Packets)

We fix the unitary Fourier transform

$$\widehat{f}(\gamma) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(u) e^{-i\gamma u} du, \quad \|f\|_{L^2} = \|\widehat{f}\|_{L^2}. \quad (6.9)$$

For the AD scale set $s(\tau) = 1 + |\tau|$, $\sigma(\tau) = \sqrt{t_0} s(\tau)$ with fixed $t_0 > 0$, and define the L^2 -normalized Gaussian packet

$$\psi_\tau(u) = \exp\left(-\frac{u^2}{2\sigma(\tau)^2}\right) e^{i\tau u} / \|\exp(-u^2/2\sigma(\tau)^2)\|_2, \quad \|\psi_\tau\|_2 = 1. \quad (6.10)$$

Then

$$\widehat{\psi}_\tau(\gamma) = \pi^{-1/4} \sigma(\tau)^{1/2} \exp\left(-\frac{\sigma(\tau)^2}{2}(\gamma - \tau)^2\right), \quad \int_{\mathbb{R}} |\widehat{\psi}_\tau|^2 = 1. \quad (6.11)$$

Consequently, the zero-side diagonal contributes $\frac{1}{2\pi} \log(1+|\tau|)$ up to a bounded edge constant $C_{\text{edge}}(t_0)$, and the Zero $\rightarrow$ Prime bridge A3 yields

$$\Gamma(K) \geq \left(\frac{1}{2\pi} - \Lambda_0(t_0, \kappa)\right) \log(1+K) - C_{\text{edge}}(t_0), \quad (6.12)$$

with $\Lambda_0(t_0, \kappa) = 2 \sum_{m \geq 1} e^{-t_0 \kappa^2 m^2 / 8}$.

7 Local Density (A1')

We work on $C_{\text{even}}^+([-K, K])$ with the uniform norm $\|\cdot\|_\infty$. Convolution with the Fejér kernel and subsequent heat smoothing preserve evenness and nonnegativity.

Theorem 7.1 (A1' — density). *For every compact $[-K, K]$ the cone $\{\text{Fejér} * \text{heat approximants}\}$ is dense in $C_{\text{even}}^+([-K, K])$ in $\|\cdot\|_\infty$.*

Proof. This is an immediate corollary of Theorem 7.3, which provides the explicit uniform approximation on $[-K, K]$ by finite nonnegative combinations of Fejér $\times$ heat atoms. $\square$

Remark (PW reinforcement). On $[-K, K]$ the heat kernel satisfies $\hat{\rho}_t(s) = e^{-4\pi^2 ts^2} \geq e^{-4\pi^2 tK^2} > 0$, hence the convolution with ρ_t is invertible on the compact in the PW metric. Together with the Fejér (positive) hat interpolation and a Weierstrass/Fejér–Riesz approximation step in $\|\cdot\|_\infty$, this yields the cone density in $W_{\text{PW}, K}$ with explicit error control; the constants enter only via $e^{-4\pi^2 tK^2}$ and the mesh parameter in the hat partition of unity.

Lemma 7.2 (Compact support convolution reduction). *Let $f \in C_c(\mathbb{R})$ be compactly supported with $\text{supp}(f) \subseteq [-L, L]$, and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be any function. Then for all $x \in \mathbb{R}$,*

$$(f * g)(x) = \int_{\mathbb{R}} f(y) g(x - y) dy = \int_{-L}^L f(y) g(x - y) dy.$$

Proof. Since $f(y) = 0$ for $|y| > L$, the integrand vanishes outside $[-L, L]$. $\square$

Theorem 7.3 (A1' (mainline $\tau = 0$ specialization)). *Let $K = [-R, R]$ with $R > 0$. Fix the single heat scale $t = t_{\text{crit}}$ and, for $B > 0$, define the even nonnegative frequency window*

$$\Phi_{B,t}(\xi) := \Lambda_B(\xi) \rho_t(\xi),$$

where $\Lambda_B(x) = (1 - |x|/B)_+$ and $\rho_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$ (so $\int_{\mathbb{R}} \rho_t = 1$, $\rho_t \geq 0$). Let $\mathcal{C}$ be the closed convex cone generated by finite nonnegative combinations of $\{\Phi_{B,t}\}$ with B sufficiently large (depending on R). Then $\mathcal{C}$ is dense in $C_{\text{even}}^+([-R, R])$ in the uniform norm.

Proof. The full shifted-atom density argument is archived in Appendix G.1. The mainline statement is its $\tau = 0$ specialization at the fixed scale $t = t_{\text{crit}}$. $\square$

Remark (Parameter scaling in A1'). The bandwidth B required in Theorem 7.3 depends on the compact $[-R, R]$ and the target accuracy ε , while the heat scale is fixed at $t = t_{\text{crit}}$ throughout the main chain. Accordingly, only $K \mapsto B(K)$ varies; no K -dependent heat schedule is used.

Remark (Legacy density with shifts). The shifted-atom cone and the fixed- t_0 density statement are recorded only in the legacy appendix (Appendix G.1). The mainline proof keeps $\tau = 0$ throughout.

8 Continuity of Q on Compacts (A2)

Lemma 8.1 (Local finiteness of the prime sampler). *Fix $K > 0$ and $t > 0$. For every even $\Phi \in C_c(\mathbb{R})$ with $\text{supp } \Phi \subset [-K, K]$, the prime part of $Q^*(t; \Phi)$,*

$$\sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n), \quad \xi_n := \frac{\log n}{2\pi},$$

is a finite sum: only finitely many terms are non-zero.

Proof. Under the T0 normalization (Section 4) prime nodes sit at $\xi_n = \log n / (2\pi)$ and

$$Q^*(t; \Phi) = \int_{\mathbb{R}} a^*(\xi) \Phi(\xi) d\xi - \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n), \quad a^*(\xi) = 2\pi(\log \pi - \Re \psi(\tfrac{1}{4} + i\pi\xi)).$$

If $\text{supp } \Phi \subset [-K, K]$, then $\Phi(\xi_n) = 0$ whenever $|\xi_n| > K$. The inequality $|\xi_n| \leq K$ is equivalent to $n \leq \lfloor e^{2\pi K} \rfloor$, so only finitely many indices contribute to the sum. In particular the active nodes in $[-K, K]$ have a positive minimum spacing

$$\delta_K := \min_{m \neq n} |\xi_m - \xi_n| \geq \frac{1}{2\pi(\lfloor e^{2\pi K} \rfloor + 1)},$$

which records the lack of accumulation points, although this bound is not needed for finiteness. $\square$

Corollary 8.2 (Lipschitz continuity on a compact window). *Let $\Phi_1, \Phi_2 \in C_c([-K, K])$ be even. Then*

$$|Q^*(t; \Phi_1) - Q^*(t; \Phi_2)| \leq \|a^*\|_{L^\infty([-K, K])} 2K \|\Phi_1 - \Phi_2\|_\infty + \left(\sum_{\xi_n \in [-K, K]} \frac{2\Lambda(n)}{\sqrt{n}} \right) \|\Phi_1 - \Phi_2\|_\infty.$$

In particular $Q^(t; \cdot)$ is Lipschitz on $C_c([-K, K])$ with the stated explicit constant.*

Proof. The Archimedean term is continuous in Φ in $L^1([-K, K])$ because a^* is bounded on the compact, while the prime term is a finite sum of point evaluations by Lemma 8.1. The bound follows by estimating each piece separately. $\square$

Lemma 8.3 (A2). *Fix $t > 0$ and a compact $K = [-R, R]$. For even nonnegative Φ supported in K define*

$$Q^*(t; \Phi) := \int_{-R}^R a_*(\xi) \Phi(\xi) d\xi - \sum_{\xi_n \in K} w_Q((n)) \Phi(\xi_n), \quad (8.1)$$

where $a_(\xi) = 2\pi(\log \pi - \Re \psi(\tfrac{1}{4} + i\pi\xi))$ and $w(p^m) = \frac{2\log p}{p^{m/2}}$ (doubled from evenization: $2\Lambda(n)/\sqrt{n}$ at positive nodes $\equiv \Lambda(n)/\sqrt{n}$ at $\pm$ nodes for even tests), $\xi_n = \frac{\log n}{2\pi}$. Then $Q^*(t; \cdot)$ is Lipschitz on $C_{\text{even}}^+(K)$ in $\|\cdot\|_\infty$:*

$$|Q^*(t; \Phi_1) - Q^*(t; \Phi_2)| \leq \left(\|a_*\|_{L^1(K)} + \sum_{\xi_n \in K} |w(n)| \right) \|\Phi_1 - \Phi_2\|_\infty. \quad (8.2)$$

If a construction uses Fejér $\times$ heat with small leakage outside K , then for any cutoff $N \geq e^2$ the tail satisfies

$$\text{Tail}(t; N) := \sum_{\xi_n \notin K, n > N} w_Q((n)) \Phi(\xi_n) \leq \frac{e^{-t(\log N)^2}}{t}, \quad (8.3)$$

so the leakage is explicitly controlled by the Gaussian tail.

Remark. Throughout this section the shorthand $w_Q((n)) = 2\Lambda(n)/\sqrt{n}$ denotes the evenized weights used in the Weil functional $Q^*(t; \cdot)$. In RKHS or operator bounds (Sections 10.5–9) we instead use the undoubled weights $w_{\text{RKHS}}((n)) = \Lambda(n)/\sqrt{n}$, so that $w_{\text{max}}^{\text{RKHS}} := \sup_n w_{\text{RKHS}}((n)) \leq 2/e$.

Proof. The Lipschitz bound follows from Lemma 8.1. Indeed,

$$\left| \int_{-R}^R a_*(\xi) (\Phi_1 - \Phi_2)(\xi) d\xi \right| \leq \|a_*\|_{L^1(K)} \|\Phi_1 - \Phi_2\|_\infty, \quad (8.4)$$

and since $\{\xi_n \in K\}$ is finite ($n \leq e^{2\pi R}$),

$$\left| \sum_{\xi_n \in K} w_Q((n)) (\Phi_1 - \Phi_2)(\xi_n) \right| \leq \left(\sum_{\xi_n \in K} w_Q((n)) \right) \|\Phi_1 - \Phi_2\|_\infty. \quad (8.5)$$

For the tail, note $\Phi(\xi) \leq e^{-4\pi^2 t \xi^2}$ and $\xi_n = \frac{\log n}{2\pi}$, hence

$$\sum_{n > N} w_Q((n)) \Phi(\xi_n) \leq \sum_{n > N} \frac{2 \log n}{\sqrt{n}} e^{-t(\log n)^2}. \quad (8.6)$$

Estimating the sum by an integral with the change of variables $y = \log x$ yields

$$\begin{aligned} \sum_{n > N} \frac{\log n}{\sqrt{n}} e^{-t(\log n)^2} &\leq \int_{N-1}^{\infty} \frac{\log x}{\sqrt{x}} e^{-t(\log x)^2} dx = \int_{\log(N-1)}^{\infty} y e^{-ty^2} e^{-y/2} dy \\ &\leq \int_{\log N}^{\infty} y e^{-ty^2} dy = \frac{1}{2t} e^{-t(\log N)^2}. \end{aligned} \quad (8.7)$$

Multiplying by the factor 2 from (8.6) gives $\text{Tail}(t; N) \leq t^{-1} e^{-t(\log N)^2}$ for $N \geq e^2$. This bound is independent of R once K is fixed and $B \gg R$; if Φ is strictly supported in K the tail vanishes. $\square$

Remark (Leakage control). When Fejér $\times$ heat windows are used on $[-K, K]$, the Gaussian factor produces exponentially small leakage outside the compact. The tail bound (8.3) therefore contributes at most $t^{-1} e^{-t(\log N)^2}$ to $Q^*(t; \Phi)$, which is absorbed in the A2 continuity budget.

Corollary 8.4 (Explicit Lipschitz modulus for $Q^*(t; \cdot)$). *Fix $K = [-R, R]$ and set*

$$L_Q(K) := \|a_*\|_{L^1(K)} + \sum_{\xi_n \in K} \frac{2\Lambda(n)}{\sqrt{n}}.$$

Then for all even, nonnegative $\Phi_1, \Phi_2 \in C_c(K)$ one has

$$|Q^*(t; \Phi_1) - Q^*(t; \Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_\infty.$$

In particular, if Φ is supported in K and is Fejér $\times$ heat with parameters (B, t) , the tail estimate (8.3) shows that extending Φ by zero outside K alters $Q^(t; \Phi)$ by at most $t^{-1} e^{-t(\log N)^2}$ once N truncates the prime sum (for $N \geq e^2$).*

Proof. Combine Corollary 8.2 with the evenization convention $w(n) = 2\Lambda(n)/\sqrt{n}$. The tail clause follows from Lemma 8.3. $\square$

9 Toeplitz–Symbol Bridge (A3)

9.1 A3 Calibration: The Constant $\kappa_{A3}(t)$

See also. Normalization T0 Lemma 6.2, Toeplitz bridge A3 Theorem 9.35.

Lemma 9.1 (Period-1 normalization audit). *Let $g \in L^1(\mathbb{R})$ be even and define the period-1 symbol*

$$P_A(\theta) := 2\pi \sum_{m \in \mathbb{Z}} g(\theta + m), \quad \theta \in [-\frac{1}{2}, \frac{1}{2}].$$

Then

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta = 2\pi \int_{\mathbb{R}} g(\xi) d\xi,$$

and the Fourier coefficients with respect to the basis $e^{2\pi i k \theta}$ satisfy

$$A_k = 2\pi \int_{\mathbb{R}} g(\xi) e^{-2\pi i k \xi} d\xi, \quad P_A(\theta) = A_0 + 2 \sum_{k \geq 1} A_k \cos(2\pi k \theta).$$

In particular, with $g = a \Phi$ and $a_(\xi) = 2\pi a(\xi)$, the Rayleigh pairing matches the T0-normalized Weil functional Q^* without further rescaling.*

Proof. By Fubini and the change of variables $\xi = \theta + m$,

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta = 2\pi \sum_{m \in \mathbb{Z}} \int_{-\frac{1}{2}}^{\frac{1}{2}} g(\theta + m) d\theta = 2\pi \int_{\mathbb{R}} g(\xi) d\xi.$$

The Fourier coefficient computation is identical, yielding the stated A_k and cosine series. $\square$

Lemma 9.2 (Calibration of κ_{A3}). *Let $\Phi(\xi) = (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2}$ be an even Fejér $\times$ heat window. Define the Arch coefficients*

$$A_k := 2\pi \int_{\mathbb{R}} a(\xi) \Phi(\xi) \cos(2\pi k \xi) d\xi, \quad P_A(\theta) := A_0 + 2 \sum_{k \geq 1} A_k \cos(2\pi k \theta), \quad (9.1)$$

with $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi \xi)$, and let $T_P^{\text{Ray}}(t)$ be the even prime sampling operator with weights $w(n) = \frac{2\Lambda(n)}{\sqrt{n}}$ at nodes $\xi_n = \frac{\log n}{2\pi}$. Then, in the Rayleigh identification of Theorem 9.35, at the constant test $p \equiv 1$ one has

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta - \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n) = \underbrace{\int_{\mathbb{R}} a_*(\xi) \Phi(\xi) d\xi}_{= A_0} - \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n). \quad (9.2)$$

By the T0 normalization (Lemma 6.2), the Weil functional on our axis is

$$Q^*(t; \Phi) = \int_{\mathbb{R}} a_*(\xi) \Phi(\xi) d\xi - \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n), \quad a_*(\xi) := 2\pi a(\xi). \quad (9.3)$$

Therefore

$$Q^*(t; \Phi) = \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta - \sum_{n \geq 2} \frac{2\Lambda(n)}{\sqrt{n}} \Phi(\xi_n), \quad (9.4)$$

and the bridge A3 introduces the fixed scale factor

$$\boxed{\kappa_{A3}(t) = 1} \quad (\text{independent of } t). \quad (9.5)$$

Equivalently, the normalization in (9.1) absorbs the Jacobian 2π into the symbol coefficients, so $\kappa_{A3} \equiv 1$.

Lemma 9.3 (Rayleigh identification (infinite-dimensional)). *For every even Fejér×heat window Φ the operator form and the Weil functional satisfy*

$$\left\langle (T_M[P_A] - T_P^{\text{Ray}}(t))p, p \right\rangle = Q^*(t; \Phi)$$

whenever p corresponds to Φ via the standard Dirichlet sampling operator. Note: This is the infinite-dimensional idealization. For the finite-dimensional compression to $\mathcal{P}_M$, see Theorem 9.10 where the normalization of vectors introduces a factor $(2M+1)$.

Proof. Write the Fejér×heat window as

$$\Phi(\xi) = \sum_{k \in \mathbb{Z}} \widehat{\Phi}(k) e^{2\pi i k \xi}, \quad \widehat{\Phi}(k) = \int_{\mathbb{R}} \Phi(\xi) e^{-2\pi i k \xi} d\xi.$$

The Dirichlet sampling operator maps $p(\theta) = \sum_{k \in \mathbb{Z}} \widehat{\Phi}(k) e^{2\pi i k \theta}$ to Φ ; hence

$$\left\langle T_M[P_A]p, p \right\rangle = \sum_{k \in \mathbb{Z}} A_k |\widehat{\Phi}(k)|^2 = A_0 |\widehat{\Phi}(0)|^2 + 2 \sum_{k \geq 1} A_k |\widehat{\Phi}(k)|^2,$$

where A_k are the Arch coefficients from (9.1). Likewise, the prime operator contributes

$$\left\langle T_P^{\text{Ray}}(t)p, p \right\rangle = \sum_{n \geq 2} \frac{2 \Lambda(n)}{\sqrt{n}} \Phi(\xi_n) \overline{\Phi(\xi_n)}.$$

Subtracting and recalling $Q^*(t; \Phi)$ from (9.3) gives

$$\left\langle (T_M[P_A] - T_P^{\text{Ray}}(t))p, p \right\rangle = Q^*(t; \Phi),$$

which is the desired identity. $\square$

Proposition 9.4 (Bridge margin calibration). *Under the uniform floor $c_* > 0$ from Lemma 9.19 and the prime cap $\rho(t) \leq c_*/4$, the mixed Toeplitz block satisfies*

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t)) \geq \frac{c_*}{4}$$

for every $M \geq M_0^{\text{unif}}$ in Theorem 9.35.

Proof. Theorem 9.35 yields $\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t)) \geq c_* - C_{\text{SB}\omega_{P_A}}(1/(2M)) - \|T_P^{\text{Ray}}(t)\|_{\text{op}}$. For $M \geq M_0^{\text{unif}}$, Corollary 9.21 ensures $C_{\text{SB}\omega_{P_A}}(1/(2M)) \leq c_*/2$. Corollary 9.22 gives $\|T_P^{\text{RKHS}}(t)\|_{\text{op}} \leq \rho(t) \leq c_*/4$. Thus $\lambda_{\min} \geq c_* - c_*/2 - c_*/4 = c_*/4$. $\square$

Remark (Evenization does not increase C_0). In the T0 normalization we already place symmetric prime weights at $\pm \xi_n$ and integrate the zero counting measure $dN(\gamma)$ over the full real line. The diagonal constant on the zero side is therefore $C_0 = \frac{1}{2\pi}$, not $\frac{1}{\pi}$. Passing to an evenized basis (replacing $\{+\tau, -\tau\}$ by a single cosine packet) redistributes mass within each pair but does not create an additional factor 2: the same symmetry is already built into T0 and into the A3 calibration. Consequently, with $\kappa_{\text{A3}} = 1$ the asymptotic PG–LS slope in Road A is $1 - \Lambda_0 \nearrow 1^-$ as $\Lambda_0 \downarrow 0$.

Remark (Consequence for the PG–LS slope). Let the zero-side packet Gram lower bound be normalized as $\sum_{\rho} |\sum_j c_j \widehat{g}_{\tau_j}(\gamma_{\rho})|^2 \geq (\frac{1}{2\pi} - \Lambda_0) \log(1+K) \sum_j |c_j|^2 - C_{\text{edge}} \sum_j |c_j|^2$. Under A3 and T0 the prime-side gain is

$$\Gamma(K) \geq \kappa_{\text{A3}} \left(\frac{1}{2\pi} - \Lambda_0 \right) \log(1+K) - \kappa_{\text{A3}} C_{\text{edge}} = \left(\frac{1}{2\pi} - \Lambda_0 \right) \log(1+K) - C_{\text{edge}}, \quad (9.6)$$

so the asymptotic slope approaches 1^- as $\Lambda_0 \rightarrow 0$. Hence a strict > 1 cannot be achieved within Road A by only shrinking Λ_0 ; one needs an amplifier (e.g. Road B/C) or a different normalization.

Formal Arch bounds (symbol side)

Lemma 9.5 (Lipschitz modulus for the periodized symbol). *Let*

$$g_{B,t}(\xi) := a(\xi) (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2},$$

and define the 1-periodic symbol $P_A(\theta) := 2\pi \sum_{m \in \mathbb{Z}} g_{B,t}(\theta + m)$. Then $P_A \in \text{Lip}(1)$ with

$$\omega_{P_A}(h) \leq L_A(B, t) h, \quad L_A(B, t) := 2\pi \sup_{\theta \in [-\frac{1}{2}, \frac{1}{2}]} \sum_{m \in \mathbb{Z}} |g'_{B,t}(\theta + m)|. \quad (9.7)$$

Proof. Lemma 9.11 applies verbatim. $\square$

Proposition 9.6 (Mean minus modulus). *Let $A_0 = 2\pi \int_{-B}^B a(\xi) (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2} d\xi$. If $P_A \in \text{Lip}(1)$ with modulus $\omega_{P_A}(h) \leq L_A h$, then*

$$\min_{\theta \in \mathbb{T}} P_A(\theta) \geq A_0 - \frac{1}{2} L_A. \quad (9.8)$$

Lemma 9.7 (Core/off-core lower bound for A_0). *Fix $r \in (0, B)$. Suppose there is $m_r > 0$ such that $a(\xi) \geq m_r$ for $|\xi| \leq r$. Then*

$$A_0 \geq \underbrace{2\pi m_r \int_{-r}^r (1 - |\xi|/B) e^{-4\pi^2 t \xi^2} d\xi}_{\text{core mass}} - \underbrace{2\pi \int_{|\xi| > r} |a(\xi)| (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2} d\xi}_{\text{off-core tail}}. \quad (9.9)$$

Moreover, the core mass admits the explicit lower bound

$$\int_{-r}^r (1 - |\xi|/B) e^{-4\pi^2 t \xi^2} d\xi \geq 2r \left(1 - \frac{r}{B}\right) \exp(-4\pi^2 t r^2), \quad (9.10)$$

and the off-core tail obeys

$$\int_{|\xi| > r} (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2} d\xi \leq 2 \int_r^\infty e^{-4\pi^2 t \xi^2} d\xi \leq \frac{1}{4\pi^2 t r} e^{-4\pi^2 t r^2}. \quad (9.11)$$

Thus, if $\|a\|_\infty \leq A_0$ then

$$A_0 \geq 4\pi m_r r \left(1 - \frac{r}{B}\right) e^{-4\pi^2 t r^2} - A_0 \frac{1}{2\pi t r} e^{-4\pi^2 t r^2}. \quad (9.12)$$

Optimizing r within $(0, B)$ yields an explicit positive lower bound $A_{0,\text{lo}}(B, t)$ whenever m_r is known.

Usage. Combine Lemma 9.5 and Lemma 9.7 to obtain $L_A(B, t)$ and $A_{0,\text{lo}}(B, t)$. Then Proposition 9.6 gives

$$\min P_A \geq A_{0,\text{lo}}(B, t) - \frac{1}{2} L_A(B, t), \quad (9.13)$$

which is the symbol margin used in the A3 bridge. All inequalities are analytic and require no floating point.

9.2 Rayleigh Identification for the Toeplitz Bridge

Throughout we fix a single heat scale $t > 0$ and a Fejér×heat window

$$\Phi_{B,t}(\xi) := \left(1 - \frac{|\xi|}{B}\right)_+ e^{-4\pi^2 t \xi^2},$$

and write P_A for the associated Archimedean symbol obtained by smoothing the T0 density $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$ with the Fejér and heat kernels on $[-B, B]$. The prime weights are $w(n) = \frac{2\Lambda(n)}{\sqrt{n}}$ located at the nodes $\xi_n = \frac{\log n}{2\pi}$, as fixed in Section 6. The corresponding Weil functional is denoted $Q^*(t; \Phi_{B,t})$.

Let $\mathcal{P}_M := \{p(\theta) = \sum_{|k| \leq M} c_k e^{2\pi i k \theta}\}$ denote the trigonometric polynomials of degree at most M , equipped with the $L^2(\mathbb{T})$ inner product, and let $\iota_M : \mathcal{P}_M \hookrightarrow L^2(\mathbb{T})$ be the canonical inclusion with adjoint ι_M^* equal to the orthogonal projection onto $\mathcal{P}_M$.

Lemma 9.8 (Model-space restriction). *The Toeplitz operator $T_M[P_A]$ acts on $\mathcal{P}_M$, is self-adjoint and satisfies*

$$\langle T_M[P_A] p, p \rangle_{L^2(\mathbb{T})} = \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) |p(\theta)|^2 d\theta, \quad p \in \mathcal{P}_M.$$

Moreover, the symmetrised Rayleigh prime operator

$$T_P^{\text{Ray}}(t, M) := \sum_{\substack{n \geq 2 \\ |\xi_n| \leq B}} w(n) \Phi_{B,t}(\xi_n) |v_n^{(M)} \rangle \langle v_n^{(M)}|, \quad v_n^{(M)}(\theta) := \frac{1}{\sqrt{2M+1}} \sum_{|k| \leq M} e^{2\pi i k(\theta - \xi_n)},$$

is the normalized compression of the global Rayleigh operator

$$T_P^{\text{Ray}}(t) := \sum_{n \geq 2} w(n) \Phi_{B,t}(\xi_n) |e^{2\pi i(\cdot)\xi_n} \rangle \langle e^{2\pi i(\cdot)\xi_n}|$$

to $\mathcal{P}_M$, in the sense that

$$\iota_M^* T_P^{\text{Ray}}(t) \iota_M = (2M+1) T_P^{\text{Ray}}(t, M).$$

It is positive semidefinite with

$$\|T_P^{\text{Ray}}(t, M)\| \leq \sum_{\substack{n \geq 2 \\ |\xi_n| \leq B}} w(n) \Phi_{B,t}(\xi_n).$$

Proof. The Toeplitz matrix $T_M[P_A]$ is the compression of the Fourier multiplier with symbol P_A to $\mathcal{P}_M$; the stated quadratic form is the standard representation of Toeplitz forms (see, e.g., [14, Chapter 1]). For the prime operator note that $T_P^{\text{Ray}}(t)$ is a finite-rank positive operator on $L^2(\mathbb{T})$, hence $\iota_M^* T_P^{\text{Ray}}(t) \iota_M$ is self-adjoint and positive semidefinite. Since $\iota_M^* e^{2\pi i(\cdot)\xi_n} = \sum_{|k| \leq M} e^{2\pi i k(\cdot - \xi_n)} = \sqrt{2M+1} v_n^{(M)}$, the displayed identity $\iota_M^* T_P^{\text{Ray}}(t) \iota_M = (2M+1) T_P^{\text{Ray}}(t, M)$ follows. The displayed norm bound is immediate from the triangle inequality applied to the sum of rank-one projections $|v_n^{(M)} \rangle \langle v_n^{(M)}|$. $\square$

Lemma 9.9 (Rayleigh pairing). *For every $p \in \mathcal{P}_M$ one has*

$$\begin{aligned} \langle (T_M[P_A] - T_P^{\text{Ray}}(t, M)) p, p \rangle_{L^2(\mathbb{T})} &= \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) |p(\theta)|^2 d\theta - \sum_{\substack{n \geq 2 \\ |\xi_n| \leq B}} w(n) \Phi_{B,t}(\xi_n) |\langle p, v_n^{(M)} \rangle|^2 \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) |p(\theta)|^2 d\theta - \frac{1}{2M+1} \sum_{\substack{n \geq 2 \\ |\xi_n| \leq B}} w(n) \Phi_{B,t}(\xi_n) |p(\xi_n)|^2. \end{aligned}$$

Proof. Combine Lemma 9.8 with the definition of $T_P^{\text{Ray}}(t, M)$ and the identities $p(\xi_n) = \sqrt{2M+1} \langle p, v_n^{(M)} \rangle$ and $\|v_n^{(M)}\| = 1$. $\square$

Theorem 9.10 (Rayleigh identification for the Fejér×heat window). *Let $\Phi_{B,t}$ and P_A be as above, and let $p \equiv 1$ be the constant polynomial. Then*

$$\langle T_M[P_A] 1, 1 \rangle_{L^2(\mathbb{T})} - (2M+1) \left\langle T_P^{\text{Ray}}(t, M) 1, 1 \right\rangle_{L^2(\mathbb{T})} = \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta - \sum_{\substack{n \geq 2 \\ |\xi_n| \leq B}} w(n) \Phi_{B,t}(\xi_n) = Q^*(t; \Phi_{B,t}),$$

where Q^* is the Weil functional in the $T0$ normalization (Lemma 6.2). The factor $(2M+1)$ arises because $|\langle 1, v_n^{(M)} \rangle|^2 = 1/(2M+1)$ while $|1(\xi_n)|^2 = 1$. Equivalently, one may use unnormalized vectors $\tilde{v}_n := \sqrt{2M+1} v_n^{(M)}$ and write the Rayleigh identity without the scaling factor. In particular, $Q^*(t; \Phi_{B,t}) \geq 0$ if and only if

$$\left\langle (T_M[P_A] - (2M+1)T_P^{\text{Ray}}(t, M)) 1, 1 \right\rangle_{L^2(\mathbb{T})} \geq 0.$$

Proof. Applying Lemma 9.9 with $p \equiv 1$ yields

$$\left\langle T_P^{\text{Ray}}(t, M) 1, 1 \right\rangle_{L^2(\mathbb{T})} = \frac{1}{2M+1} \sum_{n \geq 2} w(n) \Phi_{B,t}(\xi_n),$$

where the prime sum is finite because $\Phi_{B,t}$ is supported in $[-B, B]$. By definition of P_A and the normalization fixed in Section 6 one has

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta = \int_{\mathbb{R}} a_*(\xi) \Phi_{B,t}(\xi) d\xi,$$

and Lemma 6.2 gives $Q^*(t; \Phi_{B,t}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) d\theta - \sum_{n \geq 2} w(n) \Phi_{B,t}(\xi_n)$. Multiplying the displayed identity for $\langle T_P^{\text{Ray}}(t, M) 1, 1 \rangle$ by $(2M+1)$ yields the prime sum, proving the claim. $\square$

9.3 Embedding bridge: Rayleigh vs. RKHS

We fix a single heat scale t and work with the Rayleigh operator $T_P^{\text{Ray}}(t, M)$ on $\mathbb{C}^{2M+1}$ and the RKHS operator $T_P^{\text{RKHS}}(t)$ on $\mathcal{H}_t$. The comparison

$$\|T_P^{\text{Ray}}(t, M)\| \leq \|T_P^{\text{RKHS}}(t)\|$$

is enforced by an explicit embedding $\iota_{t,M}$ and the compression formula

$$T_P^{\text{Ray}}(t, M) := \iota_{t,M}^* T_P^{\text{RKHS}}(t) \iota_{t,M}.$$

We record two concrete constructions of $\iota_{t,M}$ that each yield the same compression inequality; the choice is driven by formalization cost.

C1 micro-lemmas (shared).

1. **Spaces:** set $E := \mathbb{C}^{2M+1}$ and $H := \mathcal{H}_t$ as Hilbert spaces.
2. **Embedding:** construct a linear isometry $\iota : E \hookrightarrow H$.
3. **Compression:** define $T_P^{\text{Ray}} := \iota^* T_P^{\text{RKHS}} \iota$.
4. **Isometry norm:** $\|\iota\| = \|\iota^*\| = 1$.
5. **Submultiplicativity:** $\|AB\| \leq \|A\| \|B\|$.
6. **Conclusion:** $\|T_P^{\text{Ray}}\| \leq \|T_P^{\text{RKHS}}\|$.

Option A: analytic feature-map embedding. Choose orthonormal feature vectors $\{\psi_k\}_{k=-M}^M \subset \mathcal{H}_t$ compatible with the heat kernel (Fourier–Gaussian features). Define

$$\iota_{t,M}(v) := \sum_{k=-M}^M v_k \psi_k,$$

so $\iota_{t,M}$ is a linear isometry. The compression matrix entries satisfy

$$\langle T_P^{\text{RKHS}}(t) \psi_j, \psi_i \rangle = \sum_n w_{\text{RKHS}}(n) \Phi_{B,t}(\xi_n) \overline{\psi_j(\xi_n)} \psi_i(\xi_n),$$

which reproduces the Rayleigh matrix once the feature evaluations are matched to the trigonometric packet basis. This is the mathematically clean bridge, but it requires explicit RKHS feature evaluations in the formal proof.

Option B: pragmatic dictionary compression. Let $\{d_1, \dots, d_N\} \subset \mathcal{H}_t$ be a finite dictionary (e.g. kernel sections $k_t(\cdot, \xi_n)$ at the prime nodes) and set $V := \text{span}\{d_i\}$. Choose an orthonormal basis $\{e_1, \dots, e_N\}$ of V and define the isometric inclusion

$$\iota_{t,M} : \mathbb{C}^N \rightarrow \mathcal{H}_t, \quad \iota_{t,M}(c) := \sum_{j=1}^N c_j e_j.$$

Then $\iota_{t,M}$ is a linear isometry and the compression $\iota_{t,M}^* T_P^{\text{RKHS}}(t) \iota_{t,M}$ yields a finite-dimensional operator dominated by $\|T_P^{\text{RKHS}}(t)\|$. This route avoids explicit kernel feature computations and is typically Lean-friendly.

9.4 Symbol Regularity and Archimedean Floor

We now record explicit regularity and lower bounds for the Archimedean symbol P_A attached to a Fejér×heat window. Throughout we fix parameters $B > 0$ and $t > 0$, set

$$\Phi_{B,t}(\xi) = \left(1 - \frac{|\xi|}{B}\right)_+ e^{-4\pi^2 t \xi^2},$$

and define

$$g_{B,t}(\xi) := a(\xi) \Phi_{B,t}(\xi), \quad P_A(\theta) := 2\pi \sum_{m \in \mathbb{Z}} g_{B,t}(\theta + m).$$

For $k \geq 0$ set

$$A_k := 2\pi \int_{\mathbb{R}} g_{B,t}(\xi) \cos(2\pi k\xi) d\xi,$$

where $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$ is the normalized Archimedean density fixed in Section 6. The function $g_{B,t}$ is compactly supported on $[-B, B]$, so the periodization sum defining P_A is finite for each θ .

Lemma 9.11 (Periodized symbol and Lipschitz modulus). *The symbol P_A is 1-periodic and admits the cosine series*

$$P_A(\theta) = A_0 + 2 \sum_{k \geq 1} A_k \cos(2\pi k\theta).$$

Moreover, for every $h \geq 0$ one has

$$\omega_{P_A}(h) \leq L_A(B, t) h,$$

where

$$L_A(B, t) := 2\pi \sup_{\theta \in [-\frac{1}{2}, \frac{1}{2}]} \sum_{m \in \mathbb{Z}} |g'_{B,t}(\theta + m)|.$$

In particular $P_A \in \text{Lip}(1)$ on the unit torus $\mathbb{T}$.

Proof. Since $g_{B,t} \in L^1(\mathbb{R})$ and has compact support, Fubini and the change of variables $\xi = \theta + m$ give

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} P_A(\theta) e^{-2\pi i k \theta} d\theta = 2\pi \int_{\mathbb{R}} g_{B,t}(\xi) e^{-2\pi i k \xi} d\xi.$$

Because $g_{B,t}$ is even, the Fourier coefficients are real and coincide with A_k , yielding the cosine series.

For the modulus, write

$$P_A(\theta + h) - P_A(\theta) = 2\pi \sum_{m \in \mathbb{Z}} (g_{B,t}(\theta + h + m) - g_{B,t}(\theta + m)).$$

The function $g_{B,t}$ is Lipschitz on $\mathbb{R}$ (it is the product of a smooth function and a compactly supported Lipschitz cutoff), so for each m the mean-value bound yields $|g(\theta + h + m) - g(\theta + m)| \leq h \sup_{s \in [\theta, \theta+h]} |g'(s + m)|$. Summing over the finitely many m with $\theta + m \in [-B, B]$ gives $\omega_{P_A}(h) \leq L_A(B, t)h$. $\square$

Next we quantify the symbol floor by splitting the integral into a “core” region $[-r, r]$ and its complement.

Lemma 9.12 (Core contribution). *Let $0 < r < B$. Set*

$$m_r := \inf_{|\xi| \leq r} a(\xi), \quad M_B := \|a\|_{L^\infty([-B, B])}.$$

Then

$$A_0 \geq 4\pi m_r r \left(1 - \frac{r}{B}\right) e^{-4\pi^2 t r^2} - \frac{M_B}{2\pi t r} e^{-4\pi^2 t r^2}.$$

Proof. Split the integral defining A_0 into $[-r, r]$ and its complement. On $[-r, r]$ we lower bound $a(\xi)$ by m_r , and on $|\xi| \in [r, B]$ we bound $|a(\xi)|$ by M_B . The integral of $\Phi_{B,t}$ over each region is computed explicitly, giving the stated inequality. $\square$

Lemma 9.13 (Unshifted core mass). *Let $0 < r < B$. Then the Fejér hat satisfies*

$$\int_{-r}^r \Lambda_B(x) dx \geq \frac{2r^2}{B}.$$

Consequently, for every $t > 0$,

$$\int_{\mathbb{R}} \Lambda_B(x) e^{-4\pi^2 t x^2} dx \geq \frac{2r^2}{B} e^{-4\pi^2 t r^2}.$$

Proof. The function Λ_B is linear on each of the intervals $[-B, 0]$ and $[0, B]$ with slope magnitude $1/B$. A direct calculation gives $\int_{-r}^r \Lambda_B(x) dx = 2r - \frac{r^2}{B} \geq \frac{2r^2}{B}$. For the Gaussian factor we use the pointwise bound $e^{-4\pi^2 t x^2} \geq e^{-4\pi^2 t r^2}$ whenever $|x| \leq r$. $\square$

Lemma 9.14 (Archimedean floor). *With notation as above define*

$$\begin{aligned} L_A^{\text{up}}(B, t) &:= L_A(B, t), \\ \underline{A}_0(B, r, t) &:= 2m_r r \left(1 - \frac{r}{B}\right) e^{-4\pi^2 t r^2} - \frac{M_B}{4\pi^2 t r} e^{-4\pi^2 t r^2}. \end{aligned}$$

Then

$$\min_{\theta \in \mathbb{T}} P_A(\theta) \geq \underline{A}_0(B, r, t) - \frac{1}{2} L_A^{\text{up}}(B, t).$$

Proof. For any θ choose a point θ_0 at which P_A attains its mean value and apply the mean-value inequality $P_A(\theta) \geq A_0 - \omega_{P_A}(|\theta - \theta_0|)$. Since $|\theta - \theta_0| \leq \frac{1}{2}$, the Lipschitz bound and Lemma 9.12 give the claimed inequality. $\square$

Lemma 9.15 (Core slope bound). *For $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$ and every $r > 0$,*

$$\inf_{|\xi| \leq r} a(\xi) \geq a(0) - L_a r, \quad L_a \leq 20\pi,$$

where $a(0) = \gamma + \frac{\pi}{2} + \log \pi + 3 \log 2 > 0$.

Proof. Differentiating a yields $a'(\xi) = \pi \Im \psi'(\frac{1}{4} + i\pi\xi)$. The trigamma admits the convergent series $\psi'(z) = \sum_{n \geq 0} (n+z)^{-2}$ for $\Re z > 0$ (see, e.g., [31, Ch. 2]), so

$$|\psi'(\frac{1}{4} + i\pi\xi)| \leq \sum_{n \geq 0} \frac{1}{|n + \frac{1}{4} + i\pi\xi|^2} \leq \sum_{n \geq 0} \frac{1}{(n + \frac{1}{4})^2} \leq \frac{1}{(\frac{1}{4})^2} + \int_0^\infty \frac{dx}{(x + \frac{1}{4})^2} = 16 + 4 = 20.$$

Therefore $|a'(\xi)| \leq 20\pi$ for all ξ , and the mean-value theorem gives $a(\xi) \geq a(0) - 20\pi|\xi|$.

The identity $\psi(\frac{1}{4}) = -\gamma - \frac{\pi}{2} - 3 \log 2$ is recorded in Appendix G.3, equation (G.25) (see also [31, Ch. 2]). Together with equation (G.28) it implies $a(0) = \log \pi - \Re \psi(\frac{1}{4}) = \gamma + \frac{\pi}{2} + \log \pi + 3 \log 2$, which is positive. Substituting this identity into the mean-value estimate completes the proof. $\square$

Lemma 9.16 (Digamma monotonicity). *For $\xi > 0$ the Archimedean density satisfies*

$$a'(\xi) = -2\pi^2 \xi \sum_{n \geq 0} \frac{n + \frac{1}{4}}{((n + \frac{1}{4})^2 + \pi^2 \xi^2)^2},$$

hence $a'(\xi) < 0$ and a is even and strictly decreasing on $[0, \infty)$. Moreover, for $\xi \geq 1$,

$$|a'(\xi)| \leq \frac{1}{|\xi|} + \frac{1}{2\pi^2 |\xi|^3} \leq \frac{11}{10} \cdot \frac{1}{|\xi|}.$$

Proof. From the trigamma series $\psi'(z) = \sum_{n \geq 0} (n+z)^{-2}$ and $\Im(x+iy)^{-2} = -2xy/(x^2+y^2)^2$ we obtain

$$a'(\xi) = \pi \Im \psi'(\tfrac{1}{4} + i\pi\xi) = -2\pi^2\xi \sum_{n \geq 0} \frac{n + \frac{1}{4}}{((n + \frac{1}{4})^2 + \pi^2\xi^2)^2},$$

which is negative for $\xi > 0$ because every term in the sum is positive. For $\xi \geq 1$, bounding the sum by the first term plus the integral $\int_0^\infty \frac{x + \frac{1}{4}}{((x + \frac{1}{4})^2 + \pi^2\xi^2)^2} dx = \frac{1}{2(\pi^2\xi^2 + \frac{1}{16})}$ gives

$$|a'(\xi)| \leq 2\pi^2|\xi| \left(\frac{1}{2\pi^2\xi^2} + \frac{1}{2\pi^4\xi^4} \right) = \frac{1}{|\xi|} + \frac{1}{2\pi^2|\xi|^3},$$

and the last inequality follows since $\xi \geq 1$. $\square$

Lemma 9.17 (Logarithmic growth bound). *For $\xi \geq 1$ one has*

$$|a(\xi)| \leq a(0) + \frac{11}{10} \log(1 + \xi).$$

Proof. By Lemma 9.16, $|a'(\xi)| \leq \frac{11}{10}\xi^{-1}$ for $\xi \geq 1$. Integrating from 1 to ξ yields $|a(\xi)| \leq |a(1)| + \frac{11}{10} \log \xi \leq a(0) + \frac{11}{10} \log(1 + \xi)$, since a is decreasing and $|a(1)| \leq a(0)$. $\square$

Lemma 9.18 (Sample-point bounds for a). *The Archimedean density satisfies*

$$a\left(\frac{1}{2}\right) \geq \frac{29}{50}, \quad a\left(\frac{3}{2}\right) \geq -\frac{3}{5}, \quad a\left(\frac{5}{2}\right) \geq -\frac{11}{10}.$$

Proof. Write $y = \pi\xi$ and

$$t_n(y) := \frac{1}{n+1} - \frac{n + \frac{1}{4}}{(n + \frac{1}{4})^2 + y^2}, \quad \Re \psi\left(\frac{1}{4} + iy\right) = -\gamma + \sum_{n \geq 0} t_n(y).$$

For $\xi = \frac{1}{2}$ one has $y^2 = \pi^2/4$ and $t_n(y) \leq 0$ for $n \geq 4$, so

$$\Re \psi\left(\frac{1}{4} + \frac{i\pi}{2}\right) \leq -\gamma + \sum_{n=0}^3 t_n\left(\frac{\pi}{2}\right).$$

Using $\pi^2 < 10$ gives the explicit upper bounds

$$\begin{aligned} t_0 &\leq 1 - \frac{1/4}{1/16 + 5/2} = \frac{37}{41}, & t_1 &\leq \frac{1}{2} - \frac{5/4}{25/16 + 5/2} = \frac{5}{26}, \\ t_2 &\leq \frac{1}{3} - \frac{9/4}{81/16 + 5/2} = \frac{13}{363}, & t_3 &\leq \frac{1}{4} - \frac{13/4}{169/16 + 5/2} = \frac{1}{836}. \end{aligned}$$

Hence $\sum_{n=0}^3 t_n(\pi/2) < 1.132$. With the classical bounds $\pi > 333/106$ and $\gamma > 0.5772$ we obtain $\log \pi + \gamma > 1.72$, therefore

$$a\left(\frac{1}{2}\right) = \log \pi + \gamma - \sum_{n \geq 0} t_n(\pi/2) \geq 1.72 - 1.132 > \frac{29}{50}.$$

For $\xi \geq 1$ we use the standard digamma remainder bound

$$|\psi(z) - \log z + \frac{1}{2z}| \leq \frac{1}{12|z|^2} \quad (\Re z > 0)$$

([22, §5.11]). Taking real parts at $z = \frac{1}{4} + i\pi\xi$ yields

$$a(\xi) \geq -\log \xi - \frac{1}{2\pi\xi} - \frac{1}{12\pi^2\xi^2}.$$

For $\xi = \frac{3}{2}$, using $\pi > 3$ and $\log(3/2) < \frac{5}{12}$ (from the alternating series for $\log(1+x)$ at $x = \frac{1}{2}$) gives

$$a\left(\frac{3}{2}\right) \geq -\frac{5}{12} - \frac{1}{9} - \frac{1}{243} > -\frac{3}{5}.$$

For $\xi = \frac{5}{2}$, using $\pi > 3$ and $\log(5/2) < 1$ (since $e > 2.5$ by the series for e) gives

$$a\left(\frac{5}{2}\right) \geq -1 - \frac{1}{15} - \frac{1}{675} > -\frac{11}{10}.$$

□

Lemma 9.19 (Uniform Archimedean floor (pointwise)). *Fix $t = \frac{3}{50}$ and $B_{\min} = 3$. Then for every $B \geq B_{\min}$ and every $\theta \in \mathbb{T}$ the Archimedean symbol satisfies*

$$P_A(\theta) \geq c_* := \frac{11}{10}.$$

Proof. By Lemma 9.16 the function $a(\xi)$ is decreasing on $[0, \infty)$, and the window $w_B(\xi) := (1 - \xi/B)e^{-4\pi^2 t \xi^2}$ is decreasing for $\xi \in [0, B]$. For $\theta \in [0, \frac{1}{2}]$ and $B \geq 3$ we have

$$P_A(\theta) = 2\pi \sum_{m \in \mathbb{Z}} g_{B,t}(\theta + m) \geq 2\pi(g(\theta) + 2g(\theta + 1) + 2g(\theta + 2)) - \mathcal{T},$$

where $\mathcal{T}$ collects the $|m| \geq 3$ tail terms. Using monotonicity and the bounds from Lemma 9.18 we obtain

$$g(\theta) \geq \frac{29}{50} w_{B_{\min}}\left(\frac{1}{2}\right), \quad g(\theta + 1) \geq -\frac{3}{5} w_{B_{\min}}(1), \quad g(\theta + 2) \geq -\frac{11}{10} w_{B_{\min}}(2).$$

With $\pi < \frac{22}{7}$ we have $\pi^2 < 10$, so

$$w_{B_{\min}}\left(\frac{1}{2}\right) \geq \frac{5}{6} e^{-3/5} > \frac{9}{20}.$$

Using $\pi > \frac{333}{106}$ we obtain $\frac{6\pi^2}{25} > \frac{665334}{280900}$. The tail is bounded by

$$\mathcal{T} \leq 4\pi \int_{5/2}^{\infty} |a(\xi)| e^{-4\pi^2 t \xi^2} d\xi \leq 10^{-5},$$

using Lemma 9.17 to bound $|a(\xi)| \leq a(0) + \frac{11}{10} \log(1 + \xi) \leq 8$ for $\xi \geq \frac{5}{2}$ and the Gaussian tail bound $\int_R^{\infty} e^{-\alpha \xi^2} d\xi \leq e^{-\alpha R^2} / (2\alpha R)$. The exponential series bound $e^x \geq \sum_{j=0}^5 \frac{x^j}{j!}$ applied at $x = \frac{665334}{280900}$ gives $e^{6\pi^2/25} > 10$, hence

$$w_{B_{\min}}(1) \leq \frac{2}{3} e^{-6\pi^2/25} \leq \frac{1}{15}, \quad w_{B_{\min}}(2) \leq \frac{1}{3} e^{-24\pi^2/25} \leq \frac{1}{30000}.$$

Combining the inequalities and using $2\pi > \frac{666}{106}$ yields

$$P_A(\theta) \geq 2\pi \left(\frac{29}{50} \cdot \frac{9}{20} - \frac{2}{25} - \frac{11}{150000} \right) - 10^{-5} > \frac{11}{10}.$$

Evenness of P_A completes the bound on all of $\mathbb{T}$. □

Definition 9.20 (Uniform Lipschitz constant). For $B \geq B_{\min}$ set

$$L_A(B, t) := 2\pi \sup_{\theta \in [-\frac{1}{2}, \frac{1}{2}]} \sum_{m \in \mathbb{Z}} |g'_{B,t}(\theta + m)|,$$

and define

$$L_*(t) := \sup_{B \geq B_{\min}} L_A(B, t).$$

The constant L_* is used only for discretisation in the Szegő–Böttcher bridge.

Corollary 9.21 (Uniform discretisation threshold). Assume $c_* > 0$ in Lemma 9.19, and let $C_{\text{SB}} = 4$ be the absolute constant of Lemma 9.30. Define

$$M_0^{\text{unif}} := \left\lceil \frac{C_{\text{SB}} L_*(t)}{c_*} \right\rceil.$$

Then for every $B \geq B_{\min}$ and every $M \geq M_0^{\text{unif}}$,

$$\lambda_{\min}(T_M[P_A]) \geq \frac{1}{2} c_*.$$

Corollary 9.22 (Uniform prime cap time). Assume $c_* > 0$ in Lemma 9.19. Define

$$t_{\text{crit}} := \frac{3}{20} = 0.15.$$

Then the symmetrised prime operator satisfies

$$\|T_P^{\text{RKHS}}(t_{\text{crit}})\| \leq \rho(t_{\text{crit}}) \leq \frac{c_*}{4},$$

where $\rho(t)$ is the Gaussian norm cap of Lemma 10.24.

Definition of $B_{\min}$. We fix the minimal bandwidth parameter:

$$B_{\min} := 3. \tag{9.14}$$

This choice ensures that the Fejér kernel F_B has sufficient width to capture the essential support of the Archimedean density $a(\xi)$, while the Gaussian factor $e^{-4\pi^2 t \xi^2}$ with $t = t_{\text{crit}} = 3/20 = 0.15$ provides rapid decay beyond $|\xi| \approx 2$.

Lemma 9.23 (Analytic mean bound (auxiliary)). Let $t = t_{\text{crit}} = \frac{3}{20} = 0.15$ and $B_{\min} = 3$, and define $A_*(t) := \inf_{B \geq B_{\min}} A_0(B, t)$. Write $\alpha := 4\pi^2 t$ and let L_a be the global slope bound from Lemma 9.15. For any $r \in (0, \min\{B_{\min}, a(0)/L_a\})$ define

$$A_{\text{low}}(r) := 4\pi(a(0) - L_a r) \int_0^r \left(1 - \frac{\xi}{B_{\min}}\right) e^{-\alpha \xi^2} d\xi - 4\pi \int_r^\infty (a(0) + L_a \xi) e^{-\alpha \xi^2} d\xi.$$

Then $A_*(t) \geq A_{\text{low}}(r)$ for every $r \in (0, B_{\min})$.

Proof. By Lemma 9.15, $|a(\xi) - a(0)| \leq L_a |\xi|$ for all ξ , hence $a(\xi) \geq a(0) - L_a |\xi|$ on $|\xi| \leq r$ and $a(\xi) \geq -(a(0) + L_a |\xi|)$ for $|\xi| \geq r$. The restriction $r \leq a(0)/L_a$ ensures $a(0) - L_a \xi \geq 0$ on $[0, r]$. Since $0 \leq (1 - |\xi|/B) \leq 1$, we obtain

$$A_0(B, t) \geq 4\pi \int_0^r (a(0) - L_a \xi) \left(1 - \frac{\xi}{B}\right) e^{-\alpha \xi^2} d\xi - 4\pi \int_r^B (a(0) + L_a \xi) e^{-\alpha \xi^2} d\xi.$$

For $B \geq B_{\min}$ the first integral is bounded below by replacing B with $B_{\min}$, and the second is bounded below by extending to $[r, \infty)$. This yields the stated bound, uniformly in B , and therefore $A_*(t) \geq A_{\text{low}}(r)$. $\square$

Lemma 9.24 (Analytic Lipschitz bound (auxiliary)). *Let $t = t_{\text{crit}} = \frac{3}{20}$, $B_{\min} = 3$, and $L_*(t) = \sup_{B \geq B_{\min}} L_A(B, t)$ as in Definition 9.20. Set $\alpha := 4\pi^2 t$ and L_a as in Lemma 9.15. For $B \geq B_{\min}$ one has $\Phi_{B,t}(\xi) \leq e^{-\alpha\xi^2}$ and $|\Phi'_{B,t}(\xi)| \leq (B_{\min}^{-1} + 8\pi^2 t|\xi|)e^{-\alpha\xi^2}$. Define*

$$G(\xi) := L_a e^{-\alpha\xi^2} + (a(0) + L_a|\xi|)(B_{\min}^{-1} + 8\pi^2 t|\xi|)e^{-\alpha\xi^2},$$

and

$$L_{\text{up}} := 2\pi \left(G(0) + 2 \int_0^\infty G(\xi) d\xi \right).$$

Then $L_*(t) \leq L_{\text{up}}$.

Proof. Lemma 9.11 gives $L_A(B, t) = 2\pi \sup_{\theta \in [-\frac{1}{2}, \frac{1}{2}]} \sum_{m \in \mathbb{Z}} |g'_{B,t}(\theta + m)|$. Since $g' = a'\Phi + a\Phi'$, the bounds $|a'| \leq L_a$ and $|a| \leq a(0) + L_a|\xi|$ together with the envelope bounds on $\Phi_{B,t}$ and $\Phi'_{B,t}$ yield $|g'(\xi)| \leq G(\xi)$. The function G is even and decays Gaussianly, so for each θ the sum is dominated by a unit-step Riemann sum of G ; in particular

$$\sum_{m \in \mathbb{Z}} G(\theta + m) \leq G(0) + 2 \int_0^\infty G(\xi) d\xi.$$

Taking the supremum over θ and then $B \geq B_{\min}$ gives $L_*(t) \leq L_{\text{up}}$. $\square$

Remark (Auxiliary mean-modulus route). Lemmas 9.23 and 9.24 provide analytic estimates for $A_*(t)$ and $L_*(t)$, which may be used for coarse bounds on $\min P_A$ via the mean-modulus inequality. The main proof instead uses the direct pointwise floor from Lemma 9.19, which yields a much sharper constant.

Remark (Uniform vs. compact-dependent approach). The uniform constants (A_*, L_*, c_*) depend only on t and $B_{\min}$, not on the compact $[-K, K]$. This decouples the Archimedean floor from the parameter schedule $B(K)$: once $c_* > 0$ is established quantitatively, every compact inherits the same positive margin without requiring monotonicity arguments or numerical tables.

Remark (References). The Lipschitz estimate relies on standard Fourier analysis for compactly supported smooth kernels (see, e.g., Stein–Shakarchi [29, Ch. 2] and Zygmund [34, Ch. I]), while bounds on a and a' follow from classical properties of the digamma function ([22, §5.2]). The quantitative Toeplitz eigenvalue barrier used later takes the form $\lambda_{\min}(T_M[P]) \geq \min P - C_{\text{SB}} \omega_P(1/(2M))$ with $C_{\text{SB}} = 4$, as recorded in Böttcher–Silbermann [6, Ch. 5].

9.5 Fejér–Heat Modulus Control

Let $K > 0$ be fixed. Throughout this subsection we work on the interval $[-K, K]$ and the circle $\mathbb{T}$, and consider the Fejér kernel

$$\text{Fej}_M(\theta) := \frac{1}{M+1} \left(\frac{\sin(\pi(M+1)\theta)}{\sin(\pi\theta)} \right)^2,$$

and the heat kernel on the circle

$$h_t(\theta) := \sum_{k \in \mathbb{Z}} e^{-4\pi^2 t k^2} e^{2\pi i k \theta} = 1 + 2 \sum_{k \geq 1} e^{-4\pi^2 t k^2} \cos(2\pi k \theta).$$

Both kernels are nonnegative, even, and integrate to 1 on $\mathbb{T}$. Their convolution

$$\Xi_{M,t}(\theta) := (\text{Fej}_M * h_t)(\theta)$$

serves as the smoothing profile entering the definition of the Archimedean symbol. We record the basic bounds needed in the sequel; see, e.g., Stein–Shakarchi [29, Ch. 2] for the Fejér kernel and the classical heat kernel estimates.

Lemma 9.25 (Uniform bounds). *For every $M \in \mathbb{N}$ and $t > 0$ one has*

$$0 \leq \text{Fej}_M(\theta) \leq M + 1, \quad 0 \leq h_t(\theta) \leq \frac{C}{\sqrt{t}},$$

and therefore $0 \leq \Xi_{M,t}(\theta) \leq C \frac{\sqrt{M+1}}{\sqrt{t}}$ for an absolute constant $C > 0$.

Proof. The Fejér kernel is the Cesàro mean of Dirichlet kernels and satisfies $\text{Fej}_M(\theta) \leq M + 1$; the bound for h_t is classical (Gaussian upper bound). The convolution estimate follows from Cauchy–Schwarz. $\square$

Lemma 9.26 (Lipschitz modulus). *Let $f \in C^1([-K, K])$ with bounded derivative. Then for every $M \in \mathbb{N}$ and $t > 0$, the smoothed function*

$$f_{M,t}(x) := (f * (\text{Fej}_M * h_t))(x)$$

satisfies

$$\omega_{f_{M,t}}(\delta) \leq C \|f'\|_{L^\infty([-K, K])} \frac{\sqrt{M+1}}{\sqrt{t}} \delta,$$

for an absolute constant $C > 0$.

Proof. Differentiate under the convolution and use Lemma 9.25 to bound the L^1 -norm and the first moment of $\Xi_{M,t}$. $\square$

Corollary 9.27 (Modulus bound for the Arch symbol). *In the setting of Section 9.4, the Archimedean symbol P_A satisfies*

$$\omega_{P_A}(\delta) \leq C \left(\frac{\sqrt{M+1}}{\sqrt{t}} + 1 \right) \delta,$$

for all $\delta \geq 0$ and for an absolute constant $C > 0$ (depending on $\|a'\|_{L^\infty([-K, K])}$).

Proof. Apply Lemma 9.26 to $f = a$ and note that convolution with the Fejér–heat kernel preserves the Lipschitz modulus up to the displayed factor. $\square$

These analytic bounds will be combined with the Szegő–Böttcher barrier in the mixed bridge inequality of Theorem 9.35.

9.6 Matrix Guards (General Lemmas)

The lemmas below provide the Frobenius drift control and Szegő–Böttcher barrier used in the uniform A3 bridge (Theorem 9.35).

Lemma 9.28 (Hoffman–Wielandt and Ky Fan guard). *Let $A, B \in \mathbb{C}^{M \times M}$ be Hermitian and set $E := B - A$. Denote by $\lambda_i^\downarrow(A)$ the eigenvalues of A in non-increasing order. Then, for every $1 \leq k \leq M$,*

$$\sum_{i=1}^k |\lambda_i^\downarrow(B) - \lambda_i^\downarrow(A)| \leq \sqrt{k} \|E\|_F,$$

where $\|E\|_F = \sqrt{\text{Tr}(E^*E)}$ is the Frobenius norm. In particular

$$|\lambda_{\min}(B) - \lambda_{\min}(A)| \leq \|E\|_F.$$

Proof. The Hoffman–Wielandt inequality gives $\sum_i |\lambda_i(B) - \lambda_{\sigma(i)}(A)|^2 \leq \|E\|_F^2$ for a suitable permutation σ ; see Horn–Johnson, *Matrix Analysis* (2nd ed.), Thm. 7.4.9. Ky Fan majorisation (Cor. 7.3.5 loc. cit.) implies $\sum_{i \leq k} |\lambda_i^\downarrow(B) - \lambda_i^\downarrow(A)| \leq \sum_{i \leq k} \sigma_i(E)$, and Cauchy–Schwarz yields $\sum_{i \leq k} \sigma_i(E) \leq \sqrt{k} \|E\|_F$. $\square$

Corollary 9.29 (Frobenius slack for Toeplitz glue). *Let $T_M[P]$ be a Toeplitz matrix and ΔT a perturbation with $\|\Delta T\|_F \leq \varepsilon$. Then*

$$|\lambda_{\min}(T_M[P + \Delta P]) - \lambda_{\min}(T_M[P])| \leq \varepsilon.$$

Consequently, if $A := T_M[P_A] - T_P^{\text{cap}}(t, M)$ satisfies $\lambda_{\min}(A) \geq \delta > 0$ and $\|T_P^{\text{Ray}}(t, M) - T_P^{\text{cap}}(t, M)\|_F \leq \varepsilon$, then

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \delta - \varepsilon.$$

Lemma 9.30 (Szegő–Böttcher barrier with explicit modulus). *Let P_A be the Archimedean symbol constructed in Section 9.4. There exists an absolute constant $C_{\text{SB}} = 4$ such that for every $M \geq 1$*

$$\lambda_{\min}(T_M[P_A]) \geq \min_{\theta \in \mathbb{T}} P_A(\theta) - C_{\text{SB}} \omega_{P_A}\left(\frac{1}{2M}\right).$$

Remark (Sources and scope of C_{SB}). This is the classical Toeplitz eigenvalue stability for Lipschitz symbols. We use the version recorded in Böttcher–Silbermann’s *Introduction to Large Truncated Toeplitz Matrices* (Theorem 5.5 together with Corollary 5.7 in Chapter 5); see also Grenander–Szegő (Ch. 3) and Varga’s *Gershgorin and His Circles* (Cor. 2.5.3) for related Gershgorin-based formulations. For the Lipschitz/Hölder classes relevant here the constant in front of the modulus is $C_{\text{SB}} = 4$.

Lemma 9.30 is the only place where this numerical constant enters our treatment of A3. Coupled with the uniform RKHS cap (Corollary 9.22) and the uniform floor $c_* > 0$, it yields the mixed lower bound in Theorem 9.35.

Remark (Operator difference vs. symbol difference). When applying Lemma 9.30 we always work with the Toeplitz operators $T_M[P_A]$ and $T_M[P_A] - T_P^{\text{Ray}}(t, M)$; no “symbol minus symbol” simplification is invoked. The lower bounds track the operator difference directly, so all perturbative terms are measured in operator/Frobenius norms as mandated by Lemma 9.28.

Remark (Frobenius norm structure). For Hermitian Toeplitz matrices with first row $c_0, \dots, c_{M-1}$ and coefficients $c_{-k} = \overline{c_k}$, one has $\|T_M[P]\|_F^2 = M|c_0|^2 + 2 \sum_{k=1}^{M-1} (M-k)|c_k|^2$. Hence a split budget $\varepsilon = \varepsilon_{\text{tail}}^F + \varepsilon_{\text{grid}}^F + \varepsilon_{\text{num}}^F$ controls the total spectral drift of $T_P^{\text{Ray}}(t, M)$ relative to the capped operator.

9.7 A3 locking summary

We record how the local ingredients assembled in §9 feed the global lock:

- Lemma 9.34 supplies the bounded-overlap control on caps.
- Lemma 9.32 records the uniform single-scale coupling.
- Corollary 9.22 gives the uniform RKHS prime cap at $t = t_{\text{crit}}$.
- Theorem 9.35 combines the uniform symbol floor $c_* > 0$ with the RKHS prime cap from Corollary 9.22 and the Frobenius guard of Corollary 9.29.

Corollary 9.31 (Lock). *Under the hypotheses of Lemmas 9.34, 9.32, and Corollary 9.22 the A3 lock closes with a constant depending only on the overlap bound and the uniform prime cap.*

Proof. Lemma 9.34 gives almost orthogonality, Lemma 9.32 aligns the symbol and RKHS bounds at a single scale, Corollary 9.22 provides the uniform cap, and Theorem 9.35 supplies the quantitative margin with the uniform floor. Summing the contributions yields the stated lock. $\square$

See also. Lemmas 9.11–9.34, Uniform floor Lemma 9.19, uniform RKHS cap Lemma 10.18.

Throughout this section a denotes the Archimedean density and P_A denotes the Archimedean symbol defined in §9.4 via the Fejér×heat window. We use only this definition in the main proof chain. The arguments below sit inside the classical Toeplitz framework of Szegő and Böttcher [30, 7, 14, 6], with Fourier bounds calibrated against standard real-analytic estimates [29, 34]. The following chain of lemmas replaces all “A3 assume ...” statements by explicit estimates. An analytic proof of the Rayleigh identification is recorded in §9.2, while symbol regularity and Archimedean floors are collected in §9.4.

Remark (Single-scale alignment). We use a *single* heat scale t throughout the Toeplitz bridge and the RKHS cap. The Fejér×heat symbol P_A and the RKHS kernel $K_A^{(t)}$ are built from the same parameter, and in the main line we fix $t = t_{\text{crit}} = \frac{3}{20} = 0.15$. This is the scale at which the Rayleigh identification and the RKHS cap are compared.

Lemma 9.32 (Single-scale coupling (uniform)). *Let P_A be the Archimedean symbol from §9.4 built with the Fejér×heat parameter t , and assume $t = t_{\text{crit}} = \frac{3}{20} = 0.15$ in the main line. Then*

$$\min_{\theta \in \mathbb{T}} P_A(\theta) \geq c_*,$$

by Lemma 9.19, and the RKHS cap

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$$

follows from Corollary 9.22. The Toeplitz and RKHS bounds are thus evaluated at the same heat scale.

Proof. Immediate from Lemma 9.19 and Corollary 9.22. $\square$

Lemma 9.33 (Lipschitz symbol with positive floor implies A3 prerequisites). *Let $P_A \in \text{Lip}(1)$ with $\min_{\mathbb{T}} P_A \geq c_0 > 0$. Then the Toeplitz operator T_{P_A} satisfies*

$$T_{P_A} \succeq c_0 I, \quad \|T_{P_A}\|_{\text{op}} \leq \|P_A\|_{L^\infty}.$$

In particular, once $\rho_K \geq \|P_A\|_{L^\infty}$ the A3-lock positivity and boundedness hypotheses hold.

Proof. For any f with $\|f\|_2 = 1$ we have $\langle T_{P_A} f, f \rangle = \int_{\mathbb{T}} P_A(\theta) |f(\theta)|^2 d\theta \geq c_0$, hence $T_{P_A} \succeq c_0 I$. The $\|P_A\|_\infty$ bound is immediate from the Rayleigh quotient; see, e.g., the spectral calculus in [18, 32]. $\square$

Lemma 9.34 (Combining with the RKHS cap). *Suppose P_A is constructed as above and the RKHS cap*

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$$

holds (Corollary 9.22). Then T_{P_A} simultaneously satisfies the positivity floor and the operator-norm bound required by A3-lock.

Proof. Apply Lemmas 9.32 and 9.33, together with Corollary 9.22. $\square$

A3 input summary (uniform version).

(A3-U.1) *Uniform Arch floor.* Lemma 9.19 provides the explicit floor $c_* = \frac{11}{10}$ on $\mathbb{T}$ for all $B \geq B_{\min}$.

(A3-U.2) *Uniform prime cap.* Corollary 9.22 gives $\rho(t) \leq c_*/4$ at $t = t_{\text{crit}} = \frac{3}{20} = 0.15$.

(A3-U.3) *Uniform discretisation.* Corollary 9.21 provides M_0^{unif} such that $\lambda_{\min}(T_M[P_A]) \geq c_*/2$ for all $M \geq M_0^{\text{unif}}$.

The uniform Archimedean floor (Lemma 9.19) provides a **K-independent** lower bound

$$\min_{\theta \in \mathbb{T}} P_A(\theta) \geq c_*$$

valid for the **entire circle** $\mathbb{T}$ and all $B \geq B_{\min}$. This eliminates the need for arc-based constructions and provides a simpler, cleaner proof.

Theorem 9.35 (Uniform A3 bridge). *Assume the uniform floor $c_* > 0$ from Lemma 9.19 and fix $B \geq B_{\min}$ and $t = t_{\text{crit}} = \frac{3}{20} = 0.15$. Then for every $M \geq M_0^{\text{unif}}$ (Corollary 9.21),*

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \frac{c_*}{4} > 0,$$

and the associated Fejér×heat test functions satisfy

$$Q^*(t; \Phi_{B,t}) \geq 0.$$

Proof. The uniform floor Lemma 9.19 gives $\min_{\theta \in \mathbb{T}} P_A(\theta) \geq c_*$ for all $B \geq B_{\min}$. Corollary 9.21 provides M_0^{unif} such that $C_{\text{SB}} \omega_{P_A}(1/(2M)) \leq c_*/2$ for all $M \geq M_0^{\text{unif}}$. Corollary 9.22 gives $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq c_*/4$. Thus $\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq c_* - c_*/2 - c_*/4 = c_*/4$. Lemma 9.8 combined with Theorem 9.10 converts the matrix margin into $Q^*(t; \Phi_{B,t}) \geq 0$. $\square$

10 Prime Operator Control via RKHS

10.1 RKHS Core

Our RKHS setup follows the classical foundation laid by Aronszajn [2] and the modern expositions of Berline–Thomas–Agnan and Paulsen–Raghupathi [4, 23, 5, 24].

Let $(\mathcal{X}, \mu)$ be a measure space and let $k: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a positive definite kernel with reproducing kernel Hilbert space $(\mathcal{H}_k, \langle \cdot, \cdot \rangle_{\mathcal{H}_k})$. Denote by $T_k: L^2(\mu) \rightarrow L^2(\mu)$ the integral operator

$$(T_k f)(x) := \int_{\mathcal{X}} k(x, y) f(y) d\mu(y).$$

If $\mathcal{X}$ is represented discretely by nodes $\{x_i\}_{i=1}^N$ we write $K = [k(x_i, x_j)]_{i,j=1}^N$ for the Gram matrix.

Lemma 10.1 (Energy identity). *For $f \in \mathcal{H}_k$ supported on the closure of $\text{span}\{k(\cdot, x): x \in \mathcal{X}\}$ one has*

$$\|f\|_{\mathcal{H}_k}^2 = \langle f, T_k^\dagger f \rangle_{L^2(\mu)},$$

where $T_k^\dagger$ is the pseudoinverse on the image of T_k . In particular, if $f(x) = \sum_{i=1}^N a_i k(x, x_i)$ for a finite sample, then

$$\|f\|_{\mathcal{H}_k}^2 = a^\top K a.$$

Lemma 10.2 (Spectral floor for Gram matrices). *Assume the diagonal of K obeys $k(x_i, x_i) \geq c_0$ and the off-diagonal mass satisfies*

$$\sum_{j \neq i} |k(x_i, x_j)| \leq \rho_K \quad \text{for every } i \in \{1, \dots, N\}.$$

Then

$$\lambda_{\min}(K) \geq c_0 - \rho_K.$$

Proof. Gershgorin's circle theorem states that every eigenvalue λ of K belongs to at least one disc

$$D_i = \left\{ z \in \mathbb{C} : |z - k(x_i, x_i)| \leq \sum_{j \neq i} |k(x_i, x_j)| \right\}.$$

The hypothesis guarantees $\inf D_i \geq c_0 - \rho_K$, hence every eigenvalue lies in $[c_0 - \rho_K, \infty)$. $\square$

Proposition 10.3 (Operator sandwich). *Let T_k be positive on $\mathcal{H}_k$ with spectral bottom at least c_0 , and suppose a discretisation or truncation K satisfies the off-diagonal bound of Lemma 10.2. For $f = \sum_i a_i k(\cdot, x_i)$ we have*

$$\|f\|_{L^2(\mu)}^2 \leq \frac{1}{c_0 - \rho_K} \|f\|_{\mathcal{H}_k}^2, \quad \lambda_{\min}(K) \geq c_0 - \rho_K.$$

In particular, whenever $\rho_K \leq c_0/2$ the bridge margin $\frac{1}{2}(c_0 - \rho_K)$ of Theorem 9.35 is available.

Proof. Lemma 10.2 yields the spectral bound. Any $g = \sum_i a_i k(\cdot, x_i)$ satisfies $g^\top K g = \|g\|_{\mathcal{H}_k}^2$ by Lemma 10.1. Since $K \succeq (c_0 - \rho_K)I$, Rayleigh quotients yield $\|g\|_{L^2(\mu)}^2 \leq (c_0 - \rho_K)^{-1} \|g\|_{\mathcal{H}_k}^2$. $\square$

These statements provide the structural ingredients cited in Assumption (A3.1) and in the proof of Theorem 9.35: the diagonal floor produces c_0 , the RKHS contraction supplies ρ_K , and Lemma 10.2 transfers the margin to the finite Toeplitz block.

Lemma 10.4 (Rayleigh sampling identification). *For any Fejér $\times$ heat window $\Phi = \Phi_{B,t}$ with Dirichlet sampling polynomial $p(\theta) = \sum_{k \in \mathbb{Z}} \hat{\Phi}(k) e^{2\pi i k \theta}$, one has*

$$\langle T_M[P_A]p, p \rangle_{L^2(\mathbb{T})} - (2M+1) \langle T_P^{\text{Ray}}(t, M)p, p \rangle_{L^2(\mathbb{T})} = Q^*(t; \Phi)$$

whenever M is large enough that the Dirichlet coefficients of Φ lie in the span $\{|k_\tau\rangle\}$. In particular, any estimate on the two quadratic forms $\langle T_M[P_A]p, p \rangle$ and $\langle T_P^{\text{Ray}}(t, M)p, p \rangle$ transfers directly to $Q^*(t; \Phi)$ via the displayed identity.

Remark (Finite support and the threshold M). A Fejér window of bandwidth B has Fourier support contained in $\{|k| \leq B\}$, and the subsequent heat factor e^{-tk^2} only rescales these coefficients. Thus the Dirichlet polynomial p is already finite, and it suffices to choose $M \geq B$ to meet the span condition used in Lemma 10.4. The detailed identification is spelt out in Lemma 9.8.

10.2 RKHS Contraction Mechanism

See also. Weight cap Lemma (w_max $\leq 2/e$) 10.8, Node-gap lower bound Lemma 10.10, Single-scale alignment Corollary 10.11, Mixed lower bound in A3 Theorem 9.35.

We briefly record the RKHS framework that delivers operator positivity $T_A - T_P^{\text{RKHS}}(t) \succeq 0$ on each compact without pointwise measure domination; comprehensive background may be found in [2, 4, 23, 5, 24].

10.3 Setup

Fix $K = [-K, K]$ and let $\{\alpha_n\}$ be the active nodes on K . Let $K_A^{(t)}(\alpha, \beta)$ be the Archimedean kernel associated to the heat scale $t > 0$ (normalized $K_A^{(t)}(\alpha, \alpha) = 1$). Define the Hilbert space $\mathcal{H}_K$ as the RKHS with kernel $K_A^{(t)}$ at the single scale t (in the main line, $t = t_{\text{crit}} = \frac{3}{20} = 0.15$). In the even setting (T0) we merge the symmetric nodes $\pm\alpha_n$ into a single reproducing vector and work with the effective weights

$$w_{\text{RKHS}}((n)) := \frac{\Lambda(n)}{\sqrt{n}} \in (0, \infty), \quad \sup_n w_{\text{RKHS}}((n)) \leq \sup_{x>0} \frac{\log x}{\sqrt{x}} = \frac{2}{e} < 1. \quad (10.1)$$

(This is the *undoubled* operator weight; in $Q^*(t; \Phi)$ the evenization yields doubled weights $2\Lambda(n)/\sqrt{n}$ at positive nodes, equivalent to $\Lambda(n)/\sqrt{n}$ at $\pm$ nodes for even tests.) The prime operator is

$$T_P^{\text{RKHS}}(t) := \sum_{\alpha_n \in [-K, K]} w_{\text{RKHS}}((n)) |k_{\alpha_n}\rangle \langle k_{\alpha_n}|, \quad \|k_\alpha\|_{\mathcal{H}_K} = 1, \quad (10.2)$$

and the Archimedean operator acts via this kernel and is positive semidefinite on $\mathcal{H}_K$.

10.4 Norm bound via weighted Gram

Let G be the Gram matrix $G_{mn} = \langle k_{\alpha_m}, k_{\alpha_n} \rangle_{\mathcal{H}_K}$. With $W = \text{diag}(w_{\text{RKHS}}((n)))$ one has $\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} = \|W^{1/2} G W^{1/2}\|_{\ell^2 \rightarrow \ell^2}$. Writing δ_K for the minimal node spacing on $[-K, K]$ and setting

$$S_K(t) := \sum_{m \neq n} e^{-\frac{(\alpha_m - \alpha_n)^2}{4t}} \leq \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}} \quad (10.3)$$

one obtains the Gershgorin-type bound

$$\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} S_K(t), \quad w_{\max}^{\text{RKHS}} := \max_{\alpha_n \in [-K, K]} w_{\text{RKHS}}((n)). \quad (10.4)$$

Lemma 10.5 (Geometric tail bound for SK(t)). *For any node set with minimal spacing $\delta_K > 0$ one has*

$$S_K(t) := \sum_{m \neq n} e^{-\frac{(\alpha_m - \alpha_n)^2}{4t}} \leq 2 \sum_{j \geq 1} e^{-\frac{j^2 \delta_K^2}{4t}} \leq \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}}. \quad (10.5)$$

Proof. Fix n and order the remaining nodes by increasing distance. The j -th nearest neighbor lies at distance at least $j\delta_K$, hence the n -th row sum of off-diagonal magnitudes is bounded by $2 \sum_{j \geq 1} e^{-j^2 \delta_K^2/(4t)}$. Summing rows and using symmetry gives the first inequality. Since $j^2 \geq j$ for $j \geq 1$, $e^{-j^2 c} \leq e^{-jc}$ for $c > 0$, yielding the geometric series bound and the stated closed form. $\square$

Theorem 10.6 (Strict contraction). *If $t = t_{\min}(K)$ is chosen so that $S_K(t_{\min}) \leq \frac{1 - w_{\max} - \varepsilon_K}{\sqrt{w_{\max}}}$ for some $\varepsilon_K \in (0, 1 - w_{\max})$, then $\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq \rho_K < 1$ with $\rho_K = w_{\max} + \sqrt{w_{\max}} S_K(t_{\min})$, and hence*

$$T_A - T_P^{\text{RKHS}}(t) \succeq (1 - \rho_K) T_A \succeq 0 \quad \text{on } \mathcal{H}_K. \quad (10.6)$$

Moreover, it suffices to enforce the geometric bound of Lemma 10.5. Solving $\frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}} \leq \eta_K$ for t gives

$$\boxed{t_{\min}(K) = \frac{\delta_K^2}{4 \ln((2 + \eta_K)/\eta_K)}}, \quad \eta_K = \frac{1 - w_{\max} - \varepsilon_K}{\sqrt{w_{\max}}}. \quad (10.7)$$

Remark. Because $\delta_K \downarrow 0$ as the compact widens, the closed form (10.7) shows that $t_{\min}(K)$ is automatically chosen monotone decreasing along the chain $K \nearrow$. This supports the optional adaptive cap, but the uniform branch fixes a single scale $t = t_{\text{crit}}$.

Proposition 10.7 (Dataset-free RKHS schedule). *Let $w_{\max} = \sup \Lambda(n)/\sqrt{n} \leq 2/e$ and let δ_K denote the minimal logarithmic spacing on $[-K, K]$ (Lemma 10.13). For*

$$S_K(t) := \sum_{m \neq n} e^{-\frac{(\alpha_m - \alpha_n)^2}{4t}} \leq \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}}$$

(Lemma 10.5) choose

$$t_{\min}(K) = \frac{\delta_K^2}{4 \ln((2 + \eta_K)/\eta_K)}, \quad \eta_K \in (0, 1 - w_{\max}).$$

Then $S_K(t_{\min}(K)) \leq \eta_K$ and therefore

$$\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq w_{\max} + \sqrt{w_{\max}} S_K(t_{\min}(K)) =: \rho_K < 1,$$

so $T_A - T_P^{\text{RKHS}}(t) \succeq (1 - \rho_K) T_A$ on the RKHS.

Proof. By the Gershgorin circle theorem applied to $W^{1/2}GW^{1/2}$ (see, e.g., [18, Thm. 6.1.1]; also [32]), each eigenvalue λ of $T_P^{\text{RKHS}}(t)$ lies in a disc centered at $w_{\text{RKHS}}((n))$ with radius $\sqrt{w_{\text{RKHS}}((n)) \sum_{m \neq n} \sqrt{w_{\text{RKHS}}((m))} |G_{mn}|} \leq \sqrt{w_{\text{max}}^{\text{RKHS}} \sum_{m \neq n} |G_{mn}|}$. Using $G_{mn} = \langle k_{\alpha_m}, k_{\alpha_n} \rangle \leq e^{-(\alpha_m - \alpha_n)^2/(4t)}$ and Lemma 10.5 yields $\|T_P^{\text{RKHS}}(t)\| \leq w_{\text{max}}^{\text{RKHS}} + \sqrt{w_{\text{max}}^{\text{RKHS}}} S_K(t)$. Imposing $S_K(t_{\min}) \leq (1 - w_{\text{max}}^{\text{RKHS}} - \varepsilon_K)/\sqrt{w_{\text{max}}^{\text{RKHS}}}$ gives the claim. For the explicit $t_{\min}$, set $q := e^{-\delta_K^2/(4t)} \in (0, 1)$ and require $\frac{2q}{1-q} \leq \eta_K$, i.e. $q \leq \frac{\eta_K}{2 + \eta_K}$. This is equivalent to $t \leq \delta_K^2/(4 \ln((2 + \eta_K)/\eta_K))$. $\square$

Lemma 10.8 (Effective weight cap). *For $w(p^m) = \frac{\log p}{p^{m/2}}$ one has $0 \leq w(p^m) \leq \frac{2}{e} < \frac{3}{4}$, with the maximum attained at $p^m = e^2$ formally. Hence $w_{\max} \leq 2/e < 3/4 < 1$ on every compact. (Rational bound: $2/e \approx 0.7358 < 3/4 = 0.75$.)*

Proof. Consider $f(x) = \log x/\sqrt{x}$ on $x > 1$; $f'(x) = (1 - \frac{1}{2} \log x)/x^{3/2}$ vanishes at $x = e^2$ with $f(e^2) = 2/e$. $\square$

Lemma 10.9 (Rayleigh lower bound for $\|TP\|$). *For the prime operator $T_P^{\text{RKHS}}(t) = \sum_{\alpha_n} w_{\text{RKHS}}((n)) |k_{\alpha_n}\rangle \langle k_{\alpha_n}|$ with normalized kernel vectors $\|k_{\alpha}\| = 1$, the operator norm satisfies*

$$\|T_P^{\text{RKHS}}(t)\| \geq \sup_{n: \alpha_n \in [-K, K]} w_{\text{RKHS}}((n)) =: w_{\text{max}}^{\text{RKHS}}. \quad (10.8)$$

Proof. For any node m with $\alpha_m \in [-K, K]$, the Rayleigh quotient gives

$$\langle k_{\alpha_m}, T_P^{\text{RKHS}}(t) k_{\alpha_m} \rangle = \sum_n w_{\text{RKHS}}((n)) |\langle k_{\alpha_n}, k_{\alpha_m} \rangle|^2 \geq w_{\text{RKHS}}((m)) \|k_{\alpha_m}\|^2 = w_{\text{RKHS}}((m)). \quad (10.9)$$

Hence $\|T_P^{\text{RKHS}}(t)\| \geq w(m)$ for every active node, implying $\|T_P^{\text{RKHS}}(t)\| \geq w_{\text{max}}$. $\square$

Lemma 10.10 (Node gap on compacts). *For $\alpha_n = \frac{\log n}{2\pi}$ and fixed $K > 0$ the active set is $\{2, \dots, \lfloor e^{2\pi K} \rfloor\}$ and the minimal spacing satisfies*

$$\delta_K := \min_{m \neq n, \alpha_m, \alpha_n \in [-K, K]} |\alpha_m - \alpha_n| \geq \frac{1}{2\pi(\lfloor e^{2\pi K} \rfloor + 1)}. \quad (10.10)$$

Proof. Mean value theorem on $\log x$ between consecutive integers. $\square$

Corollary 10.11 (Single-scale alignment (uniform)). *Let $t = t_{\text{crit}}$ be the fixed heat scale used to build P_A and the RKHS kernel $K_A^{(t)}$. If $L_A(B, t) \leq L_A^*$ and $\min P_A \geq c_* > 0$, then Corollary 9.7 applies with the uniform cap $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$ and modulus L_A^* . Thus the symbol modulus and the RKHS contraction are evaluated at the same scale.*

Remark (Uniform vs. adaptive prime caps). Two complementary caps for $T_P^{\text{RKHS}}(t)$ are available. A *uniform* trace cap at $t = t_{\text{crit}}$ gives $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq c_*/4$ on every compact and suffices for the A3 budget. An *adaptive* cap at $t = t_{\min}(K)$ uses the node gap δ_K and yields $\|T_P^{\text{RKHS}}(t)\| \leq w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} \eta_K$ for a prescribed off-diagonal level η_K . The mainline proof invokes the uniform cap; the adaptive one provides additional slack when needed.

Theorem 10.12 (One-prime induction). *Upon crossing an activity threshold that introduces a single new node with weight w_{new} , the update is*

$$\|T_P^{\text{RKHS}, \text{new}}(t)\| \leq \|T_P^{\text{RKHS}, \text{old}}(t)\| + w_{\text{new}}. \quad (10.11)$$

Consequently, if $\|T_P^{\text{RKHS}, \text{old}}(t)\| \leq \rho_K^{\text{old}} < 1$ and $\rho_K^{\text{old}} + w_{\text{new}} < 1$, then $T_A - T_P^{\text{RKHS}, \text{new}}(t) \succeq 0$ on $\mathcal{H}_K$.

Remark (Boxed formulas and effective weight cap).

$$\boxed{S_K(t) = \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}}}, \quad \boxed{\rho_K = w_{\max} + \sqrt{w_{\max}} S_K(t_{\min})}. \quad (10.12)$$

In the even windowed setting the effective prime weights satisfy $0 \leq w_{\text{RKHS}}((n)) \leq 2/e$ (see Lemma G.8 in the MD appendix), hence $w_{\max}^{\text{RKHS}} \leq 2/e < 1$, ensuring feasibility of strict contraction once $t_{\min}(K) \asymp c\delta_K^2$ is small enough.

Lemma 10.13 (Node separation). *For $\alpha_n = \log n/(2\pi)$ and fixed $K > 0$ one has a finite active set $\{n : \alpha_n \in [-K, K]\} = \{2, \dots, \lfloor e^{2\pi K} \rfloor\}$ and a positive minimal gap*

$$\delta_K := \min_{m \neq n, \alpha_m, \alpha_n \in [-K, K]} |\alpha_m - \alpha_n| \geq \frac{1}{2\pi(\lfloor e^{2\pi K} \rfloor + 1)}. \quad (10.13)$$

10.5 RKHS prime contraction on compacts

Notation and standing choices

Fix a compact $[-K, K] \subset \mathbb{R}$, $K \geq 1$. Prime sample nodes (as in the normalization Lemma 6.2) are

$$\xi_n := \frac{\log n}{2\pi} \in [0, \infty), \quad n \geq 2,$$

with weights (doubling belongs only to the Weil functional $Q^*(t; \Phi)$; see Remark 10.5)

$$w_{\text{RKHS}}((n)) := \frac{\Lambda(n)}{\sqrt{n}}, \quad w_{\max}^{\text{RKHS}} := \sup_{n \geq 2} w_{\text{RKHS}}((n)) \leq \frac{2}{e}.$$

We work in the RKHS H_k of the heat kernel on $\mathbb{R}$,

$$k_t(x, y) := \exp\left(-\frac{(x-y)^2}{4t}\right), \quad t > 0,$$

and write $K_t = (k_t(\xi_m, \xi_n))_{m, n \geq 2}$ for the Gram matrix on the sample nodes.

Remark (Evenization and weights). Lemma 6.2 identifies the node set $\xi_n = \log n/(2\pi)$ and shows that $Q^*(t; \Phi)$ uses the doubled weights $2\Lambda(n)/\sqrt{n}$ on the positive half-line. Operator and RKHS estimates are performed on the symmetric node set $\{\pm\xi_n\}$ with weights $\Lambda(n)/\sqrt{n}$, which is equivalent to keeping the positive nodes and doubling the weights (the Q^* convention). All prime caps below are interpreted in this symmetric sense; no additional assumptions enter.

For separation we use the simple lower bound

$$\delta_K := \min\left\{\xi_{n+1} - \xi_n : \xi_n, \xi_{n+1} \in [-K, K]\right\} \geq \frac{1}{2\pi(\lfloor e^{2\pi K} \rfloor + 1)}. \quad (10.14)$$

Remark (Bookkeeping parameters). Fix any $\eta_K \in (0, 1 - w_{\max})$ and set

$$t_{\min}(K) := \frac{\delta_K^2}{4 \ln((2 + \eta_K)/\eta_K)}. \quad (10.15)$$

We also use the shorthand

$$S_K(t) := \sup_{x \in [-K, K]} \sum_{\substack{n \geq 2 \\ \xi_n \in [-K, K] \\ \xi_n \neq x}} \exp\left(-\frac{(x - \xi_n)^2}{4t}\right).$$

Lemma 10.14 (Shift-robust sampling window). *Let $0 < r \leq \delta_K$ and $\tau \in [-K, K]$. Then for every $t > 0$,*

$$\sum_{\xi_n \in [-K, K]} w_{\text{RKHS}}(\xi_n) \int_{\tau-r}^{\tau+r} k_t(x, \xi_n)^2 dx \leq w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} S_K(t).$$

In particular, with $t = t_{\min}(K)$ the right-hand side is at most $w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} \eta_K$, uniformly in τ .

Proof. Integrate the Schur/Gram estimate from Proposition 10.17 over $x \in [\tau - r, \tau + r]$. The diagonal contributes at most $w_{\max} \int k_t(x, x)^2 dx$, while off-diagonal terms are controlled by $\sqrt{w_{\max}} \sup_{x \in [-K, K]} \sum_{\xi_n \neq x} k_t(x, \xi_n)^2$, which is $\sqrt{w_{\max}} S_K(t)$. $\square$

Energy and Gram

Lemma 10.15 (Energy identity). *For any finite sample $x_1, \dots, x_M$ and coefficients $a \in \mathbb{R}^M$ one has*

$$\left\| \sum_{m=1}^M a_m k_t(\cdot, x_m) \right\|_{H_k}^2 = a^\top \left(k_t(x_m, x_n) \right)_{m, n=1}^M a.$$

This is the reproducing property of RKHS; see [2].

Lemma 10.16 (Off-diagonal sum bound). *For every $t > 0$ and $K \geq 1$,*

$$S_K(t) \leq \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}} \quad \text{and in particular} \quad S_K(t_{\min}(K)) \leq \eta_K,$$

with δ_K and $t_{\min}(K)$ from (10.14)–(10.15).

Proof. Enumerate the points of $\Xi_K := \{\xi_n \in [-K, K]\}$ along $\mathbb{R}$ with gaps $\geq \delta_K$. Then for any $x \in [-K, K]$ the off-diagonal sum is dominated by two geometric tails:

$$\sum_{j \geq 1} e^{-(j\delta_K)^2/(4t)} + \sum_{j \geq 1} e^{-(j\delta_K)^2/(4t)} \leq \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}},$$

giving the first claim; the second follows by the choice of $t_{\min}(K)$. $\square$

Two analytic caps for the prime operator

We view the prime sampling operator $T_P^{\text{RKHS}}(t)$ as

$$(T_P^{\text{RKHS}}(t)f)(x) := \sum_{\xi_n \in [-K, K]} w_{\text{RKHS}}((n)) f(\xi_n) k_t(x, \xi_n),$$

restricted to $H_k \upharpoonright [-K, K]$.

Proposition 10.17 (RKHS cap via Gram geometry). *For every $t > 0$ and $K \geq 1$,*

$$\|T_P^{\text{RKHS}}(t)\|_{H_k \rightarrow H_k} \leq w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} S_K(t).$$

In particular, with $t = t_{\min}(K)$ from (10.15),

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho_K := w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} \eta_K, \quad \eta_K \in (0, 1 - w_{\max}^{\text{RKHS}}). \quad (10.16)$$

Proof. Let $g_n := k_t(\cdot, \xi_n)$ so that $\|g_n\|_{H_k}^2 = k_t(\xi_n, \xi_n) = 1$. Define the unweighted frame operator

$$S_0 := \sum_{\xi_n \in [-K, K]} |g_n\rangle\langle g_n|.$$

Its Gram matrix on the span of $\{g_n\}$ is $G = (\langle g_m, g_n \rangle)_{m,n}$ with diagonal 1 and off-diagonal entries $|\langle g_m, g_n \rangle| = k_t(\xi_m, \xi_n) \leq e^{-(\xi_m - \xi_n)^2/(4t)}$. By definition of $S_K(t)$, the absolute row sum of G is bounded by $1 + S_K(t)$, hence the Schur/Gershgorin bound gives

$$\|S_0\| \leq 1 + S_K(t).$$

Since $0 \leq w_{\text{RKHS}}((n)) \leq w_{\max}^{\text{RKHS}}$, we have the PSD order inequality $T_P^{\text{RKHS}}(t) = \sum w_{\text{RKHS}}((n)) |g_n\rangle\langle g_n| \preceq w_{\max}^{\text{RKHS}} S_0$, hence

$$\|T_P^{\text{RKHS}}(t)\| \leq w_{\max}^{\text{RKHS}}(1 + S_K(t)) \leq w_{\max}^{\text{RKHS}} + \sqrt{w_{\max}^{\text{RKHS}}} S_K(t),$$

because $w_{\max}^{\text{RKHS}} \leq \sqrt{w_{\max}^{\text{RKHS}}}$ for $0 < w_{\max}^{\text{RKHS}} \leq 1$. This proves the stated bound. $\square$

Lemma 10.18 (Uniform RKHS cap). *Let*

$$\begin{aligned} \rho(t) &:= 2 \int_0^\infty y e^{y/2} e^{-4\pi^2 t y^2} dy \\ &= 2 \left[\frac{1}{8\pi^2 t} + \frac{\sqrt{\pi}}{64\pi^3 t \sqrt{t}} \exp\left(\frac{1}{64\pi^2 t}\right) \operatorname{erfc}\left(-\frac{1}{8\pi\sqrt{t}}\right) \right], \end{aligned}$$

the equality being the standard Gaussian evaluation. The function $\rho(t)$ is strictly decreasing in t and satisfies $\rho(t) \rightarrow 0$ as $t \rightarrow \infty$.

Proof. Lemma 10.26 and Lemma 10.28 give the closed form. For monotonicity, observe that the integrand $y e^{y/2} e^{-4\pi^2 t y^2}$ decreases pointwise in t , so $\rho(t)$ is decreasing by monotone convergence. The Gaussian factor forces $\rho(t) \rightarrow 0$ as $t \rightarrow \infty$. $\square$

Remark (Why uniform cap beats local bisection). Legacy per-compact bisection schedules are archived in Appendix G for provenance. The mainline selects a single scale $t = t_{\text{crit}}$ independent of K ; Corollary 9.22 then guarantees $\rho(t) \leq c_*/4$. This decouples the prime cap from local parameter tuning and keeps the bridge analytic.

Early/tail calculus (tables-free)

Lemma 10.19 (Early block). *For every $N \geq 2$,*

$$\sum_{n \leq N} \frac{\Lambda(n)}{\sqrt{n}} \leq \sum_{n \leq N} \frac{\log n}{\sqrt{n}} \leq 2\sqrt{N} \log N.$$

Proof. $\Lambda(n) \leq \log n$ is standard. For the integral bound,

$$\sum_{n \leq N} \frac{\log n}{\sqrt{n}} \leq \log N \sum_{n \leq N} \frac{1}{\sqrt{n}} \leq \log N \left(1 + \int_1^N \frac{dx}{\sqrt{x}}\right) = \log N (2\sqrt{N} - 1) \leq 2\sqrt{N} \log N.$$

$\square$

Lemma 10.20 (Log-Gaussian tail). *For every $t \geq \frac{1}{16\pi^2}$ and $N \geq 2$, set $N_0 := \max\{N, e^2\}$. Then*

$$\sum_{n > N} \frac{\Lambda(n)}{\sqrt{n}} e^{-4\pi^2 t (\log n)^2} \leq \int_{\log N_0}^{\infty} y e^{-4\pi^2 t y^2} dy = \frac{e^{-4\pi^2 t (\log N_0)^2}}{8\pi^2 t}.$$

Proof. As in Remark 10.6, for $t \geq \frac{1}{16\pi^2}$ the function $x \mapsto (\log x) x^{-1/2} e^{-4\pi^2 t (\log x)^2}$ is decreasing on $[e^2, \infty)$. Since $\Lambda(n) \leq \log n$, for $N \geq e^2$ we may bound the tail by the integral test. If $2 \leq N < e^2$, replace N by e^2 ; the right-hand side decreases with N_0 . Substitute $y = \log x$ to obtain the stated Gaussian tail bound. $\square$

Proposition 10.21 (Heat cap via early/tail split). *Define for $t > 0$ and $N \geq 2$*

$$\rho_{\text{heat}}(K; t, N) := 2 \sum_{\substack{\xi_n \in [-K, K] \\ n \leq N}} \frac{\Lambda(n)}{\sqrt{n}} e^{-4\pi^2 t (\log n)^2} + \underbrace{\sum_{\substack{\xi_n \in [-K, K] \\ n > N}} \frac{2\Lambda(n)}{\sqrt{n}} e^{-4\pi^2 t (\log n)^2}}_{\text{tail}}.$$

Assume $t \geq \frac{1}{16\pi^2}$ and set $N_0 := \max\{N, e^2\}$. Then $\|T_P^{\text{RKHS}}(t)\| \leq \rho_{\text{heat}}(K; t, N)$, and by Lemmas 10.19–10.20

$$\rho_{\text{heat}}(K; t, N) \leq 4\sqrt{N} \log N + \frac{e^{-4\pi^2 t (\log N_0)^2}}{4\pi^2 t}.$$

Analytic prime caps and the PCU theorem

Corollary 10.22 (Uniform prime cap at the analytic scale). *Let $t = t_{\text{crit}}$ as in Corollary 9.22. Then*

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq \frac{c_*}{4}.$$

Proof. Corollary 9.22 gives $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq c_*/4$ at $t = t_{\text{crit}}$. $\square$

Remark (No K-dependent parameters). The uniform prime cap uses a single scale $t = t_{\text{crit}}$ and the uniform floor $c_* > 0$. No K-dependent schedules appear in the argument.

Lemma 10.23 (RKHS–Weil Isometry). *Let $(\mathcal{X}, \mu)$ be a measure space and $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ a positive-definite kernel. Denote by $(\mathcal{H}_k, \langle \cdot, \cdot \rangle_{\mathcal{H}_k})$ its RKHS and by Φ the map that sends each kernel section $k_x := k(\cdot, x)$ to $\varphi_x \in \mathcal{W}$ via a fixed Weil representation. Then:*

1. *The map Φ is well-defined on the span of the kernel sections and preserves inner products: $\langle \Phi f, \Phi g \rangle_{\mathcal{W}} = \langle f, g \rangle_{\mathcal{H}_k}$.*
2. *Φ extends uniquely to an isometry from $\mathcal{H}_k$ into $\mathcal{W}$.*
3. *If $\{\varphi_x\}_{x \in \mathcal{X}}$ spans $\mathcal{W}$, then $\Phi(\mathcal{H}_k)$ is dense in $\mathcal{W}$.*

Lemma 10.24 (Closed-form upper bound for the prime trace). *For $t > 0$ one has*

$$\rho(t) \leq 2 \int_0^\infty y e^{y/2} e^{-4\pi^2 t y^2} dy. \quad (10.17)$$

With $a = 4\pi^2 t$ and $b = \frac{1}{2}$ this implies

$$\rho(t) \leq \frac{1}{4\pi^2 t} + \frac{\sqrt{\pi}}{2(4\pi^2 t)^{3/2}} \exp\left(\frac{1}{16\pi^2 t}\right). \quad (10.18)$$

In particular, at the fixed scale $t = t_{\text{crit}}$ this yields the cap $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$ for all compacts.

Proof. The display (10.17) is Lemma 10.26. Complete the square: $\int_0^\infty y e^{-ay^2+by} dy$ admits the identity $e^{\frac{b^2}{4a}} \frac{b\sqrt{\pi}}{4a^{3/2}} (1 + \text{erf}(\frac{b}{2\sqrt{a}})) + \frac{1}{2a}$. Using $1 + \text{erf}(x) \leq 2$ gives the upper bound (10.18). Plug $a = 4\pi^2 t$, $b = \frac{1}{2}$ and simplify. At $t = 1$, $\pi > 3$ implies $4\pi^2 > 36$ and hence $1/(4\pi^2) < 1/36$. Moreover

$$\begin{aligned} \frac{\sqrt{\pi}}{2(4\pi^2)^{3/2}} \exp\left(\frac{1}{16\pi^2}\right) &\leq \frac{\sqrt{\pi}}{2 \cdot 36^{3/2}} \exp\left(\frac{1}{144}\right) \leq \frac{2}{432} \cdot \exp\left(\frac{1}{144}\right) \\ &\leq \frac{2}{432} \left(1 + \frac{1}{144} + \frac{1}{20736}\right) < \frac{2}{432} \cdot \frac{1007}{1000} < \frac{1}{200}, \end{aligned}$$

so the total bound is $\frac{1}{36} + \frac{1}{200} = \frac{59}{1800} < \frac{72}{1800} = \frac{1}{25}$. $\square$

Lemma 10.25 (Unshifted trace cap). *Fix $K > 0$. For any $B > 0$ and $t > 0$, the symmetrized prime sampling operator satisfies*

$$\|T_P^{\text{RKHS}}(t)\|_{L^2 \rightarrow L^2} \leq \text{tr } T_P^{\text{RKHS}}(t) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-4\pi^2 t (\log n / (2\pi))^2} \leq \rho(t). \quad (10.19)$$

Proof. Start with $\|T_P^{\text{RKHS}}(t)\| \leq \text{tr } T_P^{\text{RKHS}}(t)$ (PSD, finite rank on compacts). Bound the sum by an integral of the positive integrand and apply the change $x = e^y$:

$$\int_1^\infty \frac{\log x}{\sqrt{x}} e^{-4\pi^2 t (\log x)^2} dx = \int_0^\infty y e^{y/2} e^{-4\pi^2 t y^2} dy. \quad (10.20)$$

Doubling for $\pm \xi_n$ yields the stated bound by (10.17). $\square$

10.6 Prime sampling norm bounded by $\rho(t)$

Throughout this subsection we write $\rho(t)$ for the Gaussian cap in Lemma 10.24.

Lemma 10.26 (Integral domination for the Gaussian-weighted prime sum). *Let $t > 0$ and write $t' := 4\pi^2 t$. Then*

$$\sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-t'(\log n)^2} \leq \int_1^\infty \frac{\log x}{\sqrt{x}} e^{-t'(\log x)^2} dx = \int_0^\infty y e^{y/2} e^{-t'y^2} dy. \quad (10.21)$$

Proof. Set

$$g(x) := \frac{1}{\sqrt{x}} e^{-t'(\log x)^2}, \quad h(x) := (\log x)g(x) = \frac{\log x}{\sqrt{x}} e^{-t'(\log x)^2}, \quad x > 1. \quad (10.22)$$

Differentiating g (using $u = \log x$, $du/dx = 1/x$) yields

$$g'(x) = -\frac{e^{-t'(\log x)^2}}{x^{3/2}} \left(\frac{1}{2} + 2t' \log x \right) < 0 \quad (x > 1, t' > 0), \quad (10.23)$$

so g is strictly decreasing on $[1, \infty)$. By the Chebyshev rearrangement principle (equivalently, by applying the integral test to the eventually decreasing function h ; see Remark 10.6 below) we have

$$\sum_{n \geq 2} \Lambda(n)g(n) \leq \sum_{n \geq 2} (\log n)g(n) \leq \int_1^\infty (\log x)g(x) dx, \quad (10.24)$$

because $\Lambda(n) \leq \log n$ for every n (indeed $\Lambda(p^m) = \log p \leq m \log p = \log(p^m)$). Substituting $x = e^y$ gives $dx = e^y dy$ and $x^{-1/2} e^y = e^{y/2}$, so the last integral equals $\int_0^\infty y e^{y/2} e^{-t'y^2} dy$, which is the claimed right-hand side of (10.21). $\square$

Remark (Eventual monotonicity of h). Writing $y = \log x$ and $h(x) = H(y)$ with $H(y) = y e^{-t'y^2+y/2}$, we compute $H'(y) = e^{-t'y^2+y/2} (1 - 2t'y^2 + \frac{1}{2}y)$. For $y \geq 2$ this derivative is nonpositive whenever $t' \geq \frac{1}{4}$, i.e. $t \geq t_\star := \frac{1}{16\pi^2}$. Therefore h decreases on $[e^2, \infty)$ in that regime, so the integral test gives $\sum_{n \geq [e^2]} h(n) \leq \int_{e^2}^\infty h(x) dx$; adding the finite block $2 \leq n < e^2$ yields (10.21) without further loss.

Proposition 10.27 (Norm bound for the symmetrized prime block). *Fix a compact interval $[-K, K]$. The even-symmetrized prime sampling operator $T_P^{\text{sym}}(t)$ on $[-K, K]$ is positive and of finite rank. Consequently,*

$$\|T_P^{\text{RKHS}}(t)\| = \|T_P^{\text{sym}}(t)\| \leq \text{Tr } T_P^{\text{sym}}(t) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-t'(\log n)^2} \leq \rho(t), \quad (10.25)$$

where the last inequality is Lemma 10.26.

Proof. Since $T_P^{\text{sym}}(t)$ is positive and of finite rank, its nonzero spectrum consists of finitely many eigenvalues $\lambda_j \geq 0$. The operator norm is $\|T_P^{\text{sym}}(t)\| = \max_j \lambda_j$ and the trace is $\text{Tr } T_P^{\text{sym}}(t) = \sum_j \lambda_j$, so $\|T_P^{\text{sym}}(t)\| \leq \text{Tr } T_P^{\text{sym}}(t)$. The final bound by $\rho(t)$ is Lemma 10.26. $\square$

Lemma 10.28 (Trace cap with explicit remainder via erfc). *Let $t > 0$, set $a := 4\pi^2 t$ and $b := \frac{1}{2}$, and introduce $\tilde{\mu} := \frac{1}{2a}$. For $z_0 \in \mathbb{R}$ define*

$$J_a(z_0) := e^{a\tilde{\mu}^2} \int_{z_0}^\infty z e^{-a(z-\tilde{\mu})^2} dz. \quad (10.26)$$

Then the even-symmetrized prime sampling operator on any compact $[-K, K]$ satisfies

$$\|T_P^{\text{RKHS}}(t)\| \leq 2 \sum_{2 \leq n \leq e^2} \frac{\log n}{\sqrt{n}} e^{-4\pi^2 t (\log n)^2} + 2 J_a(2) \leq 2 J_a(0). \quad (10.27)$$

Moreover J_a admits the closed form

$$J_a(z_0) = e^{a\tilde{\mu}^2} \left(\frac{\tilde{\mu}\sqrt{\pi}}{2\sqrt{a}} \operatorname{erfc}(\sqrt{a}(z_0 - \tilde{\mu})) + \frac{1}{2a} e^{-a(z_0 - \tilde{\mu})^2} \right). \quad (10.28)$$

Proof. Split the prime block into the finite range $2 \leq n \leq e^2$ and the tail $n > e^2$. For the tail consider

$$f(x) := \frac{\log x}{\sqrt{x}} e^{-a(\log x)^2 + b \log x} = h(\log x), \quad h(z) := z e^{-az^2 + bz}. \quad (10.29)$$

For $z \geq 2$ we compute $h'(z) = e^{-az^2 + bz} (1 - \frac{1}{2}z - 2az^2) \leq 0$ (for $a \geq \frac{1}{4}$), so f is nonincreasing on $[e^2, \infty)$. Therefore

$$\sum_{n > e^2} f(n) \leq \int_{e^2}^{\infty} f(x) dx. \quad (10.30)$$

Substituting $x = e^z$ transforms the integral into

$$\int_2^{\infty} z e^{-az^2 + (b + \frac{1}{2})z} dz = e^{a\tilde{\mu}^2} \int_2^{\infty} z e^{-a(z - \tilde{\mu})^2} dz, \quad (10.31)$$

because $-az^2 + (b + \frac{1}{2})z = -a(z - \tilde{\mu})^2 + a\tilde{\mu}^2$. Writing $z = \tilde{\mu} + u/\sqrt{a}$ (with $u = \sqrt{a}(z - \tilde{\mu})$) gives

$$J_a(2) = e^{a\tilde{\mu}^2} \left(\frac{\tilde{\mu}}{\sqrt{a}} \int_{u_0}^{\infty} e^{-u^2} du + \frac{1}{a} \int_{u_0}^{\infty} u e^{-u^2} du \right), \quad u_0 = \sqrt{a}(2 - \tilde{\mu}). \quad (10.32)$$

Evaluating the integrals via $\int_{u_0}^{\infty} e^{-u^2} du = \frac{\sqrt{\pi}}{2} \operatorname{erfc}(u_0)$ and $\int_{u_0}^{\infty} u e^{-u^2} du = \frac{1}{2} e^{-u_0^2}$ yields the closed form (10.28). Dropping the finite block enlarges the bound to $J_a(0)$, and positivity plus finite rank of $T_P^{\text{sym}}(t)$ supply the two displayed inequalities for $\|T_P^{\text{RKHS}}(t)\|$.

Finally, the integrand in the definition of J_a is nonnegative, so $z_0 \mapsto J_a(z_0)$ is decreasing, giving $J_a(2) \leq J_a(0)$ as claimed. $\square$

Notes.

- The choice $b = \frac{1}{2}$ exactly cancels the factor $e^{z/2}$ coming from $dx = e^z dz$ and $x^{-1/2}$, which is why the completing-the-square center is $\tilde{\mu} = \frac{1}{2a}$.
- If one prefers not to appeal to global monotonicity, the finite-block split at e^2 already isolates a region on which h is decreasing for every $a \geq \frac{1}{4}$ (equivalently $t \geq \frac{1}{16\pi^2}$), covering all parameter regimes used in the certificate.

Reproducibility. Legacy numerics for the optimisation parameter t and the resulting caps $\rho(t)$ are archived in Appendix E; they corroborate but do not enter the analytic bounds above.

10.6.1 Immediate corollaries used below

- From Proposition 10.27 we obtain the operator-norm cap $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) = 2 \int_0^\infty y e^{y/2} e^{-4\pi^2 t y^2} dy$ for every $t > 0$. Since $\rho(t)$ is decreasing (Lemma 10.18), any choice of t with $\rho(t) \leq c_*/4$ (in particular $t = t_{\text{crit}}$) yields the uniform prime cap in Corollary 9.22.
- Lemma 10.28 supplies the explicit finite-block plus tail bound $\|T_P^{\text{RKHS}}(t)\| \leq 2 \sum_{2 \leq n \leq e^2} \frac{\log n}{\sqrt{n}} e^{-4\pi^2 t (\log n)^2} + 2J_{4\pi^2 t}(2)$, where J_a is given by (10.28) in terms of elementary functions and erfc . This closed form is convenient both analytically (Gaussian tails) and numerically (stable evaluation).

11 Prime Cancellation (D3)

11.1 D3: Operator Bridge to $\|T_P^{\text{RKHS}}(t)\| \leq 1 - \delta_0$

The linear-algebraic bounds quoted here are standard consequences of Gershgorin and Rayleigh estimates [18, 32].

See also. D3 dispersion (Lemma 11.1), mixed bound (Theorem 9.35).

Remark (Legacy (not used in main proof)). This D3 route provides a compact-dependent contraction mechanism and is archived for reference only. The main proof chain uses the uniform A3 bridge and the uniform RKHS cap; no D3 assumptions are invoked.

Let $\mathcal{H}_K$ be the even RKHS on $[-K, K]$ with normalized kernels $\|k_\alpha\| = 1$, and set $T_P^{\text{RKHS}}(t) = \sum_{\alpha_n \in [-K, K]} w(n) |k_{\alpha_n}\rangle\langle k_{\alpha_n}|$ with $w(n) = \Lambda(n)/\sqrt{n}$.

Lemma 11.1 (Dispersion via A2/A3 data). *Assume the A3 hypotheses: $P_A \in \text{Lip}(1)$ with $\min P_A \geq c_0 > 0$ (Lemma 9.11 and Lemma 9.33), the uniform RKHS cap $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$ (Corollary 9.22), and the single-scale coupling of Lemma 9.32. Then at the fixed scale $t = t_{\text{crit}}$ there exists a sequence $\delta_A \rightarrow 0$ such that for every even RKHS test f supported in $[-K, K]$*

$$\left| \sum_{p \leq A} (f(p) - \mathbb{E}_{\mathcal{P} \cap [1, A]} f) \right| \leq C(K) (\omega_{P_A}(t) + \varepsilon_K(t)) =: C(K) \delta_A.$$

Consequently, $\delta_A \rightarrow 0$ as $A \rightarrow \infty$.

Proof. The Lipschitz control from Lemma 9.11 bounds the near-diagonal contribution by $\omega_{P_A}(t)$. The uniform RKHS cap (Corollary 9.22) together with Lemma 9.32 controls the RKHS tail by $\varepsilon_K(t)$. Adding the two estimates yields the desired inequality. $\square$

Theorem 11.2 (D3: Structural contraction). *If Lemma 11.1 provides a gain $\delta_* > 0$ after fixing the scales, then there exists $\delta_0 \in (0, \delta_*)$ with*

$$\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq 1 - \delta_0. \quad (11.1)$$

Moreover, there is a constant $C_{\text{D3}} > 0$ (the uniform remainder in the mixed Toeplitz bound with Lipschitz symbol P_A) such that for $M \gg K^3$,

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq (1 + \delta_0) \log(1+K) - C_{\text{D3}}. \quad (11.2)$$

Proof. In the packet basis the matrix of $T_P^{\text{RKHS}}(t)$ is $A := W^{1/2}GW^{1/2}$. The dispersion bound in Lemma 11.1 gives a uniform gain $\delta_* > 0$ after fixing the scales, so for every unit vector u one has $\langle Au, u \rangle \leq 1 - \delta_*$. Hence $\|T_P^{\text{RKHS}}(t)\| = \|A\| \leq 1 - \delta_*$; choose any $\delta_0 \in (0, \delta_*)$ to obtain (11.1). Inserting this norm bound into the mixed Toeplitz estimate with Lipschitz symbol P_A gives (11.2); the constant C_{D3} is the uniform remainder from that bound. $\square$

Corollary 11.3 (Amplitude closure). *With the auxiliary suppressors (Roads B/C) and Theorem 11.2 we obtain $\Gamma(K) \geq (1 + \delta_0) \log(1 + K) - C_{\text{D3}}$, closing the amplitude gate.*

11.2 D3: Structural PC(K) Theorem

See also. D3 dispersion (Lemma 11.1), operator bridge (§11.1).

Remark (Legacy (not used in main proof)). This D3 structural route is archived for reference only. The main proof chain uses the uniform A3 bridge and uniform RKHS cap, and does not invoke D3.

Definition 11.4 (Working space). Let $K > 0$. Denote by P_A the Archimedean symbol after the A3 smoothing, by $T_M[P_A]$ its Toeplitz truncation, and by $T_P^{\text{RKHS}}(t)$ the even prime operator on $[-K, K]$ at the fixed scale $t = t_{\text{crit}}$.

Definition 11.5 (Criteria AC–D3). We say that AC–D3.1 holds if: (i) $P_A \in \text{Lip}(1)$ and $\min P_A \geq c_0 > 0$; (ii) $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$ at $t = t_{\text{crit}}$; (iii) the single-scale coupling of Lemma 9.32 is in force. Condition AC–D3.2 demands a sequence $\delta_A \rightarrow 0$ with

$$\text{Disp}_K(\cdot)A \leq C(K) \delta_A.$$

Theorem 11.6 (Structural prime cancellation). *Under A2 and A3 the criteria AC–D3.1 hold. Furthermore AC–D3.1 $\Rightarrow$ AC–D3.2 with $\delta_A \rightarrow 0$, hence*

$$\text{Disp}_K(\cdot)A \leq C(K) \delta_A \xrightarrow{A \rightarrow \infty} 0.$$

Proof. A3 (Lemma 9.11, Lemma 9.33, Lemma 9.34, Lemma 9.32) yields (i)–(iii); Corollary 9.22 fixes the cap $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t)$. Lemma 11.1 then provides the dispersion bound. $\square$

Corollary 11.7 (D3-lock). *Under Theorem 11.6, for any normalized RKHS test f ,*

$$\left| \sum_{p \leq A} \left(f(p) - \mathbb{E}_{\mathcal{P} \cap [1, A]} f \right) \right| \leq C(K) \delta_A \xrightarrow{A \rightarrow \infty} 0.$$

Amplitude closure without D3

Proposition 11.8 (AB(K) supplied by A3). *Lemmas 9.19, 9.11, 9.34, and 9.32 ensure the AB(K) conditions with constants depending only on $(K, c_*, \rho(t))$.*

Proof. The Lipschitz floor $\min P_A \geq c_*$ gives (i), while the trace-cap and the two-scale parameters yield (ii) and (iii). $\square$

Theorem 11.9 (Amplitude gate without explicit D3 assumptions). *Under A2/A3, Proposition 11.8 and Corollary 9.31 imply*

$$\langle (T_M[P_A] - T_P^{\text{Ray}}(t, M))f, f \rangle \geq \left(\frac{c_*}{2} - \rho(t) \right) \|f\|_2^2$$

for every f supported in $[-K, K]$. In particular, if $\rho(t) < c_*/2$ the mixed lower bound is positive; density and continuity then yield $Q^*(t; \Phi) \geq 0$ on the Weil class and by Weil's positivity criterion, RH would hold..

Proof. Insert the AB(K) bounds into Theorem 9.35 and use Corollary 9.31 to control the prime term. $\square$

12 Weil Criterion Linkage and Main Theorem

12.1 Weil linkage: positivity implies the Riemann Hypothesis

Theorem 12.1 (Weil's positivity criterion, normalized). *Let $Q^*(t; \cdot)$ be the Weil functional attached to $\zeta(s)$ in the normalization of Section 6, and let $\mathcal{W}$ be the Weil cone described in Section 4. Then the following are equivalent:*

- (i) *The Riemann Hypothesis holds.*
- (ii) *$Q^*(t; \Phi) \geq 0$ for every $\Phi \in \mathcal{W}$.*

Theorem 12.2 (Riemann Hypothesis). *If (T0)+(A1')+(A2)+(A3)+(RKHS) hold, then the Riemann Hypothesis is true.*

Proof. By Theorem 12.4 we have $Q^*(t; \Phi) \geq 0$ on the Weil cone $\mathcal{W}$ in the normalization of Section 6. Applying Theorem 12.1 yields the claim. $\square$

Remark (On normalization and scope). The normalization in (T0) matches the Guinand–Weil conventions; thus Theorem 12.1 applies verbatim. No numerical tables or ATP artifacts are used anywhere in the proof of Theorem 12.2.

Remark (Dependency map). The sufficiency argument uses the following chain:

$$(T0) \implies (A1') \xRightarrow{\text{dens.}} (A2) \xRightarrow{\text{cap}} \text{RKHS} \xRightarrow{\text{bridge}} (A3) \implies Q^*(t; \Phi) \geq 0 \implies \text{RH}.$$

Refer to Theorem 6.2 for (T0), Theorem 7.3 for (A1'), Lemma 8.3 for (A2), Lemmas 10.23 and 10.4 for the RKHS/Weil transfer, and Theorem 9.35 for the uniform bridge. Every arrow is justified in the proof of Theorem 12.3.

Theorem 12.3 (Weil sufficiency pack). *Assume the hypotheses of Theorem 12.4, namely (T0), density (A1') on each compact $[-K, K]$ (Theorem 7.3), continuity (A2) (Lemma 8.3), the mixed bridge (A3) (Theorem 9.35) with uniform margin $c_* > 0$, and prime control via the uniform RKHS cap (Corollary 9.22). Then $Q^*(t; \Phi) \geq 0$ for all $\Phi \in \mathcal{W}$, and hence the Riemann Hypothesis follows from Weil's positivity criterion.*

Proof. By Lemma 10.23 the RKHS and Weil pictures are isometric on the working subspace. Together with Lemmas 10.4 and 10.4 we transfer the mixed lower bound of Theorem 9.35 to the quadratic functional $Q^*(t; \cdot)$, while Corollary 9.7 and the uniform prime cap ensure the required margin on each compact window W_K . Density (Theorem 7.3) and continuity (Lemma 8.3) upgrade positivity from the Fejér×heat cone to all of W_K , and taking the union over K gives $Q^*(t; \Phi) \geq 0$ on the Weil cone $\mathcal{W}$. Weil's criterion then yields the stated implication. $\square$

12.2 Main closure: from analytic modules to Weil positivity

Standing hypotheses (analytic chain)

Throughout this section we rely only on the following analytic inputs:

- **(T0) Normalization.** Guinand–Weil crosswalk and our conventions, cf. Proposition 6.1 (Section 6).
- **(A1') Density.** The Fejér×heat cone is dense in W_K , cf. Theorem 7.3.
- **(A2) Continuity.** The Weil functional $Q^*(t; \cdot)$ is continuous on W_K with a modulus $L_Q(K)$ (Section 8).
- **(A3) Uniform Toeplitz bridge.** Theorem 9.35 provides: for all $M \geq M_0^{\text{unif}}$,

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \frac{c_*}{4} > 0,$$

where c_* is the uniform Archimedean floor from Lemma 9.19.

Theorem 12.4 (Main positivity on $\mathcal{W}$). *Assume (T0), (A1'), (A2), and the uniform A3 bridge inequality (Theorem 9.35). Then*

$$Q^*(t; \Phi) \geq 0 \quad \text{for every even, real, compactly supported } \Phi \in \mathcal{W},$$

where $\mathcal{W} = \bigcup_{K>0} \mathcal{W}_K$ is the Weil cone from Section 4.

Proof. Fix $K > 0$. By Theorem 9.35 we have $Q^*(t; \Phi) \geq 0$ on the Fejér×heat generator cone inside $\mathcal{W}_K$. By (A1') density and (A2) continuity, this extends to all of $\mathcal{W}_K$. Since $\mathcal{W} = \bigcup_{K>0} \mathcal{W}_K$, we obtain $Q^*(t; \Phi) \geq 0$ on $\mathcal{W}$. This completes the proof of $Q^*(t; \Phi) \geq 0$ on the Weil cone; the RH follows in Section 12.1. $\square$

Remark (Uniform input only). The proof of Theorem 12.4 uses only the uniform analytic bounds built into Theorem 9.35. No K -dependent schedules or numerical certificates enter the main argument.

A Notation

We collect the notation used throughout.

Sets and measures. $A \cap B$, $A \cup B$, $A \setminus B$ are standard. $\mathbf{1}_E$ denotes the indicator of a set E . The symbol $|E|$ records measure/length in the relevant context.

Norms. $\|x\|_2$ is the Euclidean norm, $\|f\|_{L^2(\Omega)}^2 = \int_{\Omega} |f|^2$. For sequences $\|a\|_{\ell^2}^2 = \sum_k |a_k|^2$.

Operators. $\langle u, v \rangle$ is the inner product, A^* the adjoint, $\text{tr}(M)$ the trace, $\|T\|_{\text{op}}$ the operator norm.

Parameterized operator contract. We keep heat scale t explicit, with $t_0 = 1/(16\pi^2 t)$, and write the Fejér×heat windows $\Phi_{B,t}$ (with $\tau = 0$). The Weil functional is denoted $Q^*(t; \Phi)$, the Toeplitz block is $T_M[P_A](t, B, M)$ on $\mathbb{C}^{2M+1}$, the Rayleigh prime block is $T_P^{\text{Ray}}(t, M)$ on $\mathbb{C}^{2M+1}$, and the RKHS prime operator is $T_P^{\text{RKHS}}(t)$ on $\mathcal{H}_t$ with cap $\rho(t)$. These fully parameterized names are the authoritative contract; bare T_P , Q , or ρ are avoided.

Comparisons. $r \lesssim s$ means $r \leq Cs$ with an absolute constant C independent of the current parameters; $r \simeq s$ abbreviates $r \lesssim s$ and $s \lesssim r$ simultaneously.

Critical constants. $c_* = \frac{11}{10}$ is the **uniform** archimedean floor (Lemma 9.19), valid for all $K \geq 1$ on the full circle $\mathbb{T}$ (parameters $t = 3/20 = 0.15$, $B_{\min} = 3$). **Note:** The legacy value $\frac{1346209}{7168000}$ from Theorem 8.16 is obsolete (contains a scale error); the main proof uses c_* exclusively. The uniform RKHS prime cap is $\rho(t) \leq c_*/4$ for $t \geq t_{\star, \text{rkhs}}^{\text{unif}} = t_{\text{crit}} = 3/20 = 0.15$ (Corollary 9.22).

Fejér×heat atoms. The mainline proof fixes the shift parameter to $\tau = 0$ in the generator $\Phi_{B,t}$; any τ -shifted atoms are confined to the legacy appendix.

Uniform constants (mainline).

Constant	Value	Source
t	$3/20 = 0.15$	Lemma 9.19
$B_{\min}$	3	Lemma 9.19
c_*	$\frac{11}{10}$	Lemma 9.19
M_0^{unif}	$\lceil C_{\text{SB}} L_*(t)/c_* \rceil$	Corollary 9.21
$t_{\star, \text{rkhs}}^{\text{unif}}$	$t_{\text{crit}} = 3/20 = 0.15$	Corollary 9.22

B Clarifications

Remark (Nodes are not dense on compacts). On $[-K, K]$ the active set $\{\alpha_n = \frac{\log n}{2\pi}\}$ is finite: $n \leq N(K) = \lfloor e^{2\pi K} \rfloor$. The minimal gap satisfies

$$\delta_K = \min_{1 \leq n < N(K)} (\alpha_{n+1} - \alpha_n) = \frac{1}{2\pi} \min_{1 \leq n < N(K)} \log\left(1 + \frac{1}{n}\right) \geq \frac{1}{2\pi(N(K) + 1)} > 0.$$

Remark (Weight upper bound). For $w(n) = \Lambda(n)/\sqrt{n}$ we have $w(n) \leq \log n/\sqrt{n} \leq 2/e < 3/4 < 1$. Thus $w_{\max} < 1$ on every compact (numerically, $2/e \approx 0.7358$).

Remark (Finite Gram matrices). The Gram matrix G of $\{\alpha_n\}$ on $[-K, K]$ is finite dimensional and satisfies $\|T_P^{\text{RKHS}}(t)\| = \|W^{1/2}GW^{1/2}\|$.

Remark (Existence of $t_{\min}$). As $t \downarrow 0$, $S_K(t) = \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}} \downarrow 0$. Hence for any $\eta_K > 0$ there exists

$$t_{\min}(K) = \frac{\delta_K^2}{4 \ln((2 + \eta_K)/\eta_K)} \quad \text{with} \quad S_K(t_{\min}) \leq \eta_K.$$

Remark (Dictionary density). We assert ε -density of the cone $\mathcal{C}_K$ by a finite dictionary $\mathcal{G}_K$ at fixed K , not global density by a fixed finite set; cf. Theorem A1' and the T5 transfer.

Remark (Activity intervals). Setting $I_n = [B_n, B_{n+1})$ with $B_n = \frac{\log n}{2\pi}$, crossing $I_n \rightarrow I_{n+1}$ introduces the single new node α_{n+1} used in the one-prime induction.

Remark (Weil topology). Write $\mathcal{W} = \bigcup_K \mathcal{W}_K$ with the inductive-limit topology. Since $Q^*(t; \cdot)$ is continuous on each $\mathcal{W}_K$ (Lemma 8.3), it is continuous on $\mathcal{W}$; see Theorem G.23.

Remark (Link to zeta zeros). The connection to zeros of the Riemann zeta function is handled in Section 12 via the classical Weil criterion.

Remark (Example at $K = 1$). Taking $N(1) = \lfloor e^{2\pi} \rfloor$, one has $\delta_1 \geq 1/(2\pi(N(1) + 1))$. Choosing $t_{\min}(1)$ from the formula above with a concrete $\eta_1 \in (0, 1)$ yields $S_1(t_{\min})$ and ensures $\rho_1 = w_{\max} + \sqrt{w_{\max}}S_1(t_{\min}) < 1$. PSD of the small dictionary $\mathcal{G}_1$ can be checked for $M \in \{10, 20, 40\}$ directly.

Remark (Role of the Fejér factor). The Fejér factor localizes to compacts and contributes to the BV/Lipschitz regularity of the symbol; the heat factor provides smoothing and Gaussian-in-log tails. Their product preserves positivity and supplies the regularity required for A3 and the RKHS bounds.

Remark (What we do not assume). We do not model the problem via a selfadjoint operator with pure point spectrum on a Paley–Wiener space; on the Fourier side, multiplication by ξ has absolutely continuous spectrum. We do not use rigged eigenfunctions such as $e^{i\gamma\tau}$ as elements of the Hilbert space. We do not infer Weyl asymptotics from heat traces, and we do not impose determinant identities equivalent to RH.

Remark (Proof skeleton). The proof skeleton is Toeplitz + RKHS + Weil: (i) A3 handles the Archimedean symbol $P_A \in \text{Lip}(1)$ and keeps primes as a finite-rank operator; (ii) RKHS yields a strict contraction on each compact $[-K, K]$; (iii) T5 transfers positivity to the inductive limit; (iv) the Weil criterion concludes RH.

C Verification Notes

D Verification Appendix

Verification status: Conceptual components prepared for independent expert review; no numerical premise enters the logic. The checklist below records only analytic sources used in the main chain.

- **T0 (Normalization).** Guinand–Weil translation and the definitions of a , a_* , and prime weights.
- **A1' (Local density).** Mollification, positive Fejér Riemann sums, and symmetrisation on $[-K, K]$.
- **A2 (Continuity and tails).** Explicit $L_Q(K)$ and Gaussian tail control.
- **A3 (Toeplitz bridge).** SB barrier, Rayleigh identification, and the uniform Arch floor.
- **RKHS + Weil linkage.** Uniform prime cap and the Weil criterion linkage.

Elementary inequalities used in A3. The A3 floor proof (Lemma 9.19) uses only fixed constant bounds and elementary inequalities. For audit convenience we record the ones invoked explicitly in Section 9.4:

- $\pi < \frac{22}{7}$ and $\pi > 3$ (classical rational bounds).
- $\pi > \frac{333}{106}$ (classical lower bound; used for the exponentials in Lemma 9.19).
- $\log(3/2) < \frac{5}{12}$ from the alternating series $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ at $x = \frac{1}{2}$.
- $\log(5/2) < 1$ since $e > 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} = \frac{65}{24} > 2.5$.

- $e < 3$ from the series $e = \sum_{k \geq 0} \frac{1}{k!}$ and a tail bound. Using the Taylor remainder at $x = \frac{3}{5}$ gives $e^{3/5} \leq \sum_{j=0}^4 \frac{(3/5)^j}{j!} + \frac{3}{120} \left(\frac{3}{5}\right)^5 < \frac{50}{27}$, hence $e^{-3/5} > \frac{27}{50}$.
- With $x_0 = \frac{665334}{280900}$ one has $e^{x_0} \geq \sum_{j=0}^5 \frac{x_0^j}{j!} > 10$, which implies $e^{6\pi^2/25} > 10$, so $w_{B_{\min}}(1) \leq \frac{1}{15}$ and $w_{B_{\min}}(2) \leq \frac{1}{30000}$ in Lemma 9.19.
- For $0 \leq x \leq 1$, $e^x \leq 1 + x + x^2$ (apply $f(x) = 1 + x + x^2 - e^x$ with $f(0) = f'(0) = 0$ and $f'(1) = 3 - e > 0$). Hence $e^{1/144} \leq 1 + \frac{1}{144} + \frac{1}{20736} < \frac{1007}{1000}$.
- $\gamma > 0.5772$ (Euler's constant lower bound; see standard tables in [31] or [22]).

Reproducibility artifacts are archived in `docs/VERIFICATION_PACK.md` and Appendix G for reference only. The expanded legacy checklist is preserved in `appendix/verification_legacy.tex`. None of these artifacts enter the logical proof: all analytic estimates are contained in the main text and use only the constants listed above.

E Reproducibility Data for A3, RKHS, and IND/AB

The tables in this appendix reproduce the legacy certificate outputs used in the Toeplitz bridge (A3) and in the RKHS trace caps. They are not part of the analytic proof and serve only as provenance for the archived JSON logs under `cert/bridge/`.

Reproducibility archive only – not used in the proof of Theorem 12.4.

A3 lock parameters

Legacy only (not used in main proof). Arch parameters recorded by the bridge locks (`release/RH_trace_only_release/cert/bridge`) in the old 2π scale.

K	B	t	$c_0(K)$	M_{lock}	$\omega_{P_A}(\pi/M)$ (legacy)
1	0.300	0.030000	0.898623847	1	0.082383510
2	0.300	0.013333	0.902866849	1	0.083965291
3	0.300	0.007500	0.904368197	1	0.084529648
4	0.300	0.004800	0.905066004	1	0.084792781
6	0.300	0.002449	0.905675120	1	0.085022900
8	0.300	0.001481	0.905926192	1	0.085117870
10	0.300	0.000992	0.906053375	1	0.085166003
12	0.300	0.000710	0.906126551	1	0.085193706
16	0.300	0.000415	0.906203168	1	0.085222716
20	0.300	0.000272	0.906240367	1	0.085236804
24	0.300	0.000192	0.906261191	1	0.085244691
28	0.300	0.000143	0.906274010	1	0.085249546
32	0.300	0.000110	0.906282458	1	0.085252746

Source: `release/RH_trace_only_release/cert/bridge/K1_A3_lock.json`, `K2_A3_lock.json`, ..., `K32_A3_lock.json`. Numerical values are reported verbatim from the `c0`, `t_sym`, `M0`, and `omega_pi_over_M` fields (legacy 2π normalization). The mainline uses $\omega_{P_A}(1/(2M))$ and the uniform floor c_* .

Reproducibility only – analytic bounds in the main text use the uniform floor $c_* > 0$ (Lemma 9.19) and the RKHS cap from Corollary 9.22.

Note: The legacy symbolic floor $\frac{1}{7} \frac{346}{168} \frac{209}{000}$ from Theorem 8.16 is obsolete; see Lemma 9.19 for the correct uniform bound.

Module	Legacy artefact (read-only)	Primary cite	Secondary cite
A3	cert/bridge/K*_A3_lock.json; proofs/A3_global/logs/ a3_global_lock_vampire.log	Theorem 9.35	Section G.14
RKHS	cert/bridge/K*_trace.json; cert/bridge/ yes_gate_chain_report_trace. json	Lemma 10.18	Corollary 10.22
IND/AB	cert/bridge/K1_blocks.json; cert/bridge/K1_step_next.json; proofs/ABK_aggregation/tptp/ ab_lin_k11.p	Appendix E	Theorem G.11
T5	cert/bridge/K*_grid.json; proofs/T5_global_transfer/ tptp/t5_{\{compact,grid\}}.p	Theorem G.21	appendix/ T5_parameters. tex

Prime trace caps (legacy)

Historical trace-mode reports; analytic arguments use the uniform cap from Corollary 9.22.

Legacy trace-mode caps recorded in release/RH_trace_only_release/cert/bridge (not used in the analytic bound).

K	t	$\rho(t)$	Mode
1	0.137100	0.224656	target
2	0.137100	0.224656	target
3	0.137100	0.224656	target
4	0.137100	0.224656	target
6	0.137100	0.224656	target
8	0.137100	0.224656	target
10	0.137100	0.224656	target
12	0.137100	0.224656	target
16	0.137100	0.224656	target
20	0.137100	0.224656	target
24	0.137100	0.224656	target
28	0.137100	0.224656	target
32	0.137100	0.224656	target

Source: release/RH_trace_only_release/cert/bridge/K1_trace.json, K2_trace.json, ..., K32_trace.json. Each entry reproduces the fields `t`, `rho`, and `mode`.

Reproducibility only – the values above exceed the analytic cap $\rho(t) \leq c_*/4$ from Section 10.5 (Corollary 10.22) and are kept solely as historical trace-mode logs.

IND/AB schedule

Table 1: Monotone IND/AB schedule extracted from legacy certificates.

K	B	t	$c_0(K)$	$\varepsilon(K) = c_0/4$	$\rho(t)$	$c_0 - \rho$	Block mass	Residual
1	0.3	0.03	0.898624	0.224656	0.224656	0.673968	0.181352	0.005220
2	0.3	0.013333	0.902867	0.225717	0.225717	0.677150	—	—
3	0.3	0.007500	0.904368	0.226092	0.226092	0.678276	—	—
4	0.3	0.004800	0.905066	0.226267	0.226267	0.678800	—	—
6	0.3	0.002449	0.905675	0.226419	0.226419	0.679256	—	—
8	0.3	0.001481	0.905926	0.226482	0.226482	0.679445	—	—
10	0.3	0.000992	0.906053	0.226513	0.226513	0.679540	—	—
12	0.3	0.000710	0.906127	0.226532	0.226532	0.679595	—	—
16	0.3	0.000415	0.906203	0.226551	0.226551	0.679652	—	—
20	0.3	0.000272	0.906240	0.226560	0.226560	0.679680	—	—
24	0.3	0.000192	0.906261	0.226565	0.226565	0.679696	—	—
28	0.3	0.000143	0.906274	0.226569	0.226569	0.679706	—	—
32	0.3	0.000110	0.906282	0.226571	0.226571	0.679712	—	—

Source: `cert/bridge/K{K}_A3_lock.json` for c_0 , B , t ; `cert/bridge/K{K}_trace.json` (mode `target`) for $\rho(t)$. Block mass and residual for $K = 1$ come from `cert/bridge/K1_blocks.json` with log `cert/bridge/logs/K1_blocks.txt`; subsequent entries admit the same budgeting without explicit blocks. These numbers document reproducibility only and play no role in the analytic proof.

Reproducibility only – archived IND/AB schedule.

F Digamma Bounds: Analytic Framework and Optional Numerics

This appendix records the analytic framework behind the bounds for the uniform Archimedean floor, and separates any numerical checks as non-normative evidence. The main proof relies only on the analytic lemmas in Section 9.4.

F.1 Definitions

The Archimedean density is defined as

$$a(\xi) := \log \pi - \Re \psi\left(\frac{1}{4} + i\pi\xi\right),$$

where $\psi(z) = \Gamma'(z)/\Gamma(z)$ is the digamma function.

For $B > 0$ and $t > 0$, define

$$A_0(B, t) := 2\pi \int_{-B}^B a(\xi) \left(1 - \frac{|\xi|}{B}\right) e^{-4\pi^2 t \xi^2} d\xi, \quad (\text{F.1})$$

$$g_{B,t}(\xi) := a(\xi) \left(1 - \frac{|\xi|}{B}\right) e^{-4\pi^2 t \xi^2}, \quad (\text{F.2})$$

$$L_A(B, t) := 2\pi \sup_{\theta \in [-\frac{1}{2}, \frac{1}{2}]} \sum_{m \in \mathbb{Z}} |g'_{B,t}(\theta + m)|. \quad (\text{F.3})$$

With $B_{\min} = 3$ and $t = 3/20 = 0.15$, the uniform Lipschitz constant is

$$L_*(t) := \sup_{B \geq B_{\min}} L_A(B, t)$$

as in Definition 9.20. The uniform Archimedean floor $c_* > 0$ is supplied directly by Lemma 9.19.

F.2 Digamma Properties

The digamma function at $z = 1/4$ satisfies the reflection formula [31, Ch. 2]:

$$\psi\left(\frac{1}{4}\right) = -\gamma - \frac{\pi}{2} - 3 \ln 2,$$

where γ is Euler's constant. This gives:

$$a(0) = \log \pi + \gamma + \frac{\pi}{2} + 3 \ln 2.$$

For $\xi \neq 0$, the real part of $\psi(1/4 + i\pi\xi)$ can be computed via the series [31, Ch. 2]:

$$\Re\psi\left(\frac{1}{4} + i\pi\xi\right) = -\gamma + \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{n+1/4}{(n+1/4)^2 + \pi^2\xi^2} \right).$$

F.3 Analytic bound framework

Section 9.4 introduces analytic bounds for $L_*(t)$ and records the direct pointwise floor c_* (Lemma 9.19). The mean-modulus estimates for A_0 and L_A remain as auxiliary bounds but are not used to define c_* in the main proof chain.

F.4 Finite-sum bounds at sample points

For $y = \pi\xi$ set

$$t_n(y) := \frac{1}{n+1} - \frac{n + \frac{1}{4}}{(n + \frac{1}{4})^2 + y^2}, \quad \Re\psi\left(\frac{1}{4} + iy\right) = -\gamma + \sum_{n \geq 0} t_n(y).$$

The summand satisfies $t_n(y) \leq 0$ once $n \geq \lceil \frac{4}{3}y^2 - \frac{1}{4} \rceil$, so for $\xi = \frac{1}{2}$ we have $t_n(\pi/2) \leq 0$ for $n \geq 4$ and therefore

$$\Re\psi\left(\frac{1}{4} + \frac{i\pi}{2}\right) \leq -\gamma + \sum_{n=0}^3 t_n\left(\frac{\pi}{2}\right).$$

Using $\pi^2 < 10$ yields the explicit bounds

$$t_0 \leq \frac{37}{41}, \quad t_1 \leq \frac{5}{26}, \quad t_2 \leq \frac{13}{363}, \quad t_3 \leq \frac{1}{836},$$

hence $\sum_{n=0}^3 t_n(\pi/2) < 1.132$. With $\pi > 333/106$ and $\gamma > 0.5772$ this gives $a(\frac{1}{2}) = \log \pi + \gamma - \sum t_n(\pi/2) > \frac{29}{50}$, as recorded in Lemma 9.18.

For $\xi \geq 1$ the standard digamma remainder bound $|\psi(z) - \log z + \frac{1}{2z}| \leq \frac{1}{12|z|^2}$ ([22, §5.11]) implies

$$a(\xi) \geq -\log \xi - \frac{1}{2\pi\xi} - \frac{1}{12\pi^2\xi^2}.$$

Plugging $\xi = \frac{3}{2}$ and $\xi = \frac{5}{2}$ and using $\pi > 3$ together with the elementary bounds $\log(3/2) < \frac{5}{12}$ and $\log(5/2) < 1$ gives

$$a\left(\frac{3}{2}\right) \geq -\frac{5}{12} - \frac{1}{9} - \frac{1}{243} > -\frac{3}{5},$$

and

$$a\left(\frac{5}{2}\right) \geq -1 - \frac{1}{15} - \frac{1}{675} > -\frac{11}{10},$$

matching Lemma 9.18.

F.5 Optional numerical evidence (not used in the proof)

Exploratory numerical checks (high-precision quadrature and grid evaluation of the periodized derivative sum) are implemented in `scripts/digamma_bounds.py`. These computations are *not* part of the logical proof; they merely provide sanity checks on the scale of $A_0(B, t)$ and $L_A(B, t)$.

G Legacy K-dependent branch (not used in the proof)

G.1 Local density with shifted atoms (legacy/appendix)

This subsection records the shifted-atom proof that yields density on $[-R, R]$. It is kept for archival purposes; the mainline proof fixes $\tau = 0$ in the generator definition.

Theorem G.1 (A1' density with shifts (legacy)). *Let $K = [-R, R]$ with $R > 0$. For $B > 0$, $t > 0$, $\tau \in [-R, R]$ define the normalized even nonnegative frequency windows*

$$\Phi_{B,t,\tau}(\xi) := \frac{1}{2} [\Lambda_B(\xi - \tau) \rho_t(\xi - \tau) + \Lambda_B(\xi + \tau) \rho_t(\xi + \tau)],$$

where $\Lambda_B(x) = (1 - |x|/B)_+$ and $\rho_t(x) = (4\pi t)^{-1/2} e^{-x^2/(4t)}$ (so $\int_{\mathbb{R}} \rho_t = 1$, $\rho_t \geq 0$). The factor $\frac{1}{2}$ ensures that the symmetrization operator $\text{Sym}(u) := (u(\cdot) + u(-\cdot))/2$ fixes even functions; scaling generators by a positive constant does not change the generated cone. Let $\mathcal{C}$ be the closed convex cone generated by finite nonnegative combinations of $\{\Phi_{B,t,\tau}\}$ with $\tau \in [-R, R]$ and B sufficiently large (depending on R). Then $\mathcal{C}$ is dense in $C_{\text{even}}^+([-R, R])$ in the uniform norm.

Proof. Fix $f \in C_{\text{even}}^+([-R, R])$ and $\varepsilon > 0$. Extend f by zero to a compactly supported $\tilde{f} \in C_c(\mathbb{R})$ with $\tilde{f} = f$ on $[-R, R]$.

Step 1 (mollification). Since ρ_t is a positive approximate identity, there exists $t \in (0, t_0]$ such that

$$\sup_{|\xi| \leq R} |(\tilde{f} * \rho_t)(\xi) - f(\xi)| < \varepsilon/3. \quad (\text{G.1})$$

Set $g := \tilde{f} * \rho_t$. Then $g \geq 0$, $g \in C^\infty(\mathbb{R})$ and g is even.

Step 2 (positive Riemann sums). Choose a uniform partition $-R = \tau_0 < \tau_1 < \dots < \tau_N = R$ with mesh Δ small enough so that

$$g_R(\xi) := \sum_{j=0}^{N-1} g(\tau_j^*) \rho_t(\xi - \tau_j^*) (\tau_{j+1} - \tau_j) \quad (\text{G.2})$$

satisfies $\sup_{|\xi| \leq R} |g_R(\xi) - g(\xi)| < \varepsilon/3$ for some choices $\tau_j^* \in [\tau_j, \tau_{j+1}]$. Because the coefficients $g(\tau_j^*)(\tau_{j+1} - \tau_j)$ are nonnegative, g_R is a finite nonnegative combination of translates of ρ_t .

Step 3 (Fejér truncation). For any $|\xi|, |\tau| \leq R$ one has $|\Lambda_B(\xi - \tau) - 1| \leq (|\xi| + |\tau|)/B \leq 2R/B$. Choosing $B \geq B_0 := 6R/\varepsilon$ ensures

$$\sup_{|\xi| \leq R, |\tau| \leq R} |\Lambda_B(\xi - \tau) - 1| < \varepsilon/3. \quad (\text{G.3})$$

Define the symmetric Fejér×heat mixture as a sum of normalized atoms:

$$h(\xi) := \sum_{j=0}^{N-1} g(\tau_j^*) (\tau_{j+1} - \tau_j) \Phi_{B,t,\tau_j^*}(\xi). \quad (\text{G.4})$$

Since $g_R^{\text{sym}}(\xi) := (g_R(\xi) + g_R(-\xi))/2$ is the symmetrized Riemann sum, and by (G.3) the Fejér factor satisfies $|\Lambda_B(\xi \pm \tau) - 1| < \varepsilon/3$, we have for $|\xi| \leq R$:

$$|h(\xi) - g_R^{\text{sym}}(\xi)| \leq (\varepsilon/3) \sum_j g(\tau_j^*)(\tau_{j+1} - \tau_j) \leq C_f \varepsilon/3, \quad (\text{G.5})$$

with $C_f = \int_{-R}^R g(\tau) d\tau$ finite. Rescaling ε by $3 \max(1, C_f)$ if necessary, we get

$$\sup_{|\xi| \leq R} |h(\xi) - g_R^{\text{sym}}(\xi)| < \varepsilon/3. \quad (\text{G.6})$$

Step 4 (collect errors via triangle inequality). Since g is even, g_R^{sym} satisfies $|g_R^{\text{sym}}(\xi) - g(\xi)| < \varepsilon/3$ (by the Riemann sum bound plus symmetry contraction). Combined with Step 1 and Step 3:

$$\sup_{|\xi| \leq R} |h(\xi) - f(\xi)| \leq \sup |h - g_R^{\text{sym}}| + \sup |g_R^{\text{sym}} - g| + \sup |g - f| < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \quad (\text{G.7})$$

By construction, h is a finite nonnegative combination of Φ_{B,t,τ_j^*} with $\tau_j^* \in [-R, R]$, hence $h \in \text{cone}\{\Phi_{B,t,\tau}\}$. Taking closures in $\|\cdot\|_\infty$ yields density of $\mathcal{C}$ in $C_{\text{even}}^+([-R, R])$. $\square$

Lemma G.2 (Fixed- t_0 cone density (legacy)). *Fix $K > 0$ and $t_0 > 0$. Let $\mathcal{C}_K(t_0)$ be the uniform-closure on $[-K, K]$ of the conoid generated by Fejér×heat atoms $\{\Phi_{B,t_0,\tau} : B > 0, |\tau| + B \leq K\}$ and their even symmetrizations. Then $\mathcal{C}_K(t_0) = C_{\text{even}}^+([-K, K])$.*

Proof. Let $f \in C_{\text{even}}^+([-K, K])$ and $\varepsilon > 0$. Choose a uniform grid with mesh $\delta > 0$ and hats $H_j(\xi) := \Lambda_\delta(\xi - \tau_j)$. By positive piecewise-linear interpolation, $h(\xi) := \sum_j f(\tau_j) H_j(\xi)$ satisfies $\|h - f\|_\infty < \varepsilon/3$ and $h \geq 0$. Since ρ_{t_0} is positive and Lipschitz on $[-2K, 2K]$, there is $L_{t_0} > 0$ with $|\rho_{t_0}(x) - \rho_{t_0}(y)| \leq L_{t_0}|x - y|$ for $x, y \in [-2K, 2K]$. Set $g(\xi) := \sum_j c_j \Phi_{\delta,t_0,\tau_j}(\xi)$ with $c_j := f(\tau_j)/\rho_{t_0}(\tau_j)$. Then for $|\xi| \leq K$,

$$g(\xi) - h(\xi) = \sum_j c_j \Lambda_\delta(\xi - \tau_j) (\rho_{t_0}(\xi - \tau_j) - \rho_{t_0}(\tau_j)),$$

and therefore

$$|g(\xi) - h(\xi)| \leq L_{t_0} \delta \sum_j |c_j| \Lambda_\delta(\xi - \tau_j) \leq L_{t_0} \delta \|c\|_{\ell^1}.$$

Choosing δ small gives $\|g - f\|_\infty < \varepsilon$. Evenization preserves nonnegativity. Taking the conoid-closure yields the claim. $\square$

G.2 Prime Operator Control via Measure Domination and Induction (legacy)

Remark (MD_{2,3} role: optional sufficient condition). The MD_{2,3} base interval theorem is an *alternative sufficient condition* for achieving symbol floor domination over prime contribution on a small compact. It is **not** required for the main logical chain.

Two proof routes:

- **Main route (RNA gate):** A3-Lock (symbol barrier + RKHS contraction) + AB(K) aggregation + T5 transfer. Uses constructive parameter recipe (Section Parameter Recipe) with explicit formulas for $(B, t, M, \Delta, \eta_K)$. *No numerical Gold $K=1$ example needed.*
- **Alternative route (MD base):** Explicit parameter windows (B, r, t) where criterion (G.9) holds analytically on base interval $[B_3, B_4)$. Provides:

- *Constructive illustration* that feasible parameters exist;
- *QA check*: Gold K=1 numerical scan confirms parameter feasibility;
- *Fallback*: If A3-Lock slack becomes tight, MD gives certified explicit windows.

Logical necessity: MD_{2,3} is *sufficient but not necessary*. The proof chain works without it via the parameter recipe's constructive formulas. MD serves as historical context and quality assurance, not as a required step.

Remark (Weight convention). Throughout Sections 10.5 and Appendix G.2 we write $w(n)$ for the undoubled operator weight $w_{\text{RKHS}}((n) = \Lambda(n)/\sqrt{n}$; the evenized weights $w_Q((n) = 2\Lambda(n)/\sqrt{n}$ only appear inside the Weil functional $Q^*(t; \cdot)$.

Theorem G.3 (MD_{2,3}: Base interval $[B_3, B_4]$). *Let $B \in [B_3, B_4]$ with $B_3 = \frac{\log 3}{2\pi}$ and $B_4 = \frac{\log 4}{2\pi}$. Active integers are $\{2, 3\}$ with nodes $\xi_n = \frac{\log n}{2\pi}$. For the unshifted window $\Phi_{B,t}(\xi) = \Lambda_B(\xi) \rho_t(\xi)$ (even, nonnegative) where $\Lambda_B(x) = (1 - |x|/B)_+$ and ρ_t is a normalized heat kernel, define*

$$\nu_{\text{Arch}}(d\xi) = a(\xi) d\xi, \quad a(\xi) = \log \pi - \Re \psi\left(\frac{1}{4} + i\pi\xi\right), \quad \nu_P = \sum_{n \in \{2,3\}} \frac{2\Lambda(n)}{\sqrt{n}} \delta_{\xi_n}. \quad (\text{G.8})$$

For $r \in (0, B)$ and $t > 0$, set the core minimum $m_r := \inf_{|\xi| \leq r} a(\xi)$ and the offcore mass $N_{B,r} := \int_{[-B, B] \setminus [-r, r]} |a(\xi)| d\xi$. With $\rho_t(\xi) = (4\pi t)^{-1/2} e^{-(2\pi)^2 \xi^2 / t}$, write $\rho_t(r) = (4\pi t)^{-1/2} e^{-(2\pi)^2 r^2 / t}$. If

$$m_r \rho_t(r) \frac{r^2}{B} - 2(4\pi t)^{-1/2} N_{B,r} \geq \frac{\log 2}{\sqrt{2}} + \frac{\log 3}{\sqrt{3}}, \quad (\text{G.9})$$

then one has

$$\int_{-B}^B a(\xi) \Phi_{B,t}(\xi) d\xi \geq \sum_{n \in \{2,3\}} \frac{2\Lambda(n)}{\sqrt{n}} \Phi_{B,t}(\xi_n), \quad (\text{G.10})$$

equivalently $Q^*(t; \Phi_{B,t}) \geq 0$ on the base interval cone.

Remark (Constants table). Illustrative bounds supporting the sufficient condition (G.9) for sample parameters (B, r, t) are summarized in the appendix table `MD_2_3_constants_table.tex`. The proof itself is analytic and does not rely on numerics; the table serves communication only.

Proof. We prove the inequality $\int_{-B}^B a(\xi) \Phi_{B,t}(\xi) d\xi \geq \sum_{n \in \{2,3\}} \frac{2\Lambda(n)}{\sqrt{n}} \Phi_{B,t}(\xi_n)$ under condition (G.9).

Step 1 (Prime side). Since $\Lambda_B \leq 1$ and $\|\rho_t\|_\infty = (4\pi t)^{-1/2}$, one has $\Phi_{B,t}(\xi_n) \leq 2(4\pi t)^{-1/2}$. In particular, if $t \geq 1/\pi$ then $2(4\pi t)^{-1/2} \leq 1$ and $\Phi_{B,t}(\xi_n) \leq 1$ for $n \in \{2, 3\}$; hence

$$\sum_{n \in \{2,3\}} \frac{2\Lambda(n)}{\sqrt{n}} \Phi_{B,t}(\xi_n) \leq \frac{2\log 2}{\sqrt{2}} + \frac{2\log 3}{\sqrt{3}}. \quad (\text{G.11})$$

Step 2 (Core/offcore split). Decompose

$$\int_{-B}^B a \Phi_{B,t} d\xi = \int_{-r}^r a \Phi_{B,t} d\xi + \int_{[-B, B] \setminus [-r, r]} a \Phi_{B,t} d\xi. \quad (\text{G.12})$$

Step 3 (Core lower bound). On $[-r, r]$, $a \geq m_r$ and $\rho_t(\xi) \geq \rho_t(r)$, so

$$\int_{-r}^r \Lambda_B(\xi) \rho_t(\xi) d\xi \geq \rho_t(r) \int_{-r}^r \Lambda_B(\xi) d\xi \geq \rho_t(r) \frac{r^2}{B}. \quad (\text{G.13})$$

Therefore

$$\int_{-r}^r \Phi_{B,t}(\xi) d\xi \geq \rho_t(r) \frac{r^2}{B}, \quad \text{so} \quad \int_{-r}^r a \Phi_{B,t} d\xi \geq m_r \rho_t(r) \frac{r^2}{B}. \quad (\text{G.14})$$

Step 4 (Offcore upper bound). On $[-B, B] \setminus [-r, r]$, using $\Lambda_B \leq 1$ and Young's inequality for convolution (e.g. [29, Ch. 3]) in the form $\|f * \rho_t\|_\infty \leq (4\pi t)^{-1/2} \|f\|_1$ applied to $f = |a| \mathbf{1}_{[-B, B] \setminus [-r, r]}$, we obtain

$$\int_{[-B, B] \setminus [-r, r]} |a(\xi)| \Lambda_B(\xi) \rho_t(\xi) d\xi \leq (4\pi t)^{-1/2} N_{B,r}. \quad (\text{G.15})$$

This yields an offcore penalty $\leq 2(4\pi t)^{-1/2} N_{B,r}$ (a safe bound compatible with the criterion).

Step 5 (Combine). Putting pieces together,

$$\int_{-B}^B a \Phi_{B,t} d\xi \geq m_r \rho_t(r) \frac{r^2}{B} - 2(4\pi t)^{-1/2} N_{B,r}. \quad (\text{G.16})$$

By assumption (G.9) this lower bound is at least $\frac{2\log 2}{\sqrt{2}} + \frac{2\log 3}{\sqrt{3}}$, which in turn dominates the prime contribution from Step 1. Hence the claimed inequality holds. $\square$

Remark. Explicit lower bounds for m_r on small r follow from classical digamma bounds (see, e.g., [22, §5]); $N_{B,r}$ is finite for fixed B and admits explicit upper bounds via $\Re \psi(\frac{1}{4} + i\pi\xi) = \log |\pi\xi| + O(1/|\xi|)$. The core mass factor $\rho_t(r) \frac{r^2}{B}$ captures Gaussian localization and Fejér area; taking $t \geq 1/\pi$ ensures the pointwise prime contribution $\Phi_{B,t}(\xi_n) \leq 1$.

Theorem G.4 (MD_{2,3} in operator form). *Let $B \in [B_3, B_4]$ so that only $n \in \{2, 3\}$ are active on $[-K, K]$. With the RKHS normalization $\|k_\alpha\| = 1$, one has*

$$\|T_P^{\text{RKHS}}(t)\| \leq w_{\max} + \sqrt{w_{\max}} S_K(t), \quad w_{\max} = \max \left\{ \frac{\log 2}{\sqrt{2}}, \frac{\log 3}{\sqrt{3}} \right\}. \quad (\text{G.17})$$

Choosing $t = t_{\min}(K)$ so that $S_K(t_{\min}) \leq \frac{1 - w_{\max} - \varepsilon_K}{\sqrt{w_{\max}}}$ yields $\|T_P^{\text{RKHS}}(t)\| \leq \rho_K < 1$ and hence $T_A - T_P^{\text{RKHS}}(t) \succeq 0$ on $\mathcal{H}_K$.

Theorem G.5 (Block induction IND^{block}). *Suppose on a compact $[-K, K]$ one has $\|T_{P,\text{old}}^{\text{RKHS}}(t)\| \leq \rho_K^{\text{old}} < 1$. Let $\mathcal{N}$ be a finite set of newly active nodes with weights $\{w(n) : n \in \mathcal{N}\}$ and let $T_{P,\text{new}}^{\text{RKHS}}(t) = T_{P,\text{old}}^{\text{RKHS}}(t) + \sum_{n \in \mathcal{N}} w(n) |k_{\alpha_n}\rangle\langle k_{\alpha_n}|$. Then*

$$\|T_{P,\text{new}}^{\text{RKHS}}(t)\| \leq \|T_{P,\text{old}}^{\text{RKHS}}(t)\| + \sum_{n \in \mathcal{N}} w(n). \quad (\text{G.18})$$

In particular, if $\sum_{n \in \mathcal{N}} w(n) \leq \varepsilon_K$ with $\rho_K^{\text{old}} + \varepsilon_K < 1$, then $T_A - T_{P,\text{new}}^{\text{RKHS}}(t) \succeq 0$ on $\mathcal{H}_K$.

Proof. The update is a finite sum of positive rank-one operators. By the triangle inequality for the operator norm and $\| |k\rangle\langle k| \| = \|k\|^2 = 1$, we obtain $\| \sum_{n \in \mathcal{N}} w(n) |k_{\alpha_n}\rangle\langle k_{\alpha_n}| \| \leq \sum_{n \in \mathcal{N}} w(n)$. The conclusion follows. $\square$

Theorem G.6 (Block induction across early active thresholds). *Fix $K > 0$ and let $\mathcal{N}_{\leq N_0}$ be the finite set of active nodes on $[-K, K]$ up to a cutoff index $N_0 = N_0(K)$. There exist:*

- a partition $\mathcal{N}_{\leq N_0} = B_1 \sqcup B_2 \sqcup \cdots \sqcup B_J$ into consecutive blocks (in any fixed ordering),
- a number $\varepsilon(K) \in (0, 1)$ and a uniform margin $\gamma(K) > 0$,

• for each block B_j a single-scale Fejér×heat window Φ_j with parameters from Lemma 9.32, such that

$$\sum_{n \in B_j} w(n) \leq \varepsilon(K) \quad \text{for all } j, \quad (\text{G.19})$$

and the following operator inequality holds uniformly in j :

$$(T_A - T_P^{\text{RKHS}}(t))[\Phi_j; B_1 \cup \dots \cup B_j] \succeq \gamma(K) I. \quad (\text{G.20})$$

After exhausting the early blocks, the one-prime step (Theorem 10.12) applies since the remaining new weights satisfy $w_{\text{new}} \leq \varepsilon(K)$ and $\rho_K^{\text{old}} + w_{\text{new}} < 1$.

Proof. Let $c_0(K)$, t , and M_0 be as in Lemma 9.32. Choose $\varepsilon(K) := \frac{1}{4}c_0(K)$ and $\gamma(K) := \frac{1}{2}c_0(K)$. Construct blocks greedily along the chosen ordering so that each block satisfies $\sum_{n \in B_j} w(n) \leq \varepsilon(K)$ (the last block may have a strictly smaller sum). For Φ_j take the single-scale Fejér×heat window furnished by Lemma 9.32.

By that theorem, uniformly for $M \geq M_0$,

$$\lambda_{\min}(T_M[P_A[\Phi_j]] - T_P^{\text{Ray}}(t, M)[\Phi_j]) \geq \frac{1}{2}c_0(K). \quad (\text{G.21})$$

Restricting the prime sum to a subset (the cumulative blocks $\bigcup_{i \leq j} B_i$) can only decrease the prime operator in the Loewner order, hence preserves the lower bound. Equivalently, on the RKHS side one has

$$\|T_P^{\text{RKHS}}(t)[\Phi_j; B_1 \cup \dots \cup B_j]\| \leq \|T_P^{\text{RKHS}}(t)[\Phi_j]\| \leq \frac{1}{4}c_0(K) \leq \varepsilon(K), \quad (\text{G.22})$$

while the Archimedean part contributes at least $\frac{3}{4}c_0(K)$ in the mixed symbol bound. Combining these gives (G.20) with $\gamma(K) = \frac{1}{2}c_0(K)$. The tail phase follows from Theorem 10.12 because each subsequent new node has weight at most $\varepsilon(K)$ and the previously accumulated norm is bounded away from 1. $\square$

Theorem G.7 (IND^{block} (block update on activity jumps)). *Let $[-K, K]$ be fixed and suppose on some activity interval $I = [B_n, B_{n+1})$ we have the operator margin*

$$T_A - T_P^{\text{RKHS}}(t) \succeq \gamma_K T_A \quad \text{with } \gamma_K \in (0, 1]. \quad (\text{G.23})$$

Let a packet B of new prime nodes enter when crossing to the next activity interval, with cumulative weight $W_B := \sum_{n \in B} w(n)$. Then

$$T_A - (T_P^{\text{RKHS}}(t) + \Delta T_P^{\text{RKHS}}(t)) \succeq (\gamma_K - W_B) T_A, \quad (\text{G.24})$$

where $\Delta T_P^{\text{RKHS}}(t) = \sum_{n \in B} w(n) |k_{\alpha_n}\rangle \langle k_{\alpha_n}|$ in the RKHS normalization $\|k_{\alpha}\| = 1$. In particular, if $W_B \leq \varepsilon(K) < \gamma_K$, positivity persists: $T_A - (T_P^{\text{RKHS}}(t) + \Delta T_P^{\text{RKHS}}(t)) \succeq (\gamma_K - \varepsilon(K)) T_A \succeq 0$. After the block, one may continue with the one-prime step (IND').

Proof. Monotonicity in the Loewner order and the rank-one bound give $\|\Delta T_P^{\text{RKHS}}(t)\| \leq \sum_{n \in B} w(n) = W_B$. For any unit vector f , $\langle (T_A - (T_P^{\text{RKHS}}(t) + \Delta T_P^{\text{RKHS}}(t)))f, f \rangle \geq \gamma_K \langle T_A f, f \rangle - \|\Delta T_P^{\text{RKHS}}(t)\| \langle f, f \rangle \geq (\gamma_K - W_B) \langle T_A f, f \rangle$. $\square$

G.3 Explicit Constants for MD_{2,3}

We collect analytic bounds sufficient to verify the base interval MD_{2,3} without numerics in the main text. Numerical certification (interval arithmetic) may be delegated to the reproducibility appendix.

G.4 Lower bound for m_r

Define $a(\xi) = \log \pi - \Re \psi(\frac{1}{4} + i\pi\xi)$. For $r \in (0, 1]$ set

$$m_r := \inf_{|\xi| \leq r} a(\xi) = \log \pi - \sup_{|\xi| \leq r} \Re \psi\left(\frac{1}{4} + i\pi\xi\right). \quad (\text{G.25})$$

Using the integral representation (for $\Re z > 0$) from [31, Ch. 2]

$$\psi(z) = \log z - \int_0^\infty \left(\frac{1}{t} - \frac{1}{1-e^{-t}}\right) e^{-zt} dt, \quad (\text{G.26})$$

we obtain, after taking real parts at $z = \frac{1}{4} + i\pi\xi$, the bound

$$\Re \psi\left(\frac{1}{4} + i\pi\xi\right) \leq \log \sqrt{\frac{1}{16} + \pi^2 \xi^2} + C_0, \quad C_0 := \int_0^\infty \left|\frac{1}{t} - \frac{1}{1-e^{-t}}\right| e^{-t/4} dt. \quad (\text{G.27})$$

Hence

$$m_r \geq \log \pi - \log \sqrt{\frac{1}{16} + \pi^2 r^2} - C_0 = \frac{1}{2} \log \left(\frac{\pi^2}{\frac{1}{16} + \pi^2 r^2} \right) - C_0. \quad (\text{G.28})$$

This gives an explicit (computable) lower bound $m_r \downarrow 0$ as $r \downarrow 0$.

G.5 Upper bound for $N_{B,r}$

Let $N_{B,r} = \int_{[-B,B] \setminus [-r,r]} |a(\xi)| d\xi$. For $|\xi| \geq r$ and $r \in (0, 1]$ we use the asymptotic ([31, Ch. 2])

$$\Re \psi\left(\frac{1}{4} + i\pi\xi\right) = \log(\pi|\xi|) + O\left(\frac{1}{1+|\xi|}\right), \quad (\text{G.29})$$

whence $|a(\xi)| \leq |\log \pi - \log(\pi|\xi|)| + C_1 \leq \log^+ \frac{1}{|\xi|} + C_1$ for a universal C_1 . Therefore

$$N_{B,r} \leq \int_{[-B,-r] \cup [r,B]} (\log^+ \frac{1}{|\xi|} + C_1) d\xi \leq 2\left(r \log \frac{1}{r} + r + (B-r)C_1\right). \quad (\text{G.30})$$

In particular, for fixed B and small r one has $N_{B,r} = O(r \log \frac{1}{r})$.

G.6 Core mass via $\rho_t(r)$ and Fejér area

For any $r \in (0, B)$,

$$\int_{-r}^r \Lambda_B(\xi) \rho_t(\xi) d\xi \geq \rho_t(r) \int_{-r}^r \Lambda_B(\xi) d\xi \geq \rho_t(r) \frac{r^2}{B}, \quad (\text{G.31})$$

using $\rho_t(\xi) \geq \rho_t(r)$ on $[-r, r]$ and the Fejér area bound.

G.7 Sufficient criterion (reprise)

Combining the bounds gives the sufficient condition for MD_{2,3} on $[B_3, B_4]$:

$$\boxed{m_r \rho_t(r) \frac{r^2}{B} - 2(4\pi t)^{-1/2} N_{B,r} \geq \frac{\log 2}{\sqrt{2}} + \frac{\log 3}{\sqrt{3}}} \quad (\text{G.32})$$

with $m_r, N_{B,r}$ as above and $\rho_t(r) = (4\pi t)^{-1/2} e^{-(2\pi)^2 r^2/t}$. One may additionally fix $t \geq 1/\pi$ to ensure $\Phi_{B,t}(\xi_n) \leq 1$ on the prime side.

G.8 RKHS auxiliary bounds for the operator form

We record three elementary ingredients used by the RKHS contraction in the MD module.

Lemma G.8 (Effective weight cap). *For the even weighting $w(n) = \Lambda(n)/\sqrt{n}$ one has*

$$\sup_{x \geq 2} \frac{\log x}{\sqrt{x}} = \frac{2}{e} < \frac{3}{4} < 1, \quad \text{hence} \quad w_{\max} \leq \frac{2}{e} < \frac{3}{4}. \quad (\text{G.33})$$

(Rational bound: $2/e \approx 0.7358 \dots < 3/4 = 0.75$, ensuring all subsequent constraints with $w_{\max}$ use explicit rational inequalities.)

Lemma G.9 (Log-node gap on a compact). *Let $\alpha_n = \frac{\log n}{2\pi}$ and fix $K \geq 1$. Then the minimal active gap on $[-K, K]$ satisfies*

$$\delta_K := \min\{\alpha_{n+1} - \alpha_n : \alpha_n, \alpha_{n+1} \in [-K, K]\} \geq \frac{1}{4\pi e^{2\pi K}}. \quad (\text{G.34})$$

Proof. For $n \geq 1$, by convexity of $\log$ we have $\log(n+1) - \log n \geq \frac{1}{n+1}$. Hence

$$\alpha_{n+1} - \alpha_n = \frac{\log(n+1) - \log n}{2\pi} \geq \frac{1}{2\pi(n+1)}. \quad (\text{G.35})$$

On $[-K, K]$ one has $n+1 \leq \lfloor e^{2\pi K} \rfloor + 1 \leq 2e^{2\pi K}$ for $K \geq 1$, so $\alpha_{n+1} - \alpha_n \geq (4\pi e^{2\pi K})^{-1}$. Taking the minimum over active indices yields the claim. $\square$

Proposition G.10 (RKHS contraction parameter). *With $S_K(t) := \frac{2e^{-\delta_K^2/(4t)}}{1 - e^{-\delta_K^2/(4t)}}$ and any $\eta_K \in (0, 1)$ define*

$$t_{\min}(K) = \frac{\delta_K^2}{4 \ln\left(\frac{2+\eta_K}{\eta_K}\right)}. \quad (\text{G.36})$$

Then $S_K(t_{\min}) \leq \eta_K$ and the de Branges/RKHS contraction holds:

$$\|T_P^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq w_{\max} + \sqrt{w_{\max}} S_K(t_{\min}) \leq w_{\max} + \sqrt{w_{\max}} \eta_K. \quad (\text{G.37})$$

G.9 Illustrative constants for MD_{2,3}

The table below summarizes indicative bounds entering the sufficient condition $m_r \rho_t(r) \frac{r^2}{B} - 2(4\pi t)^{-1/2} N_{B,r} \geq \frac{\log 2}{\sqrt{2}} + \frac{\log 3}{\sqrt{3}}$ for sample parameters within the base interval $B \in [B_3, B_4)$. Values are computed from the explicit inequalities in `MD_2_3_constants.tex` and serve communication only (QA); they are not part of the proof.

B	r	t	lower m_r	upper $N_{B,r}$	$\rho_t(r) \frac{r^2}{B}$	$(4\pi t)^{-1/2} N_{B,r}$
0.210	0.10	$4.0 \cdot 10^{-1}$	0.557	1.101	$7.93 \cdot 10^{-3}$	0.492
0.210	0.08	$2.5 \cdot 10^{-1}$	0.683	1.084	$6.25 \cdot 10^{-3}$	0.611
0.208	0.10	$4.0 \cdot 10^{-1}$	0.557	1.093	$8.00 \cdot 10^{-3}$	0.487

Values computed from the explicit expressions in `MD_2_3_constants.tex` using conservative universal constants $C_0 = 1.5$ and $C_1 = 2.0$ (digamma bound for m_r and logarithmic tail bound for $N_{B,r}$), together with the Gaussian terms $\rho_t(r)$ and $(4\pi t)^{-1/2}$.

G.10 RKHS contraction (conservative parameters)

For convenience we also list two conservative parameter choices for the RKHS contraction used in the operator form of MD. Here $K := B/r$, $\delta_K \geq (4\pi e^{2\pi K})^{-1}$, we pick $\eta_K \in (0, 1)$ and set $t_{\min}(K) = \delta_K^2 / (4 \ln((2 + \eta_K)/\eta_K))$. This ensures $S_K(t_{\min}) \leq \eta_K$ and $\rho_K \leq w_{\max} + \sqrt{w_{\max}} \eta_K$ with $w_{\max} \leq 2/e$.

B	r	$K = B/r$	η_K	$t_{\min}(K)$	ρ_K upper bound
0.210	0.08	≈ 2.625	0.20	from δ_K	$\frac{2}{e} + \sqrt{\frac{2}{e}} \cdot 0.20 \approx 0.9084$
0.210	0.10	≈ 2.10	0.15	from δ_K	$\frac{2}{e} + \sqrt{\frac{2}{e}} \cdot 0.15 \approx 0.8645$

These rows are intentionally conservative: reviewers may re-evaluate δ_K and $t_{\min}(K)$ for tighter bounds; feasibility ($\rho_K < 1$) is already clear from the upper bounds.

Theorem G.11 (B.3: IND/AB). *On an activity interval $[B_n, B_{n+1})$ let $\|T_{P,\text{old}}^{\text{RKHS}}(t)\|_{\mathcal{H}_K} \leq \rho_K^{\text{old}} < 1$. When crossing the threshold B_{n+1} a single new node α_{new} with weight w_{new} enters. In the RKHS normalization $\|k_\alpha\| = 1$ one has*

$$\|T_{P,\text{new}}^{\text{RKHS}}(t)\| \leq \rho_K^{\text{old}} + w_{\text{new}}. \quad (\text{G.38})$$

Hence if $\rho_K^{\text{old}} + w_{\text{new}} < 1$, then $T_A - T_{P,\text{new}}^{\text{RKHS}}(t) \succeq 0$ on $\mathcal{H}_K$.

Proof. Rank-one update: $T_{P,\text{new}}^{\text{RKHS}}(t) = T_{P,\text{old}}^{\text{RKHS}}(t) + w_{\text{new}} |k_{\alpha_{\text{new}}}\rangle \langle k_{\alpha_{\text{new}}}|$ with $\|k_\alpha\| = 1$ gives the claimed norm bound; strict inequality implies the Loewner positivity. $\square$

Corollary G.12 (Gluing intervals). *Suppose MD_{2,3} holds on $[B_3, B_4)$, and across each threshold $B_n \rightarrow B_{n+1}$ the one-prime condition $\rho_K^{\text{old}} + w_{\text{new}} < 1$ is verified in the RKHS normalization on $[-K, K]$. Then $T_A - T_P^{\text{RKHS}}(t) \succeq 0$ holds on $[-K, K]$ for all $B \geq B_3$, i.e. the measure domination persists interval-by-interval.*

Lemma G.13 (Analytic bound for early blocks). *Let $\Phi_{B,t}(\xi) = (1 - |\xi|/B)_+ e^{-4\pi^2 t \xi^2}$ with $B > 0$. Then for the even setting with weights $w(n) = \Lambda(n)/\sqrt{n}$ and nodes $\alpha_n = \log n/(2\pi)$ one has the deterministic bound*

$$\sum_{\alpha_n \in [-B, B]} w(n) \Phi_{B,t}(\alpha_n) \leq \sum_{n \leq e^{2\pi B}} \frac{\Lambda(n)}{\sqrt{n}} \leq \int_1^{e^{2\pi B}} \frac{\log u}{\sqrt{u}} du = 2e^{\pi B} (2\pi B - 2) + 4. \quad (\text{G.39})$$

In particular, choosing $B = B(K) > 0$ small enough forces the early-block mass to lie below any prescribed budget $\varepsilon(K) > 0$.

Proof. Since $0 \leq \Phi_{B,t} \leq 1$ and $\Phi_{B,t}$ vanishes outside $[-B, B]$, the first inequality holds. For the second, use $\Lambda(n) \leq \log n$ and compare the sum to the integral; the evaluation follows by the substitution $u = v^2$. $\square$

AB(∞) closure: RNA gate with fixed modcap and SAFE lift

Theorem G.14 (AB ∞ closure (fixed $q = 30$, BRC-SAFE default)). *Work under the T0 normalisation $Q^*(t; \cdot) = Q_{\text{GW}}$ on the GW-axis. Fix $q_0 = 30$ and the single heat scale $t = t_{\text{crit}}$. Let $\{K_i\}_{i \geq 1}$ be an increasing chain with $\bigcup_i [-K_i, K_i] = \mathbb{R}$. For each i choose parameters (B_i, M_i) at the fixed scale $t = t_{\text{crit}}$, and work with the unshifted node $\tau = 0$ (so the shift grid is $\{0\}$). Assume for every i :*

(R) Arch floor (A3). With the unshifted Fejér×heat window $\Phi_{B_i,t}$ and Arch symbol P_A ,

$$\min_{\theta} P_A(\theta) \geq c_0(K_i). \quad (\text{G.40})$$

(N) Nyquist & Norm. The symbol modulus and the prime cap satisfy

$$C \omega_{P_A}\left(\frac{1}{2M_i}\right) \leq \frac{c_0(K_i)}{2}, \quad \|T_P^{(q_0)}(t_{\text{crit}})\| \leq \frac{c_0(K_i)}{2}, \quad (\text{G.41})$$

where $T_P^{(q_0)}(t_{\text{crit}})$ is the modular cap at modulus $q_0 = 30$ with RKHS smoothing scale $t = t_{\text{crit}}$.

(A) Grid→continuum (BRC–SAFE default). The (degenerate) grid at $\tau = 0$ is BRC–SAFE; equivalently, the resolvent certificate with Ky Fan/Hoffman–Wielandt budget holds at the unshifted node.

Then $Q^*(t; \Phi) \geq 0$ for all even Paley–Wiener tests Φ on $[-K_i, K_i]$ for every i . Consequently, $Q^*(t; \cdot) \geq 0$ on the full Weil class; by Weil’s positivity criterion, RH follows.

Proof (by plumbing). By the Toeplitz symbol bridge (A3), at the unshifted node $\tau = 0$,

$$\lambda_{\min}(T_{M_i}[P_A] - T_P^{\text{Ray}}(t, M_i)) \geq \min P_A - C \omega_{P_A}\left(\frac{1}{2M_i}\right) - \|T_P^{\text{Ray}}(t, M_i)\|. \quad (\text{G.42})$$

Assumptions **(R)–(N)** make the RHS $\geq c_0 - \frac{c_0}{2} - \frac{c_0}{2} = 0$, so nonnegativity holds on all grid nodes. By **(A)** (BRC–SAFE at $\tau = 0$) the sign is preserved on $[-K_i, K_i]$. Fejér×heat density (A1′) and Lipschitz continuity (A2) lift nonnegativity from the grid cone to all even PW tests on $[-K_i, K_i]$. Finally, along the chain $\{K_i\}$ the T5 compact limit transfers $Q^*(t; \cdot) \geq 0$ to the Weil class. $\square$

Remark (Lipschitz-lift option). Instead of BRC–SAFE one may enforce the deterministic Lipschitz lift $L_Q(K_i) L_\Phi(K_i) \Delta\tau \leq c_0(K_i)/4$; in the unshifted case $\Delta\tau = 0$ so the condition is immediate.

Remark (Monotone inheritance). It is convenient (not essential) to choose $B_i \uparrow$, $M_i \uparrow$, and nonincreasing budgets so that acceptance persists along the chain.

Remarks.

- The mod-30 cap is fixed once and for all; its early block and tail are the audited ones used throughout the acceptance pipeline (no dependence on K beyond the truncated lists).
- The SAFE lift replaces the coarse Lipschitz mesh. One may still use the deterministic bound $\Delta_K \leq c_0(K)/(4L_Q(K)L_\Phi(K))$ when convenient, but the BRC check is the primary path in our RNA gate.
- The scales t, t are bounded away from 0, so the arch-floor constants and the modular cap norms remain uniform along the AB_∞ ladder.
- **Grid → continuum → Weil transfer.** By A1′ (Theorem 9.2) the Fejér×heat cone is dense in W_K in $\|\cdot\|_\infty$; by A2 (Proposition 10.2) Q is Lipschitz on W_K . Hence the grid positivity and the SAFE/Lipschitz lift imply $Q \geq 0$ on all of W_K . With the monotone parameter schedule (Lemma 17.4), Theorem 18.2 transfers positivity to the Weil class.

G.11 IND/AB: Parameter recipe

Acceptance parameters (IND/AB)

Single-scale note. In the main line we fix the onset shift to $\tau = 0$; the general plateau formula is retained for legacy reference.

Name	Symbol	Range	Role
Pre-plateau slope	α	$[0, 1]$	Growth before the plateau
Plateau width	β	$(0, \infty)$	Length of the flat segment
Onset shift	τ	0 (fixed)	Position of the plateau window
Saturation level	γ	$(0, 1]$	Upper acceptance bound

$$\text{Plateau}(t; \alpha, \beta, \tau, \gamma) = \begin{cases} \alpha t, & t \leq \tau, \\ \gamma, & \tau < t \leq \tau + \beta, \\ \max\{\gamma - \alpha(t - \tau - \beta), 0\}, & t > \tau + \beta. \end{cases} \quad (\text{G.43})$$

Lemma G.15 (Plateau schedule is admissible). *Let $A(t) = \text{Plateau}(t; \alpha, \beta, \tau, \gamma)$ with $0 < \alpha \leq \gamma \leq 1$ and $\beta > 0$. Then A takes values in $[0, 1]$, is piecewise Lipschitz, and meets the IND/AB plateau constraints: monotonic rise before τ , a flat segment of width β , and compatible one-sided derivatives at the junctions.*

Proof. On $(-\infty, \tau]$ the function is $A(t) = \alpha t$, so it is increasing and Lipschitz with slope α . On $(\tau, \tau + \beta]$ we have $A(t) = \gamma$, so A is constant and Lipschitz with slope 0. On $(\tau + \beta, \infty)$ we have $A(t) = \max\{\gamma - \alpha(t - \tau - \beta), 0\}$, hence A is nonincreasing and Lipschitz with slope in $[-\alpha, 0]$. Since $0 < \alpha \leq \gamma \leq 1$, all three branches take values in $[0, 1]$, and the one-sided derivatives at the junctions are α and 0 at $t = \tau$, and 0 and $-\alpha$ (or 0 once the floor is active) at $t = \tau + \beta$. This matches the plateau constraints. $\square$

G.12 Compact-by-Compact Positivity and Limit (legacy)

G.13 T5: Compact-by-Compact Positivity and Limit to the Weil Class

Definition G.16 (Weil inductive-limit topology). Let $W_K := C_{\text{even}}^+([-K, K])$ with the uniform norm. Define the Weil class $W := \bigcup_{K \geq 1} W_K$ with the inductive (LF) topology: $U \subset W$ is open iff $U \cap W_K$ is open in W_K for every K . A quadratic functional $Q^*(t; \cdot) : W \rightarrow \mathbb{R}$ is (sequentially) continuous in this topology iff each restriction $Q^*(t; \cdot)|_{W_K}$ is continuous in $\|\cdot\|_\infty$.

Lemma G.17 (Local continuity suffices for T5). *If for every K the restriction $Q^*(t; \cdot)|_{W_K}$ is Lipschitz in $\|\cdot\|_\infty$ with some (possibly K -dependent) constant L_K , then the inductive-limit topology of Definition G.16 guarantees sequential continuity of $Q^*(t; \cdot)$ on W . No uniform bound $\sup_K L_K < \infty$ is required: whenever $\Phi_n \rightarrow \Phi$ in W , the convergence takes place in a single W_K , and the corresponding L_K controls $|Q^*(t; \Phi_n) - Q^*(t; \Phi)|$.*

Lemma G.18 (T5: transfer across $K \uparrow$). *If $Q^*(t; \Phi) \geq 0$ on every W_K and the family $\{Q^*(t; \cdot)|_{W_K}\}$ is compatible with the natural inclusions $W_K \hookrightarrow W_{K'}$ for $K < K'$, then $Q^*(t; \Phi) \geq 0$ on W .*

Proposition G.19 (LF-transfer of positivity). *Let $\{W_K\}_{K \in \mathbb{N}}$ be an increasing family of cones of even, nonnegative C_c tests supported in $[-K, K]$, and let $W = \varinjlim W_K$ be their LF inductive limit. Suppose: (i) for each K , the quadratic form $Q^*(t; \cdot)$ is continuous on W_K in the $\|\cdot\|_\infty$ topology; (ii) $Q^*(t; \Phi) \geq 0$ for all $\Phi \in W_K$ for every K ; and (iii) the embeddings $W_K \hookrightarrow W_{K+1}$ are continuous and compatible with $Q^*(t; \cdot)$. Then $Q^*(t; \Phi) \geq 0$ on W .*

Remark. Continuity in (i) uses the local constants L_K from Corollary 8.2. We never require a uniform bound in K : the inductive-limit topology only asks for continuity on each fixed W_K , which is provided by A2.

Remark (Uniform parameters). The uniform Archimedean floor $c_* > 0$ from Lemma 9.19 and the uniform RKHS cap $\rho(t) \leq c_*/4$ from Corollary 9.22 provide K -independent bounds that work for all compacts simultaneously. The inductive-limit transfer of Proposition G.19 propagates positivity from the uniform A3 bridge to the full Weil class.

Proof. Given $\Phi \in W$, pick K with $\text{supp } \Phi \subset [-K, K]$; then $\Phi \in W_K$ and $Q^*(t; \Phi) \geq 0$ by (ii). Compatibility and continuity ensure independence from the chosen K . $\square$

We work on each compact $[-K, K]$ with the cone $\mathcal{C}_K$ generated by unshifted Fejér $\times$ heat atoms $\Phi_{B,t}$ (i.e., $\tau = 0$) at the fixed scale $t = t_{\text{crit}}$. The uniform Archimedean floor $c_* > 0$ comes from Lemma 9.19, the uniform prime cap $\rho(t) \leq c_*/4$ from Corollary 9.22, and the Lipschitz constant $L_Q(K)$ from A2. Section G.14 records the uniform grid-lift Lemma G.20 and Theorem G.21: $Q^*(t; \Phi) \geq 0$ on each W_K . Density (A1') and continuity (A2) extend this to the full Weil class.

G.14 Compact-by-compact transfer (T5)

Standing analytic inputs (uniform version)

For each $K > 0$ we use the uniform analytic data from Sections 9 and 10.5:

(A3) Uniform Archimedean floor: $\min_{\theta \in \mathbb{T}} P_A(\theta) \geq c_*$ (Lemma 9.19), which supplies an explicit $c_* > 0$.

(A3.b) Discretization control: for all $M \geq M_0^{\text{unif}}$,

$$C_{\text{SB}} \omega_{P_A}\left(\frac{1}{2M}\right) \leq \frac{c_*}{2}$$

(Corollary 9.21).

(RKHS) Uniform prime contraction: for $t = t_{\text{crit}}$,

$$\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq \frac{c_*}{4}$$

(Corollary 9.22).

We also recall the density/continuity interface on W_K :

(A1') The Fejér $\times$ heat cone is dense in W_K .

(A2) $Q^*(t; \cdot)$ is continuous on W_K ; specifically $|Q^*(t; \Phi) - Q^*(t; \Psi)| \leq L_Q(K) \|\Phi - \Psi\|_\infty$.

Lemma G.20 (Grid-lift inequality). *Assume $c_* > 0$ and $t = t_{\text{crit}}$. For every $K > 0$ and $M \geq M_0^{\text{unif}}$,*

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq c_* - C_{\text{SB}} \omega_{P_A}\left(\frac{1}{2M}\right) - \|T_P^{\text{Ray}}(t, M)\| \geq \frac{c_*}{4}.$$

Proof. Combine the uniform Archimedean floor $\min P_A \geq c_*$ with:

- Discretization: $C_{\text{SB}} \omega_{P_A}(1/(2M)) \leq c_*/2$ for $M \geq M_0^{\text{unif}}$.

- Prime cap: $\|T_P^{\text{RKHS}}(t)\| \leq \rho(t) \leq c_*/4$.

Then $\lambda_{\min} \geq c_* - c_*/2 - c_*/4 = c_*/4$. $\square$

Theorem G.21 (T5: uniform compact transfer). *Assume $c_* > 0$ and $t = t_{\text{crit}}$. For every $K > 0$ one has*

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \frac{c_*}{4} > 0$$

for all $M \geq M_0^{\text{unif}}$. In particular, $Q^*(t; \Phi) \geq 0$ on W_K for all $K > 0$. Hence $Q^*(t; \Phi) \geq 0$ on $\bigcup_{K>0} W_K$, i.e., on the full Weil class.

Proof. By Lemma G.20, the finite Toeplitz form $T_M[P_A] - T_P^{\text{Ray}}(t, M)$ has positive minimum eigenvalue $\geq c_*/4 > 0$ for all $M \geq M_0^{\text{unif}}$. Positivity on the Fejér $\times$ heat cone follows. Then (A1')–(A2) extend $Q^*(t; \Phi) \geq 0$ from the dense cone to all of W_K . Taking the union over K gives the claim. $\square$

Remark (No K-dependent schedules). The uniform approach eliminates the need for K-dependent schedules $c_0(K)$, $t^*(K)$, $M^*(K)$. The same uniform parameters c_* , $t = t_{\text{crit}}$, and M_0^{unif} work for all compacts $[-K, K]$ simultaneously.

G.15 T5: Inductive Limit over Compacts

Let $\mathcal{W}_K = C_{\text{even}}^+([-K, K])$ with the uniform norm and let $\mathcal{W} = \bigcup_{K>0} \mathcal{W}_K$ carry the inductive limit topology.

Lemma G.22 (Nested dictionaries yield $\mathcal{W}$). *For each $K > 0$ let $\mathcal{G}_K \subset \mathcal{C}_K$ be a finite dictionary as in Theorem 7.3, constructed over a bandwidth grid at the fixed scale $t = t_{\text{crit}}$. If $K_i \nearrow \infty$ and the dictionaries are nested so that $\mathcal{G}_{K_i} \subset \mathcal{G}_{K_{i+1}}$, then*

$$\bigcup_i \overline{\text{cone}(\mathcal{G}_{K_i})}^{\|\cdot\|^\infty} = \bigcup_i \mathcal{W}_{K_i} =: \mathcal{W}. \quad (\text{G.44})$$

Proof. By Theorem A1' each $\overline{\text{cone}(\mathcal{G}_{K_i})}$ is dense in $\mathcal{W}_{K_i}$, and nestedness yields the union identity. $\square$

Theorem G.23 (Transfer of positivity to the Weil class). *Assume $Q^*(t; \Phi) \geq 0$ on $\mathcal{W}_{K_i}$ for every i , where $Q^*(t; \cdot)$ is continuous on each $\mathcal{W}_{K_i}$ (Lemma 8.3). Then $Q^*(t; \Phi) \geq 0$ on $\mathcal{W}$ in the inductive limit topology. With the normalization of Lemma 6.2 and the bridge of Theorem 9.35, this identifies the positivity domain with the Weil cone $\mathcal{W}$ used throughout Sections 2–12.*

Proof. Given $\Phi \in \mathcal{W}$, choose i with $\text{supp } \Phi \subset [-K_i, K_i]$. Then $\Phi \in \mathcal{W}_{K_i}$ and $Q^*(t; \Phi) \geq 0$ by hypothesis. Continuity on each $\mathcal{W}_{K_i}$ and Lemma G.22 pass the result to the closure and thus to $\mathcal{W}$. $\square$

Lemma G.24 (Grid-lift by Lipschitz margin). *Let $Q^*(t; \cdot)$ be Lipschitz on W_K with constant $L_Q(K)$ (A2). Suppose there exists a finite subset $\{\Phi_j\} \subset \mathcal{C}_K$ such that every $\Phi \in \mathcal{C}_K$ admits some j with $\|\Phi - \Phi_j\|_\infty \leq \Delta$, and*

$$\min_j Q^*(t; \Phi_j) \geq c_* > 0$$

where $c_* > 0$ is the uniform floor from Lemma 9.19. If $\Delta \leq c_*/(4L_Q(K))$, then $Q^*(t; \Phi) \geq \frac{1}{2} c_*$ for all $\Phi \in \mathcal{C}_K$.

Proof. Fix $\Phi \in \mathcal{C}_K$ and choose j with $\|\Phi - \Phi_j\|_\infty \leq \Delta$. By Lipschitz continuity,

$$Q^*(t; \Phi) \geq Q^*(t; \Phi_j) - L_Q(K) \|\Phi - \Phi_j\|_\infty \geq c_* - L_Q(K) \Delta \geq \frac{1}{2} c_*.$$

□

Lemma G.25 (Uniform inheritance across K). *Fix an increasing chain $K_0 < K_1 < \dots$. Using the uniform parameters $t = t_{\text{crit}}$ and $M \geq M_0^{\text{unif}}$ from Theorem 9.35, for every i we have*

$$\lambda_{\min}(T_M[P_A] - T_P^{\text{Ray}}(t, M)) \geq \frac{c_*}{4} > 0 \quad \text{on } \mathcal{W}_{K_i}, \quad (\text{G.45})$$

and the property propagates from K_i to K_{i+1} .

Proof. Lemma G.20 with uniform parameters gives the lower bound $\geq c_*/4$ for all K . Since the floor c_* and the parameters t, M_0^{unif} are K -independent, the same estimate applies at every K_i , so the chain inherits positivity uniformly. □

H ATP Notes

This section collects the auxiliary ATP remarks referenced by the main T5 text. All ATP mentions are parked here so that the core discussion stays concise.

C.1 T5 formal verification scope

The automated proofs cover the following ingredients of the T5 transfer:

- **Series/limit step.** The inductive-limit description $\bigcup_i \mathcal{W}_{K_i} = \mathcal{W}$ is well-defined, and the monotone parameter schedules respect the inclusions $\mathcal{W}_{K_i} \hookrightarrow \mathcal{W}_{K_{i+1}}$.
- **Tail control.** Fejér×heat leakage outside $[-K, K]$ is bounded by Gaussian tails at the chosen symbol scale t .
- **Grid lift.** Lipschitz continuity of P_A yields $\lambda_{\min}(T_M[P_A]) \geq c_0(K) - C_{\text{SB}} \omega_{P_A}(1/(2M))$, transferring positivity from grid points to $\mathcal{W}_K$.
- **Compact limit.** Along chains $K_1 < K_2 < \dots$ with monotone schedules, positivity is preserved and passes to the limit space.

The TPTP inputs and logs for these checks are archived in `proofs/T5_global_transfer/`.

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